

Huawei Datacom Certification Training

HCIA-Datacom

Lab Guide

Issue: 2.0



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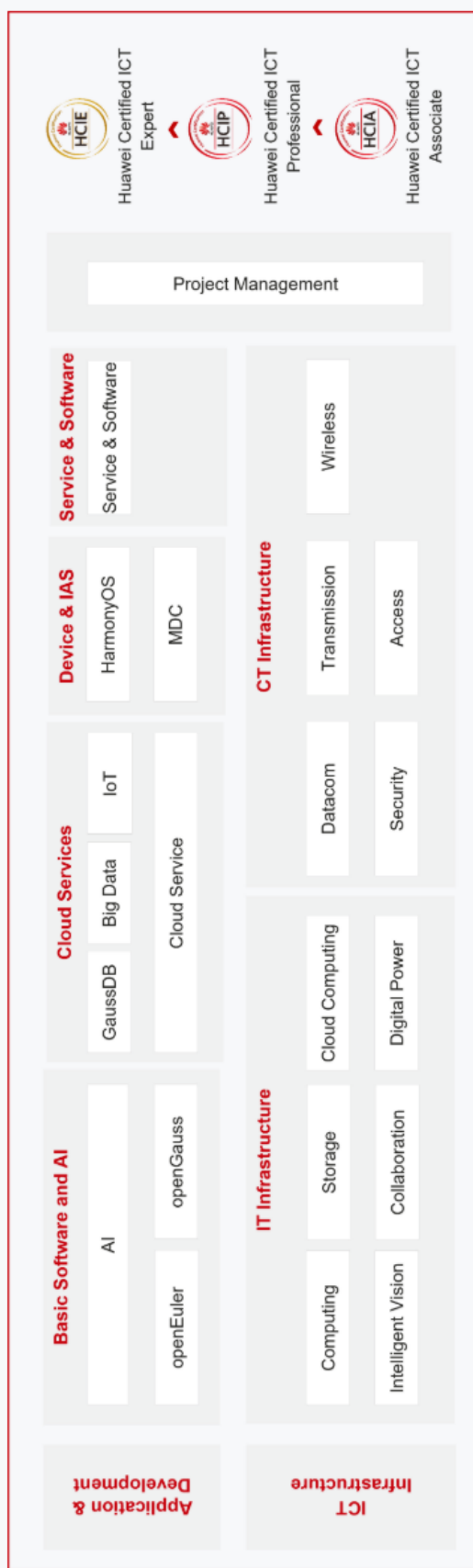
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About This Document

Overview

This document is an HCIA-Datcom training course. It is intended for trainees who are preparing for the HCIA-Datcom exam or readers who want to understand routing and switching fundamentals, data center fundamentals, WLAN fundamentals, network security and service basics, and network O&M and troubleshooting basics.

Background Knowledge Required

This course is for Huawei's basic certification. To understand the content better, ensure that you meet the following requirements:

- Have basic computer skills.
- Have basic understanding of data communication.

Common Icons



Switch



Router



PC



WAC



AP



STA

Experiment Environment Overview

This document is written based on the Huawei datacom simulator **eNSP Pro**. Datacom Simulator is a training, certification, and learning solution of Huawei datacom product line. It provides a one-stop datacom product simulator environment for users.

Simulator Name	Version
eNSP Pro	V100R002C10SPC001

Note:

1. The MAC addresses, ESNs, and IP addresses obtained through DHCP displayed in this lab manual are dynamically obtained. Please refer to the actual environment for the actual information.
2. You need to set a password when logging in to ENSP-LSW, ENSP-AR, and ENSP-AC for the first time. The password is user-defined.

Please configure the login password (8-16)

Enter Password:

Confirm Password:

Comparison with Real-Device Environment Capabilities

Checking Whether All Devices Are Available

Before starting the experiment, check whether all required devices are ready. The following table lists the required devices.

Lab Name	Real-Device Environment	Datacom Simulator
1 Basic Operations on Huawei Network Devices	Supported	Supported
2 VLAN Configuration	Supported	Supported (MAC address-based VLAN division is not supported)
3 Spanning Tree Protocol Configuration	Supported	Supported
4 Ethernet Link Aggregation Configuration	Supported	Supported
5 Direct Route Configuration	Supported	Supported

6 Static Route Configuration	Supported	Supported
7 IP Address Autoconfiguration	Supported	Supported
8 OSPF Configuration	Supported	Supported
9 IPv6 Address Configuration	Supported	Supported
10 ACL Configuration	Supported	Supported
11 File Management Configuration	Supported	Supported
12 Remote Login Configuration	Supported	Supported
13 NAT Configuration	Supported	Supported
14 WAC + Fit AP Networking Configuration	Supported	Supported
15 Network Programming and Automation	Supported	Not Supported
16 Network Troubleshooting	Supported	Supported
17 Campus Network Deployment	Supported	Supported

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1 Basic Operations on Huawei Network Devices

1.1 Introduction

1.1.1 About This Lab

In this lab activity, you will learn about basic operations of Huawei YunShan OS by configuring Huawei devices.

1.1.2 Objectives

- Understand the meaning of command views and how to access and exit command views.
- Understand common commands.
- Understand how to use the command online help.
- Learn how to undo a command.
- Learn how to use command shortcut keys.

1.1.3 Networking Topology



Router

Figure 1-1 Lab topology for basic operations on Huawei network devices

The router used in this lab is ENSP-AR.

1.1.4 Background

As shown in the networking diagram, the router is a new router without any configuration. You need to start the router and perform some initialization operations on it.

1.2 Task Configuration

1.2.1 Configuration Roadmap

1. Complete basic configurations, such as the device name and router interface IP address.
2. Save device configurations.
3. Restart the device.

1.2.2 Configuration Procedure

Step 1 Log in to the router's CLI through the console port and set a password.

Please configure the login password (8-16)

Enter Password: **Huawei@123**

Confirm Password: **Huawei@123**

Info: Save the password now. Please wait for a moment.

Info: The max number of VTY users is 21, the number of current VTY users online is 0, and total number of terminal users online is 1.

The current login time is XXXX-XX-XX XX:XX:XX.

Step 2 Display the basic device information.

Display device version information.

<HUAWEI> display version

Huawei YunShan OS

Version 1.24.1.1 (**AR6700 V600R024C10SPC100**)

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HUAWEI AR uptime is 0 day, 0 hour, 31 minutes

MPU(Master) 0 : uptime is 0 day, 0 hour, 30 minutes

StartupTime XXXX/XX/XX 03:44:27

Memory Size : 4096 M bytes

Flash Size : 0 M bytes

MPU version information:

```
1.PCB      Version : AR-SRU614H VER A
2.MAB      Version : 0
3.Board    Type    : AR
4.BIOS     Version : 0
5.CPLD     Version : 0
```

The preceding command output shows that the software version of the device is V600R024C00SPC100, the device model is AR6700, and the device has been running for 31 minutes.

Step 3 Complete basic device configurations.

Change the router name to **Datcom-Router**.

```
<HUAWEI> system-view
```

Enter system view, return user view with return command.

```
[HUAWEI]
```

You have entered the system view from the user view.

```
[HUAWEI] sysname Datcom-Router
```

```
[Datcom-Router]
```

The device name has been changed to **Datcom-Router**.

Huawei devices provide a wide variety of functions and accordingly numerous configuration and query commands. To simplify the use of such commands on the Huawei devices, these commands are registered in different command views based on their functions. To configure a function, you need to enter the required command view and run the required commands.

Enter the interface view and configure an IP address for the interface.

```
[Datcom-Router] inter //Press Tab to complete the command.
```

```
[Datcom-Router] interface //interface is the only available keyword.
```

```
[Datcom-Router] interface g //Press Tab to complete the command.
```

```
[Datcom-Router] interface GE //GE is the only available keyword.
[Datcom-Router] interface GE 0/0/1 //Complete the command.
```

Enter the first several letters of a keyword in a command and press **Tab** to display a complete keyword. The first several letters must uniquely identify the keyword. If they do not identify a specific keyword, press **Tab** continuously until the desired keyword is displayed. For example:

When you enter **inter** and press **Tab**, the command is autocompleted as **interface** since only the **interface** command starts with **inter**. The command does not change if you press **Tab** multiple times.

```
[Datcom-Router-GE0/0/1]
```

The GE 0/0/1 interface view is displayed.

```
[Datcom-Router-GE0/0/1] undo portswitch
```

The **portswitch** command changes the working mode of an Ethernet interface from Layer 3 mode to Layer 2 mode.

The **undo portswitch** command changes the working mode of an Ethernet interface from Layer 2 mode to Layer 3 mode.

By default, GE0/0/1 on the ENSP-AR is a Layer 3 interface and can be directly configured with an IP address.

```
[Datcom-Router-GE0/0/1] i?
icmp          identity
ifg           ifit
igmp          ip
ipsec         ipv6
isis
```

If you enter only the first or first several characters of a command keyword, you can use the context-sensitive help function to obtain all the keywords that begin with a character or character string. The meaning of each keyword will also be displayed. For example:

In the GE0/0/1 interface view, enter **i** and a question mark (?) to display the options of all commands starting with **i** in the current view. You can press **Tab** to complete the command or manually enter the complete command based on the help information. **icmp** and **igmp** are keywords.

[Datacom-Router-GE0/0/1] ip ?

address	Set the IP address of an interface
binding	Enable binding of an interface with a VPN instance
forward-broadcast	Specify IP directed broadcast information
option	IP option
urpf	Unicast reverse path forwarding check
verify	IP verification

When you enter some keywords of a command and a question mark (?) separated by a space, all keywords associated with this command, as well as simple descriptions, are displayed. For example:

If you enter **ip**, a space, and a question mark (?), all commands containing keyword **ip** and the corresponding descriptions are displayed.

[Datacom-Router-GE0/0/1] ip address ?

X.X.X.X	IP address
bootp-alloc	IP address allocated by BOOTP
dhcp-alloc	IP address allocated by DHCP
unnumbered	Share an address with another interface

[Datacom-Router-GE0/0/1] ip address 192.168.1.1 ?

INTEGER<0-32>	Length of the IP address mask
X.X.X.X	IP address mask

[Datacom-Router-GE0/0/1] ip address 192.168.1.1 24 ?

sub	Indicate a subordinate address
tag	Match tag of the route

<cr>

<cr> indicates that there is no keyword or parameter in this position. You can press **Enter** to run this command.

```
[Datacom-Router-GE0/0/1] dis this  
  
#  
  
interface GE0/0/1  
  
ip address 192.168.1.1 255.255.255.0  
  
#  
  
return
```

The **display this** command displays the configuration in the current view. Effective parameters set to their defaults are not displayed. Configured parameters that are not committed successfully are not displayed, either. The **display this** command is used to check the configuration.

You do not need to enter complete keywords if the entered characters can match a unique keyword in the current view. This function improves efficiency. For example:

The **dis this** command can be run on an interface because only the **display this** command matches the entered characters in the current view. Similarly, the **dis cu** or **d cu** command can also be run because they are equivalent to **display current-configuration** command.

```
[Datacom-Router-GE0/0/1] quit
```

The **quit** command returns a device from the current view to a lower-level view. If the current view is the user view, this command exits from the system.

Undo the IP address configuration because the IP address needs to be configured on GE0/0/2.

```
[Datacom-Router] interface GE 0/0/1  
  
[Datacom-Router-GE0/0/1] undo ip address  
  
[Datacom-Router-GE0/0/1] quit
```

To do so, you must delete the IP address configuration of GE0/0/1. Otherwise, an IP address conflict occurs and the configuration fails.

If a command begins with the keyword **undo**, it is an **undo** command. An **undo** command is generally used to restore a default configuration, disable a function, or delete a configuration. Almost each configuration command has a corresponding **undo** format.

```
[Datacom-Router] interface GE 0/0/2  
  
[Datacom-Router-GE0/0/2] ip address 192.168.1.1 24  
  
[Datacom-Router-GE0/0/2] quit
```

The IP address of GE 0/0/2 has been set to 192.168.1.1/24.

Display the current device configuration.

```
[Datacom-Router] display current-configuration

!Software Version V600R024C10SPC100

!Last configuration was updated at XXXX-XX-XX 6 06:33:34+00:00

!Last configuration was saved at XXXX-XX-XX 06:25:36+00:00

!ase VRPV800R006C00B016D0127-0.0.1

!autom VRPV800R006C00B016D0127-0.0.1

!fwm_ifm_mbse VRPV800R006C00B016D0127-0.0.1

!grpc VRPV800R006C00B016D0127-0.0.1

!md_tlm VRPV800R006C00B016D0127-0.0.1

!moe-objmesh VRPV800R006C00B016D0127-0.0.1

!telem VRPV800R006C00B016D0127-0.0.1

#

pki realm default

#

sysname Datacom-Router

#

undo ftp server source all-interface

undo ftp ipv6 server source all-interface

#

ssl policy default

pki-domain default

ssl minimum version tls1.2

cipher-suite exclude key-exchange rsa

cipher-suite exclude cipher mode cbc
```

---- More ----

When a command output occupies more than one screen, the system pauses after each screen, facilitating your information check. If ---- **More** ---- is displayed at the bottom of the command output, you can:

1. Press **Ctrl+C** or **Ctrl+Z** to stop the display or command execution.
2. Press the space bar to display the next screen.
3. Press **Enter** to display the next line.

Step 4 Save the current configuration of the device.

Return to the user view.

```
[Datacom-Router] quit
<Datacom-Router>
```

In addition to the **quit** command, you can also:

1. Run the **return** command to return to the user view from any view.
2. Press **Ctrl+Z** to return to the user view from any view.

Save the configuration.

```
<Datacom-Router> save

Warning: The current configuration will be written to the device. Continue? [Y/N] : y //Enter y to
confirm your operation.

Now saving the current configuration to the slot 0

Info: Save the configuration successfully.
```

Configuration changes must be saved in the configuration file to survive system restart.

You can run the **save** command to save the current configuration to the default path and overwrite the original configuration file. You can also run the **save configuration-file** command to save the current configuration to a specified file in the storage device.

Compare the current configuration with the configuration used for the next startup.

```
<Datacom-Router> compare configuration

Building configuration....

Info: The current configuration is the same as the next startup configuration file.
```

The current configuration is the same as the configuration in the next startup configuration file.

Step 5 Perform operations on the file system.

Save the current configuration and name the configuration file **test.cfg**.

```
<Datacom-Router> save test.cfg

Warning: Are you sure to save the configuration to flash:/test.cfg? [Y/N] : y

Now saving the current configuration to the slot 0

Info: Save the configuration successfully.
```

List all the files in the current directory again.

```
<Datacom-Router> dir

Directory of flash:/


```

Idx	Attr	Size(Byte)	Date	Time	FileName
0	drwx	-	Dec 26 XXXX	06:16:43	\$_autobackup
1	dr-x	-	Sep 24 XXXX	01:00:00	\$_install_mod
2	dr-x	-	Dec 26 XXXX	06:16:13	\$_license
3	dr-x	-	Dec 26 XXXX	06:16:41	\$_startup
4	dr-x	-	Dec 26 XXXX	06:17:05	\$_system
5	dr-x	-	Dec 26 XXXX	06:17:05	\$_user
6	-rw-	5,759	Dec 26 XXXX	06:37:32	1.cfg
7	-rw-	2,311,364	Dec 26 XXXX	06:17:21	application.txt
8	-rw-	2,311,364	Dec 26 XXXX	06:17:21	application_touch.txt
9	drwx	-	Dec 26 XXXX	06:17:11	default-sdb
10	-rw-	3,219	Dec 26 XXXX	06:40:31	device.sys
11	drwx	-	Dec 26 XXXX	06:17:25	dhcp
12	lrwx	16	Dec 26 XXXX	06:16:04	ensp-ar.cc -> system file
13	-rw-	2,475	Dec 26 XXXX	06:20:41	env_cfg
14	drwx	-	Dec 26 XXXX	06:17:05	full_kpi
15	drwx	-	Dec 26 XXXX	06:16:25	hips

16	drwx	-	Dec 26 XXXX 06:16:26	hrp
17	drwx	-	Dec 26 XXXX 06:17:10	idm
18	drwx	-	Dec 26 XXXX 06:19:25	idname
19	drwx	-	Dec 26 XXXX 06:16:26	loc
20	dr-x	-	Dec 26 XXXX 06:36:40	logfile
21	-rw-	296	Dec 26 XXXX 06:40:27	mem_env_cfg
22	drwx	-	Dec 26 XXXX 06:16:28	nlog
23	drwx	-	Dec 26 XXXX 06:16:27	pki
24	drwx	-	Dec 26 XXXX 06:16:27	security
25	-rw-	136	Dec 26 XXXX 06:28:20	stats_tlv.dat
26	-rw-	14,155	Dec 26 XXXX 06:17:21	subcategory.txt
27	-rw-	5,759	Dec 26 XXXX 06:40:31	test.cfg
28	drwx	-	Dec 26 XXXX 06:17:11	update
29	drwx	-	Dec 26 XXXX 06:17:11	webm
30	drwx	-	Dec 26 XXXX 06:17:05	ztp

152,579,072 KB total (143,734,040 KB free)

The configuration file is saved successfully.

Set the file as the configuration file for the next startup.

```
<Datcom-Router> startup saved-configuration test.cfg
```

Info: Succeeded in setting the configuration for booting system.

Warning: The next startup configuration file may be overwritten 30 minutes later because auto-save is enabled

The **startup saved-configuration** command specifies the system configuration file for the next startup. The configuration file for the next startup is successfully set.

Check the files used for the next startup.

```
<Datcom-Router> display startup
```



```

MainBoard:

Configured startup system software:      flash:/ensp-ar.cc
Startup system software:                  flash:/ensp-ar.cc
Next startup system software:             flash:/ensp-ar.cc
Startup saved-configuration file:         NULL
Next startup saved-configuration file:    flash:/test.cfg
Startup paf file:                         default
Next startup paf file:                    default
Startup patch package:                    NULL
Next startup patch package:               NULL
Startup feature software:                 NULL
Next startup feature software:             NULL

```

The **display startup** command displays the system software, backup system software, configuration file, license file, and patch file for the current and next startup.

Clear the configuration file to be loaded for the next startup.

```

<Datcom-Router> reset saved-configuration

Warning: The action will delete the saved configuration on the device.

The configuration will be erased to reconfigure. Continue? [Y/N] : y

Warning: Now the configuration on the device is being deleted.

Info: Succeeded in clearing the configuration on the device.

```

The **reset saved-configuration** command clears the startup configuration file on the device.

Step 6 Restart the device.

```

<Datcom-Router> reboot

MPU 0:

Next startup system software: flash:/ensp-ar.cc

Next startup saved-configuration file: NULL

Next startup paf file: default

```

Next startup patch package: NULL

Warning: The current configuration will be saved to the next startup saved-configuration file. Continue?
[Y/N]: **y**

Now saving the current configuration.

Save the configuration successfully.

Warning: The system will reboot. Continue? [Y/N]: **y**

The **reboot** command restarts a device. Before the device restarts, you are prompted to save configurations. The system is restarting. Wait for about 10 minutes. (If the device fails to restart properly, right-click the device and click **Start** to restart the device.)

Please Press ENTER.

Password: **Huawei@123**

Info: The max number of VTY users is 21, the number of current VTY users online is 0, and total number of terminal users online is 1.

The current login time is XXXX-XX-XX XX:XX:XX.

<Datcom-Router>

After the device is restarted, enter the password to log in to the device.

1.3 Quiz

1. Familiarize yourself with the function keys of Huawei YunShan OS based on the function key list in the appendix.
2. In step 5, the **reset saved-configuration** command is used to clear the configuration. Why is the configuration of the device retained after the device is restarted?

Answer:

This is because we choose to save the configuration file before the device is restarted.

Warning: The current configuration will be saved to the next startup saved-configuration file. Continue?
[Y/N] : **y**

1.4 Appendix

Table 1-1 System function keys

Key	Function
<Ctrl+A>	Moves the cursor to the beginning of the current command line.
<Ctrl+B>	Moves the cursor back one character.
<Ctrl+C>	Stops performing the current functions.
<Ctrl+D>	Deletes the character where the cursor is located.
<Ctrl+E>	Moves the cursor to the end of the current command line.
<Ctrl+F>	Moves the cursor forward one character.
<Ctrl+H>	Deletes the character before the cursor.
<Ctrl+I>	Functions the same as the Tab key.
<Ctrl+J>	Functions the same as the Enter key.
<Ctrl+K>	Terminates the connection of an outgoing call during connection establishment.
<Ctrl+M>	Functions the same as the Enter key.
<Ctrl+N>	Displays the next command in the command history.
<Ctrl+P>	Displays the previous command in the command history.
<Ctrl+Q>	Specifies the escape operation.
<Ctrl+R>	Re-displays information about the current command line.
<Ctrl+T>	Terminates the connection of an outgoing call.
<Ctrl+V>	Pastes the text of the clipboard.
<Ctrl+W>	Deletes the character string (word) to the left of the cursor.
<Ctrl+X>	Deletes all characters before the cursor.
<Ctrl+Y>	Deletes all characters following the cursor and the character where the cursor is located.
<Ctrl+Z>	Returns to the user view.
<Ctrl+J>	Terminates the connection of an incoming call or redirects the connection.
<Esc+B>	Moves the cursor back one character string (word).
<Esc+D>	Deletes the character string (word) to the right of the cursor.

Key	Function
<Esc+F>	Moves the cursor forward one character string (word).
<Esc+N>	Moves the cursor downwards a line.
<Esc+P>	Moves the cursor upwards a line.
<Esc+<>	Positions the cursor at the beginning of the text to be copied.
<Esc+>>	Positions the cursor at the end of the text to be copied.

2 VLAN Configuration

2.1 Introduction

2.1.1 About This Lab

Ethernet technology allows data communication over shared media through Carrier Sense Multiple Access with Collision Detection (CSMA/CD). When an Ethernet network has a large number of hosts, collision becomes a serious problem and can lead to broadcast storms. This can degrade network performance or even result in a complete breakdown. Using switches for connection can mitigate collisions, but broadcast packets cannot be isolated and the network quality cannot be improved.

To solve this problem, VLAN technology is introduced. This technology divides a physical LAN into multiple VLANs. Each VLAN functions as a separate broadcast domain, with hosts in the same VLAN can directly communicate with one another, while those in different VLANs cannot. In this way, broadcast packets are confined within a single VLAN.

In this lab activity, you will learn how to configure VLAN on Huawei switches.

2.1.2 Objectives

- Learn how to create a VLAN.
- Learn how to configure access, trunk, and hybrid interfaces.
- Learn how to configure interface-based VLAN assignment.
- Learn how to check the MAC address table and VLAN information.

2.1.3 Networking Topology

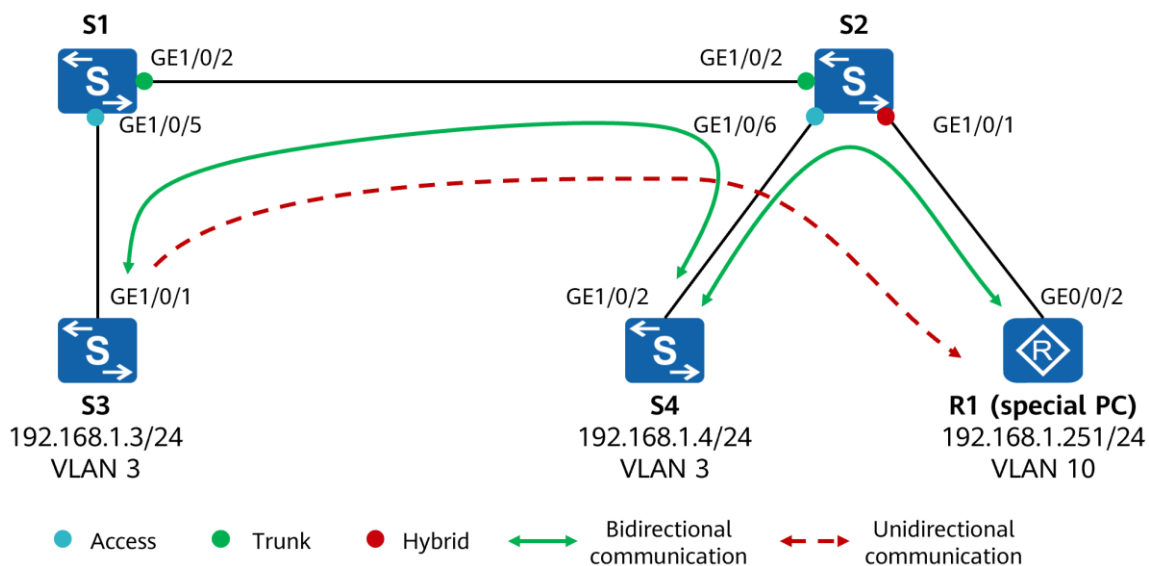


Figure 2-1 Lab topology for VLAN configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

2.1.4 Background

A company needs to divide its Layer 2 network into multiple VLANs based on service requirements. R1 simulates a special PC and can receive data packets sent by other terminals.

As shown in the lab topology, you can configure interface-based VLAN assignment on S1 and S2 to add interfaces connected to users with the same services to the same VLAN.

2.2 Task Configuration

2.2.1 Configuration Roadmap

1. Create VLANs.
2. Configure interface-based VLAN assignment on switches.

2.2.2 Configuration Procedure

Step 1 Configure device names and shut down unused interfaces.

Configure device names.

The detailed configuration is not provided here.

Switch Layer 2 interfaces to Layer 3 interfaces.

```
[S3] interface GE 1/0/1  
[S3-GE1/0/1] undo portswitch  
[S3-GE1/0/1] quit
```

```
[S4] interface GE 1/0/2  
[S4-GE1/0/2] undo portswitch  
[S4-GE1/0/2] quit
```

Step 2 Configure device IP addresses.

Configure IP addresses for interfaces.

```
[R1] interface GE0/0/2  
[R1-GE0/0/2] ip address 192.168.1.251 24  
[R1-GE0/0/2] quit
```

```
[S3] interface GE1/0/1  
[S3-GE1/0/1] ip address 192.168.1.3 24  
[S3-GE1/0/1] quit
```

```
[S4] interface GE1/0/2  
[S4-GE1/0/2] ip address 192.168.1.4 24  
[S4-GE1/0/2] quit
```

Verify the connectivity.

```
[R1] ping 192.168.1.3  
  
PING 192.168.1.3: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.1.3: bytes=56 Sequence=1 ttl=254 time=1 ms
Reply from 192.168.1.3: bytes=56 Sequence=2 ttl=254 time=1 ms
Reply from 192.168.1.3: bytes=56 Sequence=3 ttl=254 time=1 ms
Reply from 192.168.1.3: bytes=56 Sequence=4 ttl=254 time=1 ms
Reply from 192.168.1.3: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
--- 192.168.1.3 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
[R1] ping 192.168.1.4
```

```
PING 192.168.1.4: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.1.4: bytes=56 Sequence=1 ttl=254 time=1 ms
```

```
Reply from 192.168.1.4: bytes=56 Sequence=2 ttl=254 time=1 ms
```

```
Reply from 192.168.1.4: bytes=56 Sequence=3 ttl=254 time=1 ms
```

```
Reply from 192.168.1.4: bytes=56 Sequence=4 ttl=254 time=1 ms
```

```
Reply from 192.168.1.4: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
--- 192.168.1.4 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
# Check ARP entries on R1.
```

```
[R1] display arp
```


ARP Entry Types: D - Dynamic, S - Static, I - Interface

EXP: Expire-time

IP ADDRESS	MAC ADDRESS	EXPIRE(M)	TYPE	INTERFACE	VPN-INSTANCE
VLAN/CEVLAN					
192.168.1.1	facf-999b-0020		I -	MEth0/0/0	_management_vpn_
192.168.1.251	facf-999b-0022		I -	GE0/0/2	
192.168.1.3	facf-999b-0052	20	D-0	GE0/0/2	
192.168.1.4	facf-999b-0062	20	D-0	GE0/0/2	
Total:4	Dynamic:2	Static:0	Interface:2		

The command output shows that the MAC address of R1 is facf-999b-0022, the MAC address of S3 is facf-999b-0052, and the MAC address of S4 is facf-999b-0062.

Check the MAC address table on S2.

[S2] display mac-address

Flags: * - Backup

- forwarding logical interface, operations cannot be performed based on the interface.

BD : bridge-domain Age : dynamic MAC learned time in seconds

MAC Address	VLAN/VSI/BD	Learned-From	Type	Age
facf-999b-0062	1/-/-	GE1/0/6	dynamic	766
facf-999b-0052	1/-/-	GE1/0/2	dynamic	508
facf-999b-0022	1/-/-	GE1/0/1	dynamic	601

...

Total items: 4

The command output shows that the MAC address corresponding to GE1/0/6 belongs to S4, the MAC address corresponding to GE1/0/2 belongs to S3, and the MAC address corresponding to GE1/0/1 belongs to R1. The three MAC addresses are dynamically learned.

Note: If the MAC address table information is incorrect, you can run the following command to update the MAC address entries.

```
<S2> reset mac-address
```

Step 3 Create VLANs.

Create VLANs 3 and 10 on S1 and S2.

```
[S1] vlan batch 3 10
```

VLANs 3 and 10 are created successfully.

The **vlan** *vlan-id* command creates a VLAN and displays the VLAN view. If the VLAN exists, the VLAN view is displayed.

The **vlan batch** { *vlan-id1* [**to** *vlan-id2*] } command creates VLANs in batches.

```
[S2] vlan batch 3 10
```

Step 4 Configure interface-based VLAN assignment.

Configure the interfaces connecting S1 and S2 to terminals as access interfaces and add the interfaces to corresponding VLANs.

```
[S1] interface GE 1/0/5
```

```
[S1-GE1/0/5] port link-type access
```

The **port link-type** { **access** | **hybrid** | **trunk** } command configures the link type of an interface. You can set the interface type to access, trunk, or hybrid.

Supplementary note:

If the network topology changes, the link types of Ethernet interfaces need to be reconfigured, which is complex. To simplify configuration, you can configure Link-type Negotiation Protocol (LNP) on Ethernet interfaces so that they can automatically negotiate the link type as access or trunk and join corresponding VLANs.

1. When the link type of an Ethernet interface is negotiated as access, the interface is added to VLAN 1 by default.

2. When the link type of an Ethernet interface is negotiated as trunk, the interface is added to VLANs 1 to 4094 by default.

The **port link-type { negotiation-desirable | negotiation-auto }** command sets the dynamic negotiation mode of an interface. If **negotiation-desirable** is specified, the interface proactively sends negotiation packets. If **negotiation-auto** is specified, the interface does not proactively send negotiation packets.

```
[S1-GE1/0/5] port default vlan 3
[S1-GE1/0/5] quit
```

The **port default vlan *vlan-id*** command configures the default VLAN of an interface.

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] port link-type access
[S2-GE1/0/1] port default vlan 10
[S2-GE1/0/1] quit
[S2] interface GE 1/0/6
[S2-GE1/0/6] port link-type access
[S2-GE1/0/6] port default vlan 3
[S2-GE1/0/6] quit
```

Configure the interfaces connecting S1 and S2 as trunk interfaces and allow only packets from VLAN 3 and VLAN 10 to pass through.

```
[S1] interface GE1/0/2
[S1-GE1/0/2] port link-type trunk
[S1-GE1/0/2] port trunk allow-pass vlan 3 10
```

The **port trunk allow-pass vlan** command adds a trunk interface to a VLAN.

```
[S1-GE1/0/2] undo port trunk allow-pass vlan 1
[S1-GE1/0/2] quit
```

The **undo port trunk allow-pass vlan** command deletes a trunk interface from a VLAN. By default, VLAN 1 is in the list of allowed VLANs. If VLAN 1 is not used for any service, it needs to be deleted for security purposes.

```
[S2] interface GE1/0/2

[S2-GE1/0/2] port link-type trunk

[S2-GE1/0/2] port trunk allow-pass vlan 3 10

[S2-GE1/0/2] undo port trunk allow-pass vlan 1

[S2-GE1/0/2] quit
```

Verify the connectivity on S4.

```
[S4] ping 192.168.1.3

PING 192.168.1.3: 56 data bytes, press CTRL_C to break

  Reply from 192.168.1.3: bytes=56 Sequence=1 ttl=254 time=1 ms

  Reply from 192.168.1.3: bytes=56 Sequence=2 ttl=254 time=1 ms

  Reply from 192.168.1.3: bytes=56 Sequence=3 ttl=254 time=1 ms

  Reply from 192.168.1.3: bytes=56 Sequence=4 ttl=254 time=1 ms

  Reply from 192.168.1.3: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 192.168.1.3 ping statistics ---

  5 packet(s) transmitted

  5 packet(s) received

  0.00% packet loss

 round-trip min/avg/max = 1/1/1 ms
```

S3 and S4 belong to VLAN 3 and can ping each other.

```
[S4] ping 192.168.1.251

PING 192.168.1.251: 56 data bytes, press CTRL_C to break

  Request time out

  Request time out

  Request time out

  Request time out

  Request time out
```

```
--- 192.168.1.251 ping statistics ---
```

```
5 packet(s) transmitted
```

```
0 packet(s) received
```

```
100.00% packet loss
```

S4 and R1 belong to different VLANs and therefore cannot ping each other.

Clear ARP entries on S4 and R1.

```
[S4] quit
```

```
<S4> reset arp all
```

```
Warning: This operation will reset all dynamic ARP entries, and may cause services to be interrupted,
continue? [Y/N] : y
```

The **reset arp all** command clears all ARP entries.

```
[R1] quit
```

```
<R1> reset arp all
```

```
Warning: This operation will reset all dynamic ARP entries, and may cause services to be interrupted,
continue? [Y/N] : y
```

Ping 192.168.1.251 from S4 and check ARP entries on S4 and R1.

```
<S4> ping 192.168.1.251
```

```
PING 192.168.1.251: 56 data bytes, press CTRL_C to break
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
--- 192.168.1.251 ping statistics ---
```

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss

<S4> display arp

ARP Entry Types: D - Dynamic, S - Static, I - Interface, RD - Redirect

EXP: Expire-time VLAN: VLAN or Bridge Domain

IP ADDRESS	MAC ADDRESS	EXP(M)	TYPE/VLAN	INTERFACE	VPN-INSTANCE
192.168.1.4	facf-999b-0062		I	GE1/0/2	
192.168.1.3	facf-999b-0052	17	D	GE1/0/2	
Total:1	Dynamic:0	Static:0	Interface:1	Redirect:0	

The MAC address corresponding to 192.168.1.251 is not displayed on S4.

<R1> display arp

ARP Entry Types: D - Dynamic, S - Static, I - Interface

EXP: Expire-time

IP ADDRESS	MAC ADDRESS	EXP(M)	TYPE/VLAN	INTERFACE	VPN-INSTANCE
192.168.1.251	facf-999b-0022		I	GE0/0/2	
192.168.1.1	facf-999b-0020		I	MEth0/0/0	_management_vpn_
Total:2	Dynamic:0	Static:0	Interface:2		

The MAC address corresponding to 192.168.1.4 is not displayed on R1.

Analyze the problem based on ARP entries of S4 and R1. Although S4 sends an ARP request packet, the request packet is not sent to R1 because S4 and R1 belong to different VLANs and broadcast packets are blocked.

Step 5 Change GE1/0/1 on S2 to a hybrid interface.

Change the link type of GE1/0/1 on S2.

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] port link-type hybrid
[S2-GE1/0/1] port hybrid pvid vlan 10
[S2-GE1/0/1] port hybrid untagged vlan 3 10
[S2-GE1/0/1] quit
```

The **port hybrid vlan** command adds a hybrid interface to a specified existing VLAN. The **untagged/tagged** parameter specifies whether the outgoing packets of the interface carry VLAN tags.

After the configuration is complete, GE1/0/1 on S2 removes the VLAN tags from the data frames of VLAN 3 and VLAN 10 before sending them.

Ping 192.168.1.251 from S4 and check ARP entries on S4 and R1.

```
<S4> ping 192.168.1.251

PING 192.168.1.251: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

--- 192.168.1.251 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

```
<S4> display arp
```

ARP Entry Types: D - Dynamic, S - Static, I - Interface, RD - Redirect

EXP: Expire-time VLAN: VLAN or Bridge Domain

IP ADDRESS	MAC ADDRESS	EXP(M)	TYPE/VLAN	INTERFACE	VPN-INSTANCE
192.168.1.4	facf-999b-0062		I	GE1/0/2	
192.168.1.3	facf-999b-0052	1	D	GE1/0/2	
192.168.1.251	Incomplete	1	D	GE1/0/2	

Total:3	Dynamic:2	Static:0	Interface:1	Redirect:0	

The MAC address corresponding to 192.168.1.251 is not displayed on S4.

Note: After the **ping** command is run, wait for several seconds. The ARP table of S4 does not contain 192.168.1.254.

```
<R1> display arp
```

ARP Entry Types: D - Dynamic, S - Static, I - Interface

EXP: Expire-time

IP ADDRESS	MAC ADDRESS	EXP(M)	TYPE/VLAN	INTERFACE	VPN-INSTANCE
192.168.1.251	facf-999b-0022		I	GE0/0/2	
192.168.1.4	facf-999b-0062	16	D	GE0/0/2	
192.168.1.1	facf-999b-0020		I	MEth0/0/0	_management_vpn_

Total:3	Dynamic:1	Static:0	Interface:2		

On R1, you can see that the MAC address corresponding to 192.168.1.4 is facf-999b-0062. Analyze why R1 can obtain the ARP entry of S4 but S4 cannot obtain the ARP entry of R1 based on ARP entries of S4 and R1.

1. S4 broadcasts an ARP request packet. After the data frame reaches S2, S2 adds the VLAN tag of the interface, that is, VLAN 3 tag, to the data frame.
2. Then, S2 sends the broadcast packet to all interfaces that belong to VLAN 3. When sending the data frame of VLAN 3, GE1/0/1 on S2 removes the tag from the data frame. Therefore, the broadcast frame reaches R1.
3. After receiving the ARP request packet, R1 records the MAC address corresponding to 192.168.1.4 and unicasts an ARP reply packet.
4. After the reply packet from R1 reaches S2, S2 adds the VLAN 10 tag to the packet. Because S4 belongs to VLAN 3, the reply packet from R1 does not reach S4.

Check VLAN information on S2.

```
[S2] display vlan
```

The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning; STAT: Statistic;

BC: Broadcast; MC: Multicast; UC: Unknown-unicast;

FWD: Forward; DSD: Discard;

VID	Ports
1	UT:GE1/0/1(U) GE1/0/3(D) GE1/0/4(D) GE1/0/5(D) GE1/0/7(D) GE1/0/8(D) GE1/0/9(D) GE1/0/10(D)
3	UT:GE1/0/1(U) GE1/0/6(U) TG:GE1/0/2(U)
10	UT:GE1/0/1(U) TG:GE1/0/2(U)

VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description

1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
3	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0003
10	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0010

UT and **TG** before interfaces indicate that VLAN tags are removed (untagged) and added (tagged) when data frames are sent, respectively.

U and **D** after interfaces indicate that the interfaces are in Up and Down states, respectively.

The command output shows that GE1/0/1 on S2 removes VLAN tags from data frames of VLAN 3 and VLAN 10 when sending the data frames.

Check the MAC address table on S2.

[S2] display mac-address				
Flags: * - Backup				
# - forwarding logical interface, operations cannot be performed based				
on the interface.				
BD : bridge-domain Age : dynamic MAC learned time in seconds				

MAC Address	VLAN/VSI/BD	Learned-From	Type	Age

facf-999b-0062	3/-/-	GE1/0/6	dynamic	33
facf-999b-0052	3/-/-	GE1/0/2	dynamic	33
facf-999b-0022	10/-/-	GE1/0/1	dynamic	1144
...				

Total items: 4				

The command output shows that S2 learns the MAC addresses of S3 and S4 from VLAN 3 and learns the MAC address of R1 from VLAN 10.

Step 6 Change GE1/0/6 on S2 to a hybrid interface.

```
[S2] interface GE 1/0/6

[S2-GE1/0/6] port link-type hybrid

[S2-GE1/0/6] port hybrid pvid vlan 3

[S2-GE1/0/6] port hybrid untagged vlan 3 10

[S2-GE1/0/6] quit
```

Note: The interfaces here are added to VLAN 10 without tags, so that S4 and R1 can communicate with each other.

Step 7 View configuration information.

Check VLAN information on the switches.

```
[S1] display vlan

The total number of vlans is : 3

-----

U: Up;          D: Down;          TG: Tagged;      UT: Untagged;
MP: Vlan-mapping;      ST: Vlan-stacking;
#: ProtocolTransparent-vlan;  *: Management-vlan;
MAC-LRN: MAC-address learning;  STAT: Statistic;
BC: Broadcast; MC: Multicast;  UC: Unknown-unicast;
FWD: Forward;  DSD: Discard;

-----

VID          Ports
-----
1            UT:GE1/0/1(D)    GE1/0/3(D)    GE1/0/4(D)    GE1/0/6(D)
              GE1/0/7(D)    GE1/0/8(D)    GE1/0/9(D)    GE1/0/10(D)
3            UT:GE1/0/5(U)
              TG:GE1/0/2(U)
10           TG:GE1/0/2(U)
```

VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description
1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
3	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0003
10	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0010

[S2] display vlan

The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning; STAT: Statistic;

BC: Broadcast; MC: Multicast; UC: Unknown-unicast;

FWD: Forward; DSD: Discard;

VID	Ports
1	UT:GE1/0/1(U) GE1/0/3(D) GE1/0/4(D) GE1/0/5(D) GE1/0/6(U) GE1/0/7(D) GE1/0/8(D) GE1/0/9(D) GE1/0/10(D)
3	UT:GE1/0/1(U) GE1/0/4(D) GE1/0/5(D) GE1/0/6(U) TG:GE1/0/2(U)
10	UT:GE1/0/1(U) GE1/0/4(D) GE1/0/5(D) GE1/0/6(U) TG:GE1/0/2(U)

VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description

1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
3	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0003
10	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0010

2.3 Verification

Test the device connectivity and verify the VLAN configuration.

Ping S3 and R1 from S4.

```
<S4> ping 192.168.1.3

PING 192.168.1.3: 56 data bytes, press CTRL_C to break

Reply from 192.168.1.3: bytes=56 Sequence=1 ttl=254 time=1 ms

Reply from 192.168.1.3: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 192.168.1.3: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 192.168.1.3: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 192.168.1.3: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 192.168.1.3 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

```
<S4> ping 192.168.1.251

PING 192.168.1.251: 56 data bytes, press CTRL_C to break

Reply from 192.168.1.251: bytes=56 Sequence=1 ttl=255 time=1 ms

Reply from 192.168.1.251: bytes=56 Sequence=2 ttl=255 time=1 ms
```

```
Reply from 192.168.1.251: bytes=56 Sequence=3 ttl=255 time=1 ms
```

```
Reply from 192.168.1.251: bytes=56 Sequence=4 ttl=255 time=1 ms
```

```
Reply from 192.168.1.251: bytes=56 Sequence=5 ttl=255 time=1 ms
```

```
--- 192.168.1.251 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

Ping S3 from R1.

```
<R1> ping 192.168.1.3
```

```
PING 192.168.1.3: 56 data bytes, press CTRL_C to break
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
Request time out
```

```
--- 192.168.1.3 ping statistics ---
```

```
5 packet(s) transmitted
```

```
0 packet(s) received
```

```
100.00% packet loss
```

Check MAC address tables of switches.

```
[S1] display mac-address
```

```
Flags: * - Backup
```

```
# - forwarding logical interface, operations cannot be performed based
```

on the interface.

BD : bridge-domain Age : dynamic MAC learned time in seconds

MAC Address	VLAN/VSI/BD	Learned-From	Type	Age
facf-999b-0062	3/-/-	GE1/0/2	dynamic	174
facf-999b-0052	3/-/-	GE1/0/5	dynamic	174
facf-999b-0022	10/-/-	GE1/0/2	dynamic	123

...

Total items: 5

[S2] display mac-address

Flags: * - Backup

- forwarding logical interface, operations cannot be performed based

on the interface.

BD : bridge-domain Age : dynamic MAC learned time in seconds

MAC Address	VLAN/VSI/BD	Learned-From	Type	Age
facf-999b-0052	3/-/-	GE1/0/2	dynamic	191
facf-999b-0062	3/-/-	GE1/0/6	dynamic	342
facf-999b-0022	10/-/-	GE1/0/1	dynamic	2604

...

Total items: 3

2.4 Reference Configuration

2.4.1 S1 Configuration

```
#
sysname S1

#
vlan batch 3 10

#
interface GE1/0/2

    port link-type trunk

    undo port trunk allow-pass vlan 1

    port trunk allow-pass vlan 3 10

#
interface GE1/0/5

    port link-type access

    port default vlan 3

#
return
```

2.4.2 S2 Configuration

```
#
sysname S2

#
vlan batch 3 10

#
interface GE1/0/1

    port link-type hybrid

    port hybrid pvid vlan 10
```



```
port hybrid untagged vlan 3 10

#
interface GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3 10
#
interface GE1/0/6
port link-type hybrid
port hybrid pvid vlan 3
port hybrid untagged vlan 3 10
#
return
```

2.4.3 S3 Configuration

```
#
sysname S3
#
interface GE1/0/1
undo portswitch
ip address 192.168.1.3 255.255.255.0
#
return
```

2.4.4 S4 Configuration

```
#
sysname S4
```

```
#  
interface GE1/0/2  
undo portswitch  
ip address 192.168.1.4 255.255.255.0  
#  
return
```

2.4.5 R1 Configuration

```
#  
sysname R1  
#  
interface GE0/0/2  
ip address 192.168.1.251 255.255.255.0  
#  
return
```

2.5 Quiz

1. Why can R1 ping S3 and S4 in Step 2?

Answer:

By default, all interfaces on S1 and S2 belong to the same VLAN (VLAN 1). Therefore, R1 can ping S3 and S4.

2. If R1 and S3 need to ping each other, how can this requirement be met?

Answer:

You can change GE1/0/5 on S1 to a hybrid interface and configure the interface to send data frames from VLAN 10 in untagged mode.

3

Spanning Tree Protocol Configuration

3.1 Introduction

3.1.1 About This Lab

Redundant links are usually used on a switched network to implement link backup and improve network reliability. The use of redundant links, however, may cause loops on the switched network, leading to broadcast storms and unstable MAC address entries. As a result, communication quality deteriorates and communication may even be interrupted. To prevent loops on switched networks, the Spanning Tree Protocol (STP) is introduced.

Similar to the development of many other protocols, STP is updated with the development of networks. STP defined in IEEE 802.1D has evolved to the Rapid Spanning Tree Protocol (RSTP) defined in IEEE 802.1w, and then to the Multiple Spanning Tree Protocol (MSTP) defined in IEEE 802.1s.

In this lab activity, you will learn the basic STP configuration and understand its fundamentals and some features of RSTP.

3.1.2 Objectives

- Learn how to enable and disable STP or RSTP.
- Learn how to change the STP mode of a switch.
- Learn how to change bridge priorities to control the root bridge election.
- Learn how to change port priorities to control the election of the root port and designated port.
- Learn how to change port costs to control the election of the root port and designated port.
- Learn how to configure edge ports.
- Learn how to enable and disable RSTP.

3.1.3 Networking Topology

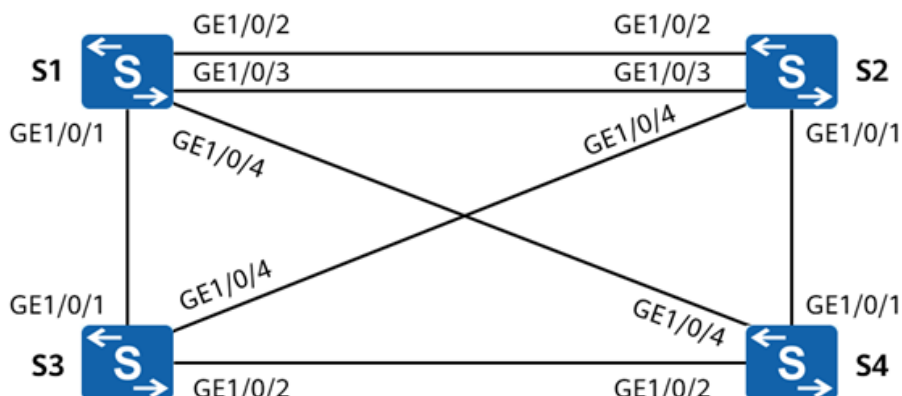


Figure 3-1 Lab topology for spanning tree protocol configuration

The switches used in this lab are ENSP-LSW.

3.1.4 Background

A company adds redundant links on its Layer 2 switched network to improve network reliability. To prevent broadcast storms and MAC address flapping caused by redundant links, a spanning tree protocol needs to be deployed on switches.

Note: The STP calculation result varies according to the MAC addresses of switches in different physical environments. The calculation result in your networking environment may be different.

3.2 Task Configuration

3.2.1 Configuration Roadmap

1. Enable STP on devices.
2. Change bridge priorities to control the root bridge election.
3. Modify port parameters to control the port role.
4. Change the spanning tree working mode to RSTP on devices.
5. Configure RSTP edge ports.

3.2.2 Configuration Procedure

- Step 1 Configure devices to run STP.

Enable STP globally.

```
[S1] stp enable
```

The **stp enable** command enables STP, RSTP, or MSTP on a switching device or a port. By default, STP is enabled on a switching device.

Change the spanning tree working mode to STP.

```
[S1] stp mode stp
```

The **stp mode {mstp | rstp | stp}** command configures the spanning tree working mode of a switching device. By default, the spanning tree working mode of a device is MSTP. The spanning tree working mode of the current device has been changed to STP.

```
[S2] stp mode stp
```

```
[S3] stp mode stp
```

```
[S4] stp mode stp
```

Check the spanning tree status. The following uses S1 as an example.

```
[S1] display stp
```

CIST Global Information:

Mode	:STP	
CIST Bridge	:32768.fa09-eea3-0011	<i>//Bridge ID of the local device</i>
Config Times	:Hello 2s MaxAge 20s FwDly 15s MaxHop 20	
Active Times	:Hello 2s MaxAge 20s FwDly 15s MaxHop 20	
CIST Root/ERPC	:32768.fa09-eea3-0011 / 0 (This bridge is the root)	<i>//Bridge ID of the root bridge and cost of the path to the root bridge</i>
CIST RegRoot/IRPC	:32768.fa09-eea3-0011 / 0 (This bridge is the root)	
CIST RootPortId	:0.0	
BPDU-Protection	:Disabled	

```

TC or TCN received :18

TC count per hello :0

STP Converge Mode :Normal

Share region-configuration :Enabled

Time since last TC :0 days 0h:0m:20s

Number of TC :19

Last TC occurred :GE1/0/4

Topo Change Flag :0

```

CIST Port Information:

```

Port Id :1

Port Name :GE1/0/1

Port State :Forwarding

Port Protocol :Enabled

```

---- More ----

The displayed information shows that the priority of the root bridge on S1 is **32768** and the MAC address is **fa09-eea3-0011**.

Note: The displayed information also includes the status of each port. In the preceding command output, the display is ended by pressing **Ctrl+C**.

Check brief spanning tree information on each switch.

[S1] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	DESI	forwarding	none	20000	disable
0	GE1/0/2	DESI	forwarding	none	20000	disable
0	GE1/0/3	DESI	forwarding	none	20000	disable
0	GE1/0/4	DESI	forwarding	none	20000	disable

[S2] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	DESI	forwarding	none	20000	disable
0	GE1/0/2	ROOT	forwarding	none	20000	disable
0	GE1/0/3	ALTE	discarding	none	20000	disable
0	GE1/0/4	DESI	forwarding	none	20000	disable

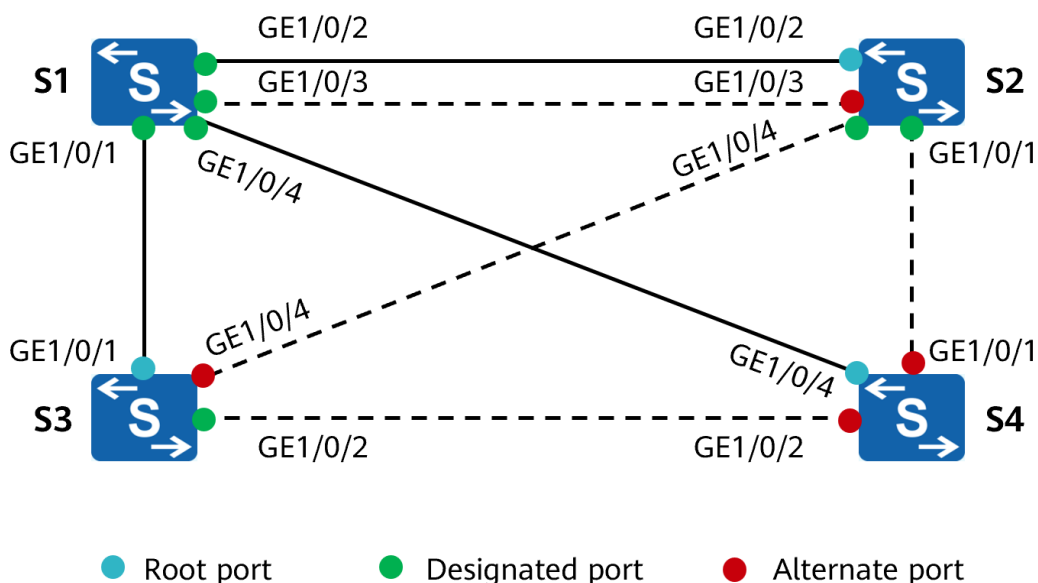
[S3] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	ROOT	forwarding	none	20000	disable
0	GE1/0/2	DESI	forwarding	none	20000	disable
0	GE1/0/4	ALTE	discarding	none	20000	disable

[S4] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	ALTE	discarding	none	20000	disable
0	GE1/0/2	ALTE	discarding	none	20000	disable
0	GE1/0/4	ROOT	forwarding	none	20000	disable

Based on the root bridge ID and port information on each switch, the current topology is as follows:



A dotted line indicates that a link does not forward service data.

Note: This topology is for reference only and may not be the same as the spanning tree topology in the actual lab environment.

Step 2 Modify device parameters to make S1 the root bridge and S2 the secondary root bridge.

Change the bridge priorities of S1 and S2.

```
[S1] stp root primary
```

Owing to the importance of the root bridge, a switching device with high performance and network hierarchy is usually elected as the root bridge. However, such a switching device may not have a high priority. You can run a command to configure the switching device as the root bridge. The **stp root** command configures the current switching device as the root bridge or secondary root bridge of a specified spanning tree.

If you run the **stp root primary** command to configure the current switching device as the root bridge in a specified spanning tree, the priority of the device in the spanning tree is 0 and cannot be changed.

Note: Although the STP calculation result in Step 1 shows that S1 is the root bridge and S2 is the secondary root bridge, to prevent the roles of the root bridge and secondary root bridge from being preempted when a switch with a smaller MAC address joins the network, you can manually specify the bridge priority to ensure that S1 is the root bridge and S2 is the secondary root bridge.

```
[S2] stp root secondary
```


If you run the **stp root secondary** command to configure the current switching device as the secondary root bridge in a specified spanning tree, the priority of the device in the spanning tree is 4096 and cannot be changed.

Check the STP status on S1.

```
[S1] display stp
```

CIST Global Information:

Mode	:STP
CIST Bridge	: 0 fa09-eea3-0011
Config Times	:Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times	:Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC	: 0 fa09-eea3-0011 / 0 (This bridge is the root)
CIST RegRoot/IRPC	:0 fa09-eea3-0011 / 0 (This bridge is the root)
CIST RootPortId	:0.0
BPDU-Protection	:Disabled
CIST Root Type	:Primary root
TC or TCN received	:18
TC count per hello	:0
STP Converge Mode	:Normal
Share region-configuration	:Enabled
Time since last TC	:0 days 0h:29m:42s
Number of TC	:19
Last TC occurred	:GE1/0/4
Topo Change Flag	:0

CIST Port Information:

Port Id	:1
Port Name	:GE1/0/1
Port State	:Learning

---- More ----

In this case, the root bridge priority of S1 is 0, and the root path cost is 0, indicating that S1 is the root bridge on the current network.

Check brief STP status information on all devices.

[S1] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	DESI	forwarding	none	20000	disable
0	GE1/0/2	DESI	forwarding	none	20000	disable
0	GE1/0/3	DESI	forwarding	none	20000	disable
0	GE1/0/4	DESI	forwarding	none	20000	disable

[S2] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	DESI	forwarding	none	20000	disable
0	GE1/0/2	ROOT	forwarding	none	20000	disable
0	GE1/0/3	ALTE	discarding	none	20000	disable
0	GE1/0/4	DESI	forwarding	none	20000	disable

[S3] display stp brief

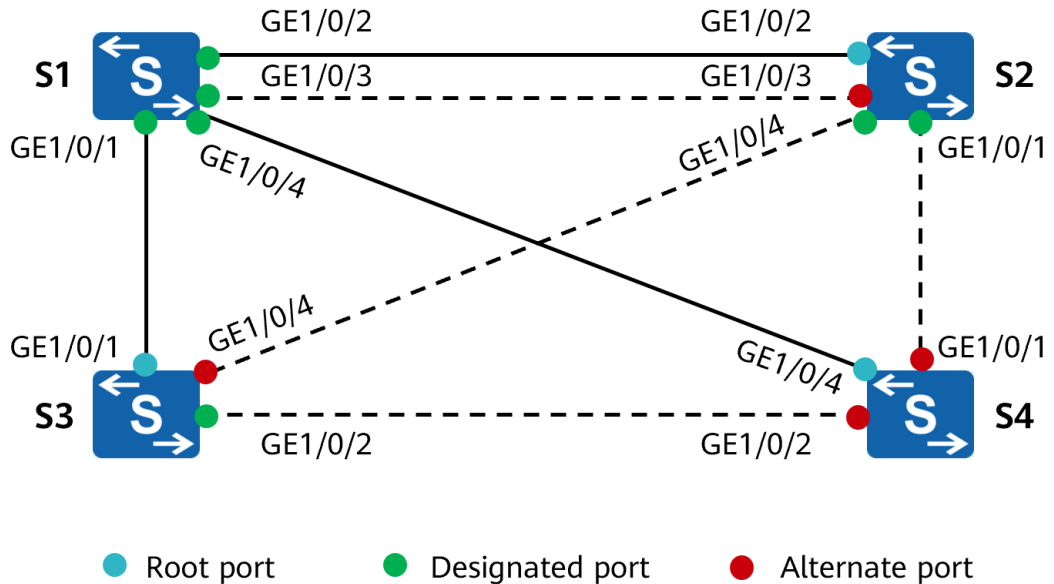
MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	ROOT	forwarding	none	20000	disable
0	GE1/0/2	DESI	forwarding	none	20000	disable
0	GE1/0/4	ALTE	discarding	none	20000	disable

[S4] display stp brief

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	ALTE	discarding	none	20000	disable

0	GE1/0/2	ALTE	discarding	none	20000	disable
0	GE1/0/4	ROOT	forwarding	none	20000	disable

Based on the root bridge ID and port information on each switch, the current topology is as follows:



Step 3 Modify device parameters so that GE1/0/1 on S4 becomes the root port.

Check STP status information on S4.

```
[S4] display stp
```

CIST Global Information:

```
Mode                :STP
CIST Bridge         :32768.fa09-eea3-0041
Config Times        :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times        :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :0.fa09-eea3-0011 / 20000
CIST RegRoot/IRPC   :32768.fa09-eea3-0041 / 0 (This bridge is the root)
CIST RootPortId     :128.4 (GE1/0/4)
BPDU-Protection     :Disabled
TC or TCN received  :102
```

```

TC count per hello :0

STP Converge Mode :Normal

Share region-configuration :Enabled

Time since last TC :0 days 0h:34m:16s

Number of TC :12

Last TC occurred :GE1/0/4

Topo Change Flag :0

```

CIST Port Information:

```

Port Id :1

Port Name :GE1/0/1

Port State :Discarding

Port Protocol :Enabled

```

---- More ----

The cost of the root path from S4 to S1 is 20000.

Change the STP cost of GE1/0/4 on S4 to 50000.

```

[S4] interface GE 1/0/4

[S4-GE1/0/4] stp cost 50000

[S4-GE1/0/4] quit

```

Check brief STP status information.

```

[S4] display stp brief

```

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	ROOT	forwarding	none	20000	disable
0	GE1/0/2	ALTE	discarding	none	20000	disable
0	GE1/0/4	ALTE	discarding	none	50000	disable

GE1/0/1 on S4 has become the root port.

Check STP status information.

[S4] display stp

CIST Global Information:

Mode :STP

CIST Bridge :32768.fa09-eea3-0041

Config Times :Hello 2s MaxAge 20s FwDly 15s MaxHop 20

Active Times :Hello 2s MaxAge 20s FwDly 15s MaxHop 20

CIST Root/ERPC :0.fa09-eea3-0011 / 40000 //The cost of the path from S4 to the root bridge is 40000.

CIST RegRoot/IRPC :32768.fa09-eea3-0041 / 0 (This bridge is the root)

CIST RootPortId :128.1 (GE1/0/1)

BPDU-Protection :Disabled

TC or TCN received :192

TC count per hello :0

STP Converge Mode :Normal

Share region-configuration :Enabled

Time since last TC :0 days 0h:0m:14s

Number of TC :18

Last TC occurred :GE1/0/1

Topo Change Flag :0

CIST Port Information:

Port Id :1

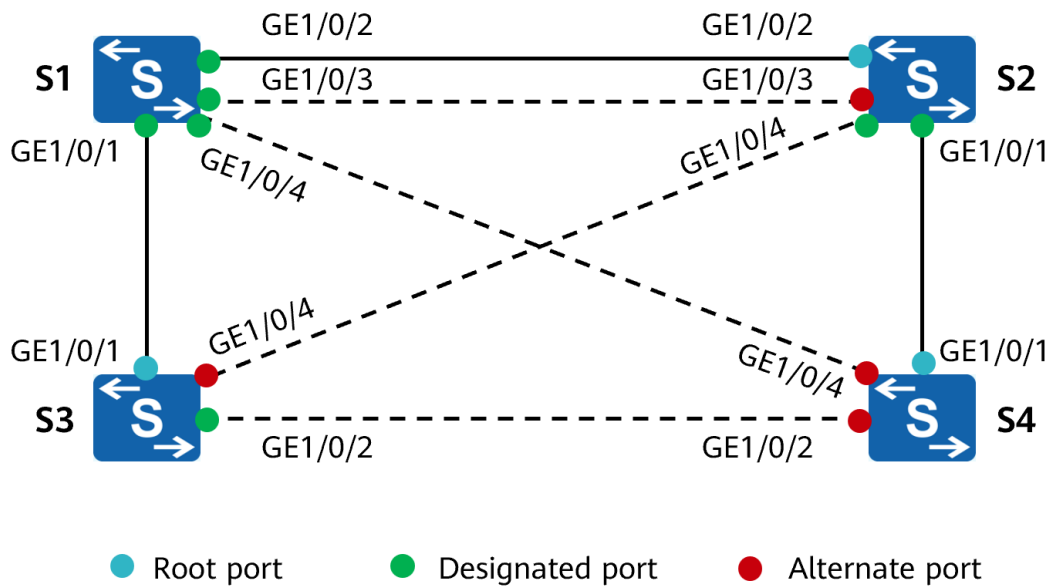
Port Name :GE1/0/1

Port State :Forwarding

Port Protocol :Enabled

---- More ----

The current topology is as follows:



Step 4 Change the spanning tree working mode to RSTP.

Change the spanning tree working mode on all devices.

```
[S1] stp mode rstp
```

```
[S2] stp mode rstp
```

```
[S3] stp mode rstp
```

```
[S4] stp mode rstp
```

Check the spanning tree status of devices. The following uses S1 as an example.

```
[S1] display stp
```

CIST Global Information:

Mode :RSTP

CIST Bridge :0 fa09-eea3-0011

```

Config Times      :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times      :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC    :0 fa09-eea3-0011 / 0 (This bridge is the root)
CIST RegRoot/IRPC :0 fa09-eea3-0011 / 0 (This bridge is the root)
CIST RootPortId   :0.0
BPDU-Protection   :Disabled
CIST Root Type    :Primary root
TC or TCN received :20
TC count per hello :0
STP Converge Mode :Normal
Share region-configuration :Enabled
Time since last TC :0 days 0h:5m:52s
Number of TC      :21
Last TC occurred  :GE1/0/2
Topo Change Flag  :0

CIST Port Information:

Port Id           :1
Port Name         :GE1/0/1
Port State        :Forwarding

---- More ----

```

After the mode is changed, the topology of the spanning tree is not affected.

Step 5 Configure edge ports.

GE1/0/5 to GE1/0/10 of S3 are connected only to terminals and need to be configured as edge ports.

```
[S3] interface range GE 1/0/5 to GE 1/0/10
```

Generally, a device provides a large number of Ethernet ports and many of them have the same configuration. Configuring these Ethernet ports one by one is time-consuming and

error-prone. To solve this problem, you can add the Ethernet ports that require the same configuration commands to a temporary port group. When commands are configured for the temporary port group, the system will automatically execute these commands on all member ports bound to the temporary port group, implementing batch configuration of the Ethernet ports.

Note: Some products may not support temporary port groups. In this case, you need to configure ports separately.

```
[S3-port-group] stp edged-port enable
```

The **stp edged-port enable** command configures a port as an edge port. If a port of a switching device receives BPDUs after being configured as an edge port, the switching device will automatically configure the edge port as a non-edge port and recalculate the spanning tree.

3.3 Verification

1. Identify the root bridge and port roles in your lab environment based on the actual network convergence.
2. Shut down any port on any switch and check whether the traffic can reach all other switches through backup links.

3.4 Reference Configuration

3.4.1 S1 Configuration

```
sysname S1
#
stp mode rstp
stp instance 0 root primary
#
```

3.4.2 S2 Configuration

```
sysname S2
```



```
#  
stp mode rstp  
stp instance 0 root secondary  
#
```

3.4.3 S3 Configuration

```
sysname S3  
  
#  
stp mode rstp  
  
#  
interface GE1/0/5  
    stp edged-port enable  
  
#  
interface GE1/0/6  
    stp edged-port enable  
  
#  
interface GE1/0/7  
    stp edged-port enable  
  
#  
interface GE1/0/8  
    stp edged-port enable  
  
#  
interface GE1/0/9  
    stp edged-port enable  
  
#  
interface GE1/0/10  
    stp edged-port enable
```

#

3.4.4 S4 Configuration

```
sysname S4
#
stp mode rstp
#
interface GE1/0/4
    stp instance 0 cost 50000
#
```

3.5 Quiz

1. Can the two links between S1 and S2 be in the forwarding state at the same time? Why?

Answer:

No. If the two links between S1 and S2 are in the forwarding state at the same time, a loop occurs. Therefore, one link must be blocked.

4 Ethernet Link Aggregation Configuration

4.1 Introduction

4.1.1 About This Lab

As networks grow in scale, users require backbone links to provide higher bandwidth and reliability. In conventional technologies, the usual way to increase the bandwidth is to use high-speed interface cards or devices that support high-speed interface cards, which is costly and inflexible.

In comparison, link aggregation technology bundles multiple physical interfaces into a logical interface to increase the link bandwidth without upgrading the hardware. In addition to increasing the bandwidth, link aggregation technology provides the link backup mechanism to improve the reliability of links between devices.

Link aggregation technology has the following advantages:

- Increased bandwidth: The maximum bandwidth of an Eth-Trunk interface is the sum of bandwidth of its member interfaces.
- Higher reliability: If an active link fails, traffic on this link can be switched to another available member link, improving reliability of the Eth-Trunk interface.
- Load balancing: Traffic is load balanced among all available active links in a link aggregation group (LAG).

In this lab activity, you will learn how to configure Ethernet link aggregation in manual and LACP modes and understand fundamentals of Ethernet link aggregation technology.

4.1.2 Objectives

- Learn how to manually configure link aggregation.
- Learn how to configure link aggregation in static LACP mode.
- Learn how to determine active links in static LACP mode.
- Learn how to configure some static LACP features.

4.1.3 Networking Topology

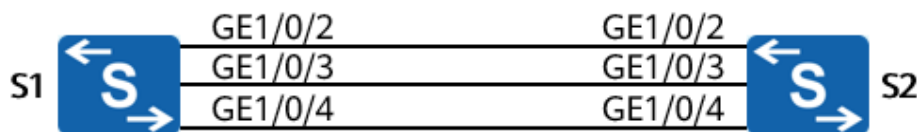


Figure 4-1 Lab topology for Ethernet link aggregation configuration

The switches used in this lab are ENSP-LSW.

4.1.4 Background

In the spanning tree protocol lab, the two links between S1 and S2 cannot be in the data forwarding state at the same time. To make full use of the bandwidth of the two links, configure Ethernet link aggregation between S1 and S2.

4.2 Task Configuration

4.2.1 Configuration Roadmap

1. Configure link aggregation in manual mode.
2. Configure link aggregation in LACP mode.
3. Modify parameters to determine active links.
4. Change the load balancing mode.

4.2.2 Configuration Procedure

Step 1 Configure link aggregation in manual mode.

Create an Eth-Trunk interface.

```
[S1] interface Eth-Trunk 1
```

The **interface eth-trunk** command displays the view of an existing Eth-Trunk interface or creates an Eth-Trunk interface and displays its view. The number **1** indicates the interface number. The interface number range varies according to devices.

```
[S2] interface Eth-Trunk 1
```

Configure the link aggregation mode of the Eth-Trunk interface.

```
[S1-Eth-Trunk1] mode manual load-balance
[S1-Eth-Trunk1] quit
```

The **mode** command configures the working mode of an Eth-Trunk interface. By default, an Eth-Trunk interface works in manual load balancing mode.

Add member interfaces to the Eth-Trunk interface.

```
[S1] interface GE 1/0/2
[S1-GE1/0/2] eth-trunk 1
[S1-GE1/0/2] quit
[S1] interface GE 1/0/3
[S1-GE1/0/3] eth-trunk 1
[S1-GE1/0/3] quit
[S1] interface GE 1/0/4
[S1-GE1/0/4] eth-trunk 1
[S1-GE1/0/4] quit
```

You can enter the interface view of interfaces and add them to an Eth-Trunk interface one by one.

You can also run the **trunkport** command in the Eth-Trunk interface view to add multiple interfaces to the Eth-Trunk interface in batches.

```
[S2] interface Eth-Trunk 1
[S2-Eth-Trunk1] trunkport GE 1/0/2 to 1/0/4
[S2-Eth-Trunk1] quit
```

When adding member interfaces to an Eth-Trunk, pay attention to the following points:

1. An Eth-Trunk interface cannot be added to another Eth-Trunk interface as a member interface.
2. An Ethernet interface can be added to only one Eth-Trunk interface. To add the Ethernet interface to another Eth-Trunk interface, delete it from the original Eth-Trunk interface first.
3. If an interface of the local device is added to an Eth-Trunk interface, an interface of the remote device directly connected to the interface of the local device must also be added to an Eth-Trunk interface so that the two ends can communicate with each other.

Check the status of the Eth-Trunk interface.

```
[S1] display eth-trunk 1
```

Eth-Trunk1's state information is:

Working Mode: **Normal** Hash Arithmetic: src-dst-ip
Least Active-linknumber: 1 Max Bandwidth-affected-linknumber: 128
Operating Status: up Number of Up Ports in Trunk: 3

PortName	Status	Weight
GE1/0/2	Up	1
GE1/0/3	Up	1
GE1/0/4	Up	1

Working mode of an Eth-Trunk interface:

1. Normal: manual load balancing mode
2. Static: static LACP mode
3. Dynamic: dynamic LACP mode

Step 2 Configure link aggregation in LACP mode.

Delete member interfaces from the Eth-Trunk interface.

```
[S1] interface Eth-Trunk 1
```

```
[S1-Eth-Trunk1] undo trunkport GE 1/0/2 to 1/0/4
```

```
[S1-Eth-Trunk1] quit
```

```
[S2] interface Eth-Trunk 1
```

```
[S2-Eth-Trunk1] undo trunkport GE 1/0/2 to 1/0/4
```

```
[S2-Eth-Trunk1] quit
```

Before changing the working mode of an Eth-Trunk interface, ensure that the Eth-Trunk interface has no member interface.

Change the working mode.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] mode lacp-static
[S1-Eth-Trunk1] quit
```

The **mode lacp-static** command configures an Eth-Trunk interface to work in static LACP mode.

```
[S2] interface Eth-Trunk 1
[S2-Eth-Trunk1] mode lacp-static
[S2-Eth-Trunk1] quit
```

Add member interfaces to the Eth-Trunk interface.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] trunkport GE 1/0/2 to 1/0/4
[S1-Eth-Trunk1] quit
```

```
[S2] interface Eth-Trunk 1
[S2-Eth-Trunk1] trunkport GE 1/0/2 to 1/0/4
[S2-Eth-Trunk1] quit
```

Check the status of the Eth-Trunk interface.

```
[S1] display eth-trunk 1

Eth-Trunk1's state information is:

(h): high priority
(r): reference port

Local:

LAG ID: 1                      Working Mode: Static
Preempt Delay: Disabled        Hash Arithmetic: src-dst-ip
System Priority: 32768          System ID: facf-999b-0071
```

Least Active-linknumber: 1		Max Active-linknumber: 128					
Operating Status: up		Number Of Up Ports In Trunk: 3					
Timeout Period: Slow							
PortKeyMode: Auto							

ActorPortName	Status	PortType	PortPri	PortNo	PortKey	PortState	Weight
GE1/0/2(hr)	Selected	1GE	32768	1	305	10111100	1
GE1/0/3(h)	Selected	1GE	32768	2	305	10111100	1
GE1/0/4(h)	Selected	1GE	32768	3	305	10111100	1
Partner:							

ActorPortName	SysPri	SystemID	PortPri	PortNo	PortKey	PortState	
GE1/0/2	32768	facf-999b-0081	32768	1	305	10111100	
GE1/0/3	32768	facf-999b-0081	32768	2	305	10111100	
GE1/0/4	32768	facf-999b-0081	32768	3	305	10111100	

Step 3 Configure the number of active interfaces.

Considering the network traffic, when the network is normal, only GE1/0/3 and GE1/0/4 need to be in the forwarding state, and GE1/0/2 functions as a standby interface. When the number of active interfaces falls below 2, the Eth-Trunk interface is shut down.

Set the LACP priority of S1 to make S1 become the Actor.

```
[S1] lacp priority 100
```

Configure the interface priority so that GE1/0/3 and GE1/0/4 are preferentially selected.

```
[S1] interface GE 1/0/2
[S1-GE1/0/2] lacp priority 40000
[S1-GE1/0/2] quit
```

LACPDUs are sent and received by devices at both ends of a LAG in LACP mode.

First, the Actor is elected.

1. The system priority field values are compared. The default priority value is 32768, and a smaller value indicates a higher priority. The device with a higher priority is elected as the LACP Actor.
2. If the devices have the same system priority, their MAC addresses are compared. The device with a smaller MAC address becomes the LACP Actor.
3. After the Actor is elected, the devices at both ends select active interfaces according to the interface priority settings on the Actor. A smaller interface priority value indicates a higher priority. The default priority value is 32768.

Set the upper and lower thresholds for the number of active interfaces in an Eth-Trunk interface.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] lacp max active-linknumber 2
[S1-Eth-Trunk1] least active-linknumber 2
[S1-Eth-Trunk1] quit
```

The bandwidth and status of an Eth-Trunk interface depend on the number of active interfaces in the Eth-Trunk interface. The bandwidth of an Eth-Trunk interface is the total bandwidth of all member interfaces in the Up state. To ensure the status and bandwidth of an Eth-Trunk interface, you can set the following thresholds to reduce the impact of member link status changes:

Lower threshold for the number of active interfaces: When the number of active interfaces falls below the configured lower threshold, the Eth-Trunk interface goes Down. This threshold is set to ensure the minimum bandwidth of the Eth-Trunk interface. The **least active-linknumber** command configures the lower threshold for the number of active interfaces in an Eth-Trunk interface.

Upper threshold for the number of active interfaces: When the number of active interfaces reaches the upper threshold, the bandwidth of the Eth-Trunk interface does not increase even if more member links go Up. This threshold is set to improve network reliability while ensuring bandwidth. The **lacp max active-linknumber** command configures the upper threshold for the number of active interfaces in an Eth-Trunk interface.

Enable the preemption function.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] lacp preempt enable
[S1-Eth-Trunk1] quit
```

In LACP mode, when an active link fails, the system selects the standby link with the highest priority to replace the faulty link. If the faulty link recovers and has a higher priority than

the standby link and the LACP priority preemption function is enabled, the link with a higher priority replaces the link with a lower priority as an active link. The **lACP preempt enable** command enables the LACP priority preemption function in LACP mode. By default, this function is disabled.

Check the status of the Eth-Trunk interface.

```
[S1] display eth-trunk 1
```

Eth-Trunk1's state information is:

(h): high priority

(r): reference port

Local:

LAG ID: 1

Working Mode: Static

Preempt Delay Time: 30

Hash Arithmetic: src-dst-ip

System Priority: 100

System ID: facf-999b-0071

Least Active-linknumber: 2

Max Active-linknumber: 2

Operating Status: up

Number Of Up Ports In Trunk: 2

Timeout Period: Slow

PortKeyMode: Auto

ActorPortName	Status	PortType	PortPri	PortNo	PortKey	PortState	Weight
GE1/0/2(h)	Unselect	1GE	40000	1	305	10100000	1
GE1/0/3(hr)	Selected	1GE	32768	2	305	10111100	1
GE1/0/4(h)	Selected	1GE	32768	3	305	10111100	1

Partner:

ActorPortName	SysPri	SystemID	PortPri	PortNo	PortKey	PortState
GE1/0/2	32768	facf-999b-0081	32768	1	305	10100000
GE1/0/3	32768	facf-999b-0081	32768	2	305	10111100
GE1/0/4	32768	facf-999b-0081	32768	3	305	10111100

Currently, GE1/0/3 and GE1/0/4 are in the active state.

Manually shut down GE1/0/4 to simulate a link fault.

```
[S1] interface GE 1/0/4
[S1-GE1/0/4] shutdown
[S1-GE1/0/4] quit
```

```
[S1] display eth-trunk 1
```

Eth-Trunk1's state information is:

(h): high priority

(r): reference port

Local:

```
LAG ID: 1                      Working Mode: Static
Preempt Delay Time: 30         Hash Arithmetic: src-dst-ip
System Priority: 100           System ID: facf-999b-0071
Least Active-linknumber: 2     Max Active-linknumber: 2
Operating Status: up           Number Of Up Ports In Trunk: 2
Timeout Period: Slow
PortKeyMode: Auto
```

ActorPortName	Status	PortType	PortPri	PortNo	PortKey	PortState	Weight
GE1/0/2(h)	Selected	1GE	40000	1	305	10111100	1
GE1/0/3(hr)	Selected	1GE	32768	2	305	10111100	1
GE1/0/4	Unselect	1GE	32768	3	305	10100010	1

Partner:

ActorPortName	SysPri	SystemID	PortPri	PortNo	PortKey	PortState
---------------	--------	----------	---------	--------	---------	-----------

GE1/0/2	32768	facf-999b-0081	32768	1	305	10111100
GE1/0/3	32768	facf-999b-0081	32768	2	305	10111100
GE1/0/4	0	0000-0000-0000	0	0	0	10100011

The status of GE1/0/2 is changed to active.

Manually shut down GE1/0/3 to simulate a link fault.

```
[S1] interface GE 1/0/3
[S1-GE1/0/3] shutdown
[S1-GE1/0/3] quit
```

```
[S1] display eth-trunk 1
```

Eth-Trunk1's state information is:

(h): high priority

(r): reference port

Local:

LAG ID: 1

Working Mode: Static

Preempt Delay Time: 30

Hash Arithmetic: src-dst-ip

System Priority: 100

System ID: facf-999b-0071

Least Active-linknumber: 2

Max Active-linknumber: 2

Operating Status: down

Number Of Up Ports In Trunk: 0

Timeout Period: Slow

PortKeyMode: Auto

ActorPortName	Status	PortType	PortPri	PortNo	PortKey	PortState	Weight
GE1/0/2(h)	Unselect	1GE	40000	1	305	10100000	1
GE1/0/3	Unselect	1GE	32768	2	305	10100010	1
GE1/0/4	Unselect	1GE	32768	3	305	10100010	1

Partner:

ActorPortName	SysPri	SystemID	PortPri	PortNo	PortKey	PortState
GE1/0/2	32768	facf-999b-0081	32768	1	305	10110000
GE1/0/3	0	0000-0000-0000	0	0	0	10100011
GE1/0/4	0	0000-0000-0000	0	0	0	10100011

The lower threshold for the number of active interfaces in the Eth-Trunk interface is set to 2. Therefore, when the number of active interfaces in the Eth-Trunk interface is less than 2, the Eth-Trunk interface is shut down. Although GE1/0/2 is Up, it is still in the Unselect state.

Step 4 Change the load balancing mode.

Enable the interfaces that are shut down in the previous step.

```
[S1] interface GE 1/0/3
[S1-GE1/0/3] undo shutdown
[S1-GE1/0/3] quit
[S1] interface GE 1/0/4
[S1-GE1/0/4] undo shutdown
[S1-GE1/0/4] quit
```

After about 30 seconds, check the status of Eth-Trunk 1.

```
[S1] display eth-trunk 1

Eth-Trunk1's state information is:

(h): high priority
(r): reference port

Local:

LAG ID: 1                Working Mode: Static
Preempt Delay Time: 30    Hash Arithmetic: src-dst-ip
System Priority: 100       System ID: facf-999b-0071
```

Least Active-linknumber: 2		Max Active-linknumber: 2					
Operating Status: up		Number Of Up Ports In Trunk: 2					
Timeout Period: Slow							
PortKeyMode: Auto							

ActorPortName	Status	PortType	PortPri	PortNo	PortKey	PortState	Weight
GE1/0/2(h)	Unselect	1GE	40000	1	305	10100000	1
GE1/0/3(hr)	Selected	1GE	32768	2	305	10111100	1
GE1/0/4(h)	Selected	1GE	32768	3	305	10111100	1
Partner:							

ActorPortName	SysPri	SystemID	PortPri	PortNo	PortKey	PortState	
GE1/0/2	32768	facf-999b-0081	32768	1	305	10110	
GE1/0/3	32768	facf-999b-0081	32768	2	305	10111	
GE1/0/4	32768	facf-999b-0081	32768	3	305	10111	

The preemption function is enabled on the Eth-Trunk interface. Therefore, when GE1/0/3 and GE1/0/4 go Up, they have higher priorities than GE1/0/2. As a result, GE1/0/2 enters the **Unselect** state. In addition, to ensure link stability, the default preemption delay is 30 seconds. Therefore, preemption occurs 30 seconds later.

Change the load balancing mode of the Eth-Trunk interface to destination IP address-based load balancing.

```
[S1] interface Eth-Trunk 1
[S1-Eth-Trunk1] load-balance dst-ip
[S1-Eth-Trunk1] quit
```

To distribute traffic on an Eth-Trunk interface to different links and ensure that the traffic can reach the same destination, run the **load-balance** command to configure a load balancing mode for the Eth-Trunk interface. This ensures that outgoing traffic is properly load balanced among physical links, preventing link congestion. Load balancing is valid only for outgoing traffic; therefore, the load balancing modes on the interfaces at both ends of an Eth-Trunk can be different and do not affect each other.

Table 4-1 Description of parameters in the load-balance command

Parameter	Description
dst-ip	Indicates that load balancing is performed based on the destination IP address.
src-ip	Indicates that load balancing is performed based on the source IP address.
src-dst-ip	Indicates that load balancing is performed based on the Exclusive-OR result of the source and destination IP addresses.
dst-mac	Indicates that load balancing is performed based on the destination MAC address.
src-mac	Indicates that load balancing is performed based on the source MAC address.
src-dst-mac	Indicates that load balancing is performed based on the Exclusive-OR result of the source and destination MAC addresses.
round-robin	Indicates that per-packet load balancing is performed in round-robin mode.
enhanced	Indicates that load balancing is performed based on an enhanced load balancing profile.
profile <i>profile-name</i>	Specifies the name of a load balancing profile.

4.3 Reference Configuration

4.3.1 S1 Configuration

```

sysname S1

#

lacp priority 100

#

interface Eth-Trunk1

 mode lacp-static

 least active-linknumber 2

 lacp preempt enable

```

```
lacp max active-linknumber 2

load-balance dst-ip

#

interface GE1/0/2

eth-trunk 1

lacp priority 40000

#

interface GE1/0/3

eth-trunk 1

#

interface GE1/0/4

eth-trunk 1

#
```

4.3.2 S2 Configuration

```
sysname S2

#

interface Eth-Trunk1

mode lacp-static

#

interface GE1/0/2

eth-trunk 1

#

interface GE1/0/3

eth-trunk 1

#

interface GE1/0/4
```



```
eth-trunk 1
```

```
#
```

4.4 Quiz

1. What are the requirements for the values of the **least active-linknumber** and **lACP max active-linknumber** parameters?

Answer:

The value of the **least active-linknumber** parameter must be less than or equal to that of the **lACP max active-linknumber** parameter.

5 Direct Route Configuration

5.1 Introduction

5.1.1 About This Lab

Once an IP address is configured for a Layer 3 interface, the device automatically generates a direct route based on this IP address if both the physical and protocol states of the interface are Up. Direct routes require no manual configuration and are assigned the highest priority. In certain scenarios, routing devices can implement communication between terminals on different network segments based on direct routes, for example, to enable inter-VLAN communication.

After VLANs are configured, users in different VLANs cannot communicate at Layer 2, effectively isolating broadcast domains. However, in real-world applications, users in different VLANs often require mutual access, making inter-VLAN communication necessary.

Huawei provides multiple technologies to implement inter-VLAN communication. The following two technologies are commonly used:

- **Dot1q termination sub-interface:** Such sub-interfaces are Layer 3 logical interfaces. Similar to a VLANIF interface, after a dot1q termination sub-interface and its IP address are configured, the device generates the corresponding MAC address entry and sets the Layer 3 forwarding flag to enable Layer 3 communication between VLANs. Dot1q termination sub-interfaces are typically used when a Layer 3 Ethernet interface needs to connect to multiple VLANs.
- **VLANIF interface:** VLANIF interfaces are Layer 3 logical interfaces. After a VLANIF interface and its IP address are configured, the device adds the MAC address and VLAN ID of the VLANIF interface to the MAC address table and sets the Layer 3 forwarding flag of the MAC address entry. When the destination MAC address of a packet matches this entry, the packet is forwarded at Layer 3, enabling communication between VLANs.

In this lab activity, you will use these two methods to implement inter-VLAN communication.

5.1.2 Objectives

- Learn how to use dot1q termination sub-interfaces to implement inter-VLAN communication.
- Learn how to use VLANIF interfaces to implement inter-VLAN communication.
- Understand the forwarding process of inter-VLAN communication.

5.1.3 Networking Topology

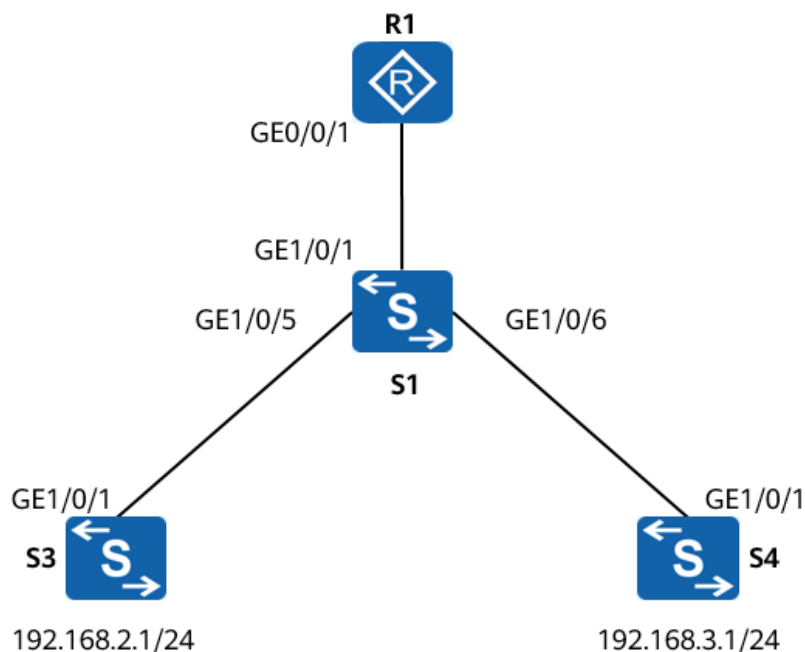


Figure 5-1 Lab topology for direct route configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

1. Simulate terminal users on S3 and S4, and assign IP addresses 192.168.2.1/24 and 192.168.3.1/24 to interfaces.
2. Set gateway addresses to 192.168.2.254 and 192.168.3.254 for S3 and S4, respectively.
3. On S1, assign GE1/0/5 and GE1/0/6 to VLAN 2 and VLAN 3, respectively.

5.1.4 Background

S3 and S4 belong to different VLANs. You need to configure dot1q termination sub-interfaces and VLANIF interfaces to enable communication between S3 and S4.

5.2 Task Configuration

5.2.1 Configuration Roadmap

1. Configure dot1q termination sub-interfaces to implement inter-VLAN communication.
2. Configure VLANIF interfaces to implement inter-VLAN communication.

5.2.2 Configuration Procedure

Step 1 Complete basic device configuration.

Configure IP addresses and gateway addresses for S3 and S4.

```
<S3> system-view  
  
Enter system view, return user view with Ctrl+Z.  
  
[S3] interface GE 1/0/1  
  
[S3-GE1/0/1] undo port link-type  
  
[S3-GE1/0/1] undo portswitch  
  
[S3-GE1/0/1] ip address 192.168.2.1 24  
  
[S3-GE1/0/1] quit  
  
[S3] ip route-static 0.0.0.0 0 192.168.2.254
```

A default route is configured, which is equivalent to configuring a gateway for the device.

```
<S4> system-view  
  
Enter system view, return user view with Ctrl+Z.  
  
[S4] interface GE 1/0/1  
  
[S4-GE1/0/1] undo port link-type  
  
[S4-GE1/0/1] undo portswitch  
  
[S4-GE1/0/1] ip address 192.168.3.1 24  
  
[S4-GE1/0/1] quit  
  
[S4] ip route-static 0.0.0.0 0 192.168.3.254
```

On S1, assign S3 and S4 to different VLANs.

```
[S1] vlan batch 2 3  
  
[S1] interface GE 1/0/5  
  
[S1-GE1/0/5] port link-type access  
  
[S1-GE1/0/5] port default vlan 2  
  
[S1-GE1/0/5] quit
```

```
[S1] interface GE 1/0/6

[S1-GE1/0/6] port link-type access

[S1-GE1/0/6] port default vlan 3

[S1-GE1/0/6] quit
```

Step 2 Configure dot1q termination sub-interfaces to implement inter-VLAN communication.

On S1, configure a trunk interface.

```
[S1] interface GE 1/0/1

[S1-GE1/0/1] port link-type trunk

[S1-GE1/0/1] port trunk allow-pass vlan 2 3

[S1-GE1/0/1] quit
```

Because R1 needs to terminate VLAN tags to enable inter-VLAN communication, the link between S1 and R1 must allow packets from VLAN 2 and VLAN 3 to pass through.

On R1, configure dot1q termination sub-interfaces.

```
[R1] interface GE 0/0/1.2
```

A sub-interface is created and its view is displayed. In this example, **2** is the sub-interface number. It is recommended that a sub-interface number be the same as the corresponding VLAN ID.

```
[R1-GE0/0/1.2] dot1q termination vid 2
```

The **dot1q termination vid *vlan-id*** command configures a VLAN ID for dot1q termination on a sub-interface.

With the preceding configuration, when GE0/0/1 receives traffic tagged with VLAN 2, it sends the data to sub-interface 2 for VLAN termination and further processing. The traffic sent from sub-interface 2 is also tagged with VLAN 2.

```
[R1-GE0/0/1.2] ip address 192.168.2.254 24

[R1-GE0/0/1.2] quit

[R1] interface GE 0/0/1.3
```

```
[R1-GE0/0/1.3] dot1q termination vid 3
[R1-GE0/0/1.3] ip address 192.168.3.254 24
[R1-GE0/0/1.3] quit
```

Check connectivity between VLANs.

```
<S3> ping 192.168.3.1

PING 192.168.3.1: 56 data bytes, press CTRL_C to break

Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=254 time=60 ms
Reply from 192.168.3.1: bytes=56 Sequence=2 ttl=254 time=40 ms
Reply from 192.168.3.1: bytes=56 Sequence=3 ttl=254 time=110 ms
Reply from 192.168.3.1: bytes=56 Sequence=4 ttl=254 time=70 ms
Reply from 192.168.3.1: bytes=56 Sequence=5 ttl=254 time=100 ms

--- 192.168.3.1 ping statistics ---

5 packet(s) transmitted
5 packet(s) received
0.00% packet loss

round-trip min/avg/max = 40/76/110 ms
```

```
<S3> traceroute 192.168.3.1

traceroute to 192.168.3.1 (192.168.3.1), max hops: 30 ,packet length: 40,press CTRL_C to break

 1 192.168.2.254      30 ms  50 ms  50 ms
 2 192.168.3.1       70 ms  60 ms  60 ms
```

The command output shows that VLAN 2 and VLAN 3 can successfully communicate with each other.

Step 3 Configure VLANIF interfaces to implement inter-VLAN communication.

Delete the configuration in the previous step.

```
[S1] interface GE 1/0/1

[S1-GE1/0/1] undo port trunk allow-pass vlan 2 3

[S1-GE1/0/1] quit
```

```
[R1] undo interface GE 0/0/1.2

[R1] undo interface GE 0/0/1.3
```

On S1, create VLANIF interfaces.

```
[S1] interface Vlanif 2
```

The **interface vlanif** *vlan-id* command creates a VLANIF interface and enters its view. You can run the **interface vlanif** command to create a VLANIF interface only after the corresponding VLAN has been created.

```
[S1-Vlanif2] ip address 192.168.2.254 24

[S1-Vlanif2] quit

[S1] interface Vlanif 3

[S1-Vlanif3] ip address 192.168.3.254 24

[S1-Vlanif3] quit
```

Check connectivity between VLANs.

```
<S3> ping 192.168.3.1

PING 192.168.3.1: 56 data bytes, press CTRL_C to break

Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=254 time=100 ms

Reply from 192.168.3.1: bytes=56 Sequence=2 ttl=254 time=50 ms

Reply from 192.168.3.1: bytes=56 Sequence=3 ttl=254 time=50 ms

Reply from 192.168.3.1: bytes=56 Sequence=4 ttl=254 time=60 ms

Reply from 192.168.3.1: bytes=56 Sequence=5 ttl=254 time=70 ms

--- 192.168.3.1 ping statistics ---

5 packet(s) transmitted
```

```
5 packet(s) received  
0.00% packet loss  
round-trip min/avg/max = 50/66/100 ms
```

```
<S3> tracert 192.168.3.1  
  
tracert to 192.168.3.1(192.168.3.1), max hops: 30 ,packet length: 40,press CTRL_C to break  
  
1 192.168.2.254 40 ms 30 ms 20 ms  
2 192.168.3.1 40 ms 30 ms 40 ms
```

The command output shows that VLAN 2 and VLAN 3 can successfully communicate with each other.

5.3 Reference Configuration

5.3.1 S1 Configuration

```
sysname S1  
  
#  
  
vlan batch 2 to 3  
  
#  
  
interface Vlanif2  
  
ip address 192.168.2.254 255.255.255.0  
  
#  
  
interface Vlanif3  
  
ip address 192.168.3.254 255.255.255.0  
  
#  
  
interface GE1/0/5  
  
port link-type access  
  
port default vlan 2  
  
#
```



```
interface GE1/0/6  
  
port link-type access  
  
port default vlan 3  
  
#
```

5.3.2 S3 Configuration

```
sysname S3  
  
#  
  
interface GE1/0/1  
  
undo portswitch  
  
ip address 192.168.2.1 255.255.255.0  
  
#  
  
ip route-static 0.0.0.0 0.0.0.0 192.168.2.254  
  
#
```

5.3.3 S4 Configuration

```
sysname S4  
  
#  
  
interface GE1/0/1  
  
undo portswitch  
  
ip address 192.168.3.1 255.255.255.0  
  
#  
  
ip route-static 0.0.0.0 0.0.0.0 192.168.3.254  
  
#
```

5.4 Quiz

1. In which situations does a VLANIF interface go Up?

Answer:

A VLANIF interface goes Up when any physical interface that allows packets from the corresponding VLAN to pass through is Up.

6 Static Route Configuration

6.1 Introduction

6.1.1 About This Lab

Internet Protocol version 4 (IPv4) is a core protocol of the TCP/IP suite. It operates at the Internet layer of the TCP/IP model, which corresponds to the network layer in the Open System Interconnection (OSI) model. The network layer provides connectionless data transmission, meaning each IP datagram is sent independently without requiring a prior connection.

Routing is a fundamental function of data communication networks. Routing information defines the path used to forward IP packets, and the routing process itself is the packet forwarding process.

In this lab activity, you will gain a clear understanding of basic routing principles by configuring IPv4 addresses and static routes.

6.1.2 Objectives

- Learn how to configure IPv4 addresses for interfaces.
- Understand the functions and purposes of loopback interfaces.
- Understand how direct routes are generated.
- Learn how to configure static routes and understand the conditions for them to take effect.
- Understand how to use the ping tool to verify network connectivity.
- Learn how to configure special static routes and understand their application scenarios.

6.1.3 Networking Topology

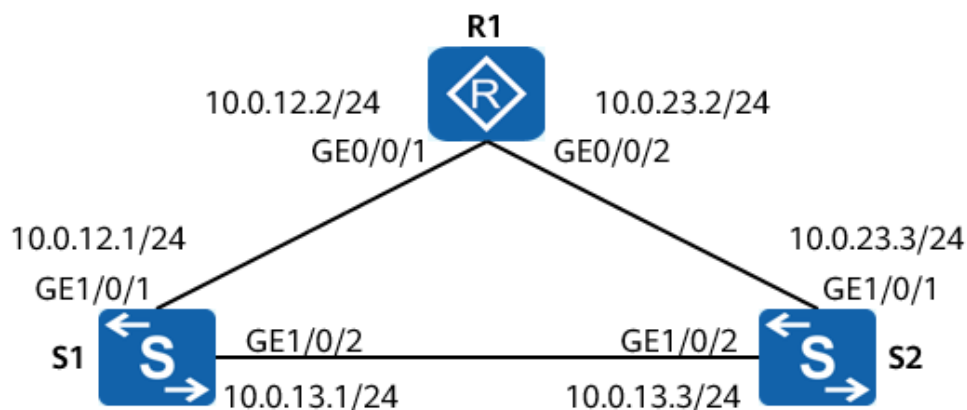


Figure 6-1 Lab topology for static route configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

6.1.4 Background

R1, S1, and S2 serve as gateways for their respective networks. You need to configure the gateways to enable connectivity among these networks.

6.2 Task Configuration

6.2.1 Configuration Roadmap

1. Configure IP addresses for interfaces on network devices.
2. Configure static routes to enable connectivity.

6.2.2 Configuration Procedure

Step 1 Complete basic device configuration.

Name the devices.

The detailed configuration is not provided here.

Initialize the interface configuration.

```
[S1] interface GE 1/0/1
```

```
[S1-GE1/0/1] undo portswitch
[S1-GE1/0/1] quit
[S1] interface GE 1/0/2
[S1-GE1/0/2] undo portswitch
[S1-GE1/0/2] quit
```

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] undo portswitch
[S2-GE1/0/1] quit
[S2] interface GE 1/0/2
[S2-GE1/0/2] undo portswitch
[S2-GE1/0/2] quit
```

Step 2 Check the current interface IP address configuration and routing table on each network device.

Check the interface status on network devices. R1 is used as an example.

```
[R1] display ip interface brief

*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(ed): error down

The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 8
The number of interface that is UP in Protocol is 2
The number of interface that is DOWN in Protocol is 10
```

Interface	IP Address/Mask	Physical	Protocol	VPN
GE0/0/0	unassigned	down	down	--
GE0/0/1	unassigned	up	down	--
GE0/0/2	unassigned	up	down	--
GE0/0/3	unassigned	down	down	--
GE0/0/4	unassigned	down	down	--
GE0/0/5	unassigned	down	down	--
GE0/0/6	unassigned	down	down	--
GE0/0/7	unassigned	down	down	--
GE0/0/8	unassigned	down	down	--
GE0/0/9	unassigned	down	down	--
MEth0/0/0	192.168.1.1/24	up	up	_management...
NULL0	unassigned	up	up(s)	--

The **display ip interface brief** command displays brief information about interface IP addresses, including IP addresses, subnet masks, physical status, protocol status, and number of interfaces in different states.

On R1, GE0/0/1 and GE0/0/2 are not assigned IP addresses. As a result, the **IP Address/Mask** field is **unassigned**, the **Physical** field is **up**, and the **Protocol** field is **down**.

Check the routing tables on network devices. R1 is used as an example.

```
[R1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : _public_

          Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre  Cost           Flags    NextHop         Interface
-----
127.0.0.0/8         Direct   0    0              D    127.0.0.1       InLoopBack0
```

127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

InLoopBack0 is the default loopback interface on a device. It is a special, fixed interface that uses the loopback address 127.0.0.1/8 to receive all data packets destined for the local device. The IP address of this interface cannot be changed and is not advertised through routing protocols.

Step 3 Configure IP addresses for physical interfaces.

Configure IP addresses for physical interfaces of network devices according to the following table.

Table 6-1 IP addresses of physical interfaces

Network Device	Interface	IP Address/Mask
R1	GE0/0/1	10.0.12.2/24
	GE0/0/2	10.0.23.2/24
S1	GE1/0/1	10.0.12.1/24
	GE1/0/2	10.0.13.1/24
S2	GE1/0/1	10.0.23.3/24
	GE1/0/2	10.0.13.3/24

```
[R1] interface GE0/0/1
[R1-GE0/0/1] ip address 10.0.12.2 24
[R1-GE0/0/1] quit
[R1] interface GE0/0/2
[R1-GE0/0/2] ip address 10.0.23.2 24
[R1-GE0/0/2] quit
```

```
[S1] interface GE1/0/1
[S1-GE1/0/1] ip address 10.0.12.1 24
[S1-GE1/0/1] quit
```

```
[S1] interface GE1/0/2
[S1-GE1/0/2] ip address 10.0.13.1 24
[S1-GE1/0/2] quit
```

```
[S2] interface GE1/0/1
[S2-GE1/0/1] ip address 10.0.23.3 24
[S2-GE1/0/1] quit
[S2] interface GE1/0/2
[S2-GE1/0/2] ip address 10.0.13.3 24
[S2-GE1/0/2] quit
```

Use the ping tool to verify connectivity.

```
[R1] ping 10.0.12.1

PING 10.0.12.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=254 time=11 ms

Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=254 time=9 ms

Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=254 time=11 ms

Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=254 time=9 ms

Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=254 time=9 ms

--- 10.0.12.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 9/9/11 ms
```

```
[R1] ping 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break
```



```

Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=11 ms

Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=7 ms

Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=11 ms

Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=10 ms

Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=8 ms

```

```

--- 10.0.23.3 ping statistics ---

```

```

5 packet(s) transmitted

```

```

5 packet(s) received

```

```

0.00% packet loss

```

```

round-trip min/avg/max = 7/9/11 ms

```

Check the routing table of R1.

```

[R1] display ip routing-table

```

```

Proto: Protocol          Pre: Preference

```

```

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

```

```

-----
Routing Table : _public_

```

```

Destinations : 10      Routes : 10

```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2

127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that three direct routes are automatically generated for each interface after IP addresses are configured for the interfaces. These direct routes include:

1. A route to the network segment where the interface resides
2. A host route to the interface IP address
3. A host route to the broadcast address of the network segment where the interface resides

Note: A host route is a route with a 32-bit mask.

Step 4 Create and configure loopback interfaces.

Configure loopback interfaces according to the following table.

Table 6-2 IP addresses of loopback interfaces

Network Device	Interface	IP Address/Mask
R1	LoopBack0	10.0.2.2/32
S1	LoopBack0	10.0.1.1/32
S2	LoopBack0	10.0.3.3/32

Loopback interfaces are logical interfaces that are manually configured and do not physically exist. They can be used to exchange data. Unless manually shut down, the physical and protocol states of a loopback interface remain Up. Typically, a loopback interface uses a 32-bit mask and serves the following purposes:

1. Its IP address is used as the management address of the network device.
2. Its IP address is used as the router ID in dynamic routing protocols.
3. It enhances network reliability.

In this lab activity, loopback interfaces are used to simulate client devices.

```
[R1] interface LoopBack0
[R1-LoopBack0] ip address 10.0.2.2 32
[R1-LoopBack0] quit
```

```
[S1] interface LoopBack0
```

```
[S1-LoopBack0] ip address 10.0.1.1 32
```

```
[S1-LoopBack0] quit
```

```
[S2] interface LoopBack0
```

```
[S2-LoopBack0] ip address 10.0.3.3 32
```

```
[S2-LoopBack0] quit
```

Check the routing table. R1 is used as an example.

```
[R1] display ip routing-table
```

```
Proto: Protocol      Pre: Preference
```

```
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
```

```
-----
```

```
Routing Table : _public_
```

```
Destinations : 11      Routes : 11
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that a direct route to the loopback interface is generated.

Verify connectivity between loopback interfaces.

```
[R1] ping -a 10.0.2.2 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

--- 10.0.1.1 ping statistics ---

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss
```

The **ping -a *source-ip-address destination-ip-address*** command specifies the source IP address of ICMP Echo Request packets. If no source IP address is specified, the IP address of the outbound interface is used as the source IP address of ICMP Echo Request packets. Because R1 does not have a route to the destination IP address, the ping operation fails.

Step 5 Configure static routes.

On R1, configure routes to LoopBack0 interfaces of S1 and S2.

```
[R1] ip route-static 10.0.1.1 32 10.0.12.1

[R1] ip route-static 10.0.3.3 32 10.0.23.3
```

Check the routing table of R1.

```
[R1] display ip routing-table

Proto: Protocol      Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

-----
```

Routing Table : _public_

Destinations : 13

Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Static	60	0	RD	10.0.12.1	GE0/0/1
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	Static	60	0	RD	10.0.23.3	GE0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the configured static routes are installed into the IP routing table.

Verify connectivity.

[R1] ping -a 10.0.2.2 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

```
--- 10.0.1.1 ping statistics ---
```

```
5 packet(s) transmitted
```

```
0 packet(s) received
```

```
100.00% packet loss
```

Because S1 does not have a route to R1's LoopBack0, R1 still cannot ping S1's LoopBack0.

On S1, configure a route to LoopBack0 of R1.

```
[S1] ip route-static 10.0.2.2 32 10.0.12.2
```

Verify connectivity.

```
[R1] ping -a 10.0.2.2 10.0.1.1
```

```
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=254 time=10 ms
```

```
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=254 time=10 ms
```

```
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=254 time=11 ms
```

```
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=254 time=3 ms
```

```
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=254 time=12 ms
```

```
--- 10.0.1.1 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 3/9/12 ms
```

The ping result shows that R1's Loopback0 can communicate with S1's Loopback0.

Configure other routes.

```
[S1] ip route-static 10.0.3.3 32 10.0.13.3
```

```
[S2] ip route-static 10.0.1.1 32 10.0.13.1
```

```
[S2] ip route-static 10.0.2.2 32 10.0.23.2
```

After the configuration is complete, verify connectivity between LoopBack0 interfaces of network devices.

Step 6 Configure a path from R1 to S1 via S2 as the backup path from R1's LoopBack0 to S1's LoopBack0.

Configure static routes on R1 and S1.

```
[R1] ip route-static 10.0.1.1 32 10.0.23.3 preference 100
```

```
[S1] ip route-static 10.0.2.2 32 10.0.13.3 preference 100
```

Check the routing table of R1.

```
[R1] display ip routing-table
```

Proto: Protocol Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

Routing Table : _public_

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Static	60	0	RD	10.0.12.1	GE0/0/1
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	Static	60	0	RD	10.0.23.3	GE0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1

10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the configured static route with preference 100 is not installed into the IP routing table.

Shut down GE0/0/1 on R1 to invalidate the higher-priority route.

```
[R1] interface GE0/0/1
[R1-GE0/0/1] shutdown
[R1-GE0/0/1] quit
```

Check the routing table of R1.

```
[R1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : _public_

          Destinations : 10          Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface

 10.0.1.1/32 Static   100    0        RD  10.0.23.3      GE0/0/2
 10.0.2.2/32 Direct    0      0        D   127.0.0.1      LoopBack0
 10.0.3.3/32 Static   60     0        RD  10.0.23.3      GE0/0/2
 10.0.23.0/24 Direct    0      0        D   10.0.23.2      GE0/0/2
```


10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

As the link between R1 and S1 is disconnected, the higher-priority static route becomes invalid and the lower-priority static route is installed.

Verify connectivity.

```
[R1] ping -a 10.0.2.2 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=253 time=10 ms

Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=253 time=11 ms

Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=253 time=20 ms

Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=253 time=11 ms

Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=253 time=10 ms

--- 10.0.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 10/12/20 ms
```

Trace the path of data packets.

```
[R1] traceroute -a 10.0.2.2 10.0.1.1

traceroute to 10.0.1.1(10.0.1.1), max hops: 64, packet length: 40, press CTRL_C to break

1 10.0.23.3 15 ms 1 ms 9 ms

2 10.0.1.1 10 ms 10 ms 22 ms
```

The **tracert** command is mainly used to display the path information of packets from the source to the destination.

The command output shows that data packets are forwarded to S1 through S2.

Note: In some lab environments, devices may not respond to ICMP packets for security purposes. As a result, the tracert output may vary. You can press **Ctrl+C** to stop the tracert operation.

Step 7 Configure a default route to enable communication between R1's LoopBack0 and S1's LoopBack0.

Enable the interface and delete the configured routes.

```
[R1] interface GE0/0/1
[R1-GE0/0/1] undo shutdown
[R1-GE0/0/1] quit
[R1] undo ip route-static 10.0.1.1 255.255.255.255 10.0.12.1
[R1] undo ip route-static 10.0.1.1 255.255.255.255 10.0.23.3
```

Check the routing table of R1.

```
[R1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : _public_

          Destinations : 12          Routes : 12

Destination/Mask    Proto    Pre    Cost           Flags NextHop          Interface
-----
10.0.2.2/32        Direct   0       0              D    127.0.0.1             LoopBack0
10.0.3.3/32        Static   60       0              RD   10.0.23.3             GE0/0/2
10.0.12.0/24       Direct   0       0              D    10.0.12.2             GE0/0/1
10.0.12.2/32       Direct   0       0              D    127.0.0.1             GE0/0/1
```

10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that R1 does not have a route to S1's LoopBack0 (10.0.1.1/32).

Configure a default route on R1.

```
[R1] ip route-static 0.0.0.0 0 10.0.12.1
```

Check the routing table of R1.

```
[R1] display ip routing-table
```

Proto: Protocol Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

Routing Table : _public_

Destinations : 13 Routes : 13

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	10.0.12.1	GE0/0/1
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	Static	60	0	RD	10.0.23.3	GE0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1

10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the default route is active.

Verify connectivity between R1's LoopBack0 and S1's LoopBack0.

```
[R1] ping -a 10.0.2.2 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=254 time=13 ms

Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=254 time=11 ms

Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=254 time=8 ms

Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=254 time=12 ms

Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=254 time=11 ms

--- 10.0.1.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 8/11/13 ms
```

The command output shows that R1's LoopBack0 can communicate with S1's LoopBack0.

6.3 Verification

Run the ping and tracet commands to check connectivity between LoopBack0 interfaces of devices.

6.4 Reference Configuration

6.4.1 R1 Configuration

```
sysname R1

#
interface GE0/0/1
 ip address 10.0.12.2 255.255.255.0
#
interface GE0/0/2
 ip address 10.0.23.2 255.255.255.0
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.255
#
ip route-static 0.0.0.0 0.0.0.0 10.0.12.1
ip route-static 10.0.3.3 255.255.255.255 10.0.23.3
#
```

6.4.2 S1 Configuration

```
sysname S1

#
interface GE1/0/1
 undo portswitch
```

```
ip address 10.0.12.1 255.255.255.0

#

interface GE1/0/2

undo portswitch

ip address 10.0.13.1 255.255.255.0

#

interface LoopBack0

ip address 10.0.1.1 255.255.255.255

#

ip route-static 10.0.2.2 255.255.255.255 10.0.12.2

ip route-static 10.0.2.2 255.255.255.255 10.0.13.3 preference 100

ip route-static 10.0.3.3 255.255.255.255 10.0.13.3
```

6.4.3 S2 Configuration

```
sysname S2

#

interface GE1/0/1

undo portswitch

ip address 10.0.23.3 255.255.255.0

#

interface GE1/0/2

undo portswitch

ip address 10.0.13.3 255.255.255.0

#

interface LoopBack0

ip address 10.0.3.3 255.255.255.255

#
```

```
ip route-static 10.0.1.1 255.255.255.255 10.0.13.1
```

```
ip route-static 10.0.2.2 255.255.255.255 10.0.23.2
```

6.5 Quiz

1. In what situations will a configured static route be installed into the IP routing table?
Can a route be installed into the IP routing table if the configured next hop is unreachable?

Answer:

A static route is installed into the IP routing table only when both of the following conditions are met:

- (a) The next hop of the static route is reachable.
- (b) The static route is the optimal path to the destination network or host.

Therefore, a static route with an unreachable next hop will not be installed into the IP routing table.

7 IP Address Autoconfiguration

7.1 Introduction

7.1.1 About This Lab

The Dynamic Host Configuration Protocol (DHCP) dynamically manages and configures user IP addresses in a centralized manner. It simplifies network deployment and scale-out, even for small networks.

DHCP is defined in RFC 2131 and uses the client/server model. A DHCP client requests configuration information from a DHCP server, and the server returns the configuration information allocated to the client.

DHCP provides two address allocation mechanisms. The network administrator can use either of the two mechanisms to allocate IP addresses to hosts based on network requirements.

- Dynamic allocation mechanism: The DHCP server allocates an IP address with a limited validity period (known as a lease) to a host. This mechanism applies to scenarios where hosts temporarily access the network and the number of idle IP addresses is less than the total number of hosts.
- Static allocation mechanism: The network administrator allocates a fixed IP address to a specified host through DHCP. Compared with manual IP address configuration, the DHCP static allocation mechanism prevents manual configuration errors and helps the network administrator to perform unified maintenance and management.

7.1.2 Objectives

- Learn how to configure a DHCP global address pool.
- Learn how to use DHCP to allocate static IP addresses.

7.1.3 Networking Topology

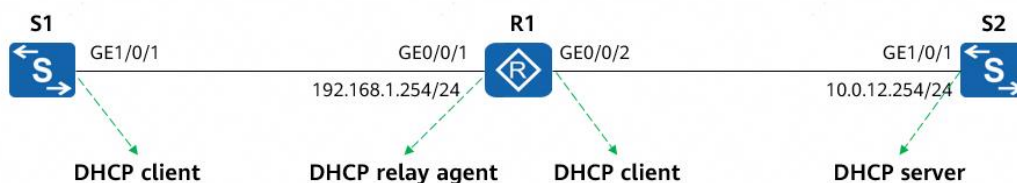


Figure 7-1 Lab topology for IP address autoconfiguration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

1. GE1/0/1 on S1 and GE0/0/2 on R1 function as DHCP clients.
2. S2 functions as a DHCP server to allocate IP addresses to S1 and R1.
3. GE0/0/1 on R1 functions as a DHCP relay agent.

7.1.4 Background

An enterprise plans to deploy DHCP on its network to reduce the workload of IP address maintenance and improve IP address utilization.

7.2 Task Configuration

7.2.1 Configuration Roadmap

1. Configure a DHCP server.
2. Configure DHCP clients.
3. Configure a DHCP relay agent.

7.2.2 Configuration Procedure

Step 1 Complete basic configurations.

Delete the default settings of the MEth interface.

The default settings of the device management interface conflict with the settings required in this lab. Therefore, you need to delete the default settings.

```
[R1] interface MEth0/0/0
[R1-METh0/0/0] undo ip address
[R1-METh0/0/0] quit
```

Configure an IP address for an interface.

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] undo portswitch
[S2-GE1/0/1] ip address 10.0.12.254 24
```

```
[S2-GE1/0/1] quit
```

Step 2 Enable the DHCP function.

```
[S2] dhcp enable
```

The **dhcp enable** command is the prerequisite for configuring DHCP-related functions including the DHCP client and DHCP server functions. These functions take effect only after the DHCP function is enabled using the **dhcp enable** command.

```
[S1] dhcp enable
```

```
[R1] dhcp enable
```

Step 3 Configure address pools.

Configure a global address pool, which is used to allocate an IP address to R1.

```
[S2] ip pool for_r1
```

Create an address pool named **for_r1**.

```
[S2-ip-pool-for_r1] network 10.0.12.0 mask 24
```

The **network** command configures a network segment address for a global address pool.

```
[S2-ip-pool-for_r1] dns-list 10.0.12.254
```

The **dns-list** command configures a DNS server address for DHCP clients.

```
[S2-ip-pool-for_r1] gateway-list 10.0.12.254
```

The **gateway-list** command configures an egress gateway address for DHCP clients. After a DHCP client obtains an IP address, a default route is generated.

```
[S2-ip-pool-for_r1] lease day 2 hour 2
```

```
[S2-ip-pool-for_r1] quit
```

The **lease** command configures the lease of IP addresses in an address pool. If the lease is set to **unlimited**, the lease is not limited. By default, the lease of IP addresses is one day.

Configure a global address pool, which is used to allocate a fixed IP address to S1.
Check the MAC address of GE1/0/1 on S1.

```
[S1] interface GE 1/0/1  
  
[S1-GE1/0/1] undo portswitch  
  
[S1-GE1/0/1] quit
```

```
[S1] display interface GE 1/0/1  
  
GE1/0/1 current state : UP (ifindex: 2)  
  
Line protocol current state : DOWN  
  
Description:  
  
Route Port,The Maximum Transmit Unit is 1500,The Maximum Frame Length is 9216  
  
Internet protocol processing : disabled  
  
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is fa28-5ddf-0022  
  
Port Mode:      COMMON COPPER,      Port Split:      -  
  
Speed:          1000,      LoopBack:          NONE  
  
Duplex:         FULL,      Negotiation:      ENABLE  
  
Input Flow-control: DISABLE,      Output Flow-control:      DISABLE  
  
Mdi:           AUTO
```

```
[S2] ip pool for_s1  
  
[S2-ip-pool-for_s1] network 192.168.1.0 mask 24  
  
[S2-ip-pool-for_s1] gateway-list 192.168.1.254  
  
[S2-ip-pool-for_s1] static-bind ip-address 192.168.1.253 mac-address fa28-5ddf-0022  
  
[S2-ip-pool-for_s1] quit
```

The **static-bind** command binds an IP address in the global address pool of the DHCP server to a MAC address. fa28-5ddf-0022 is the MAC address of GE1/0/1 on S1 in the current lab

environment. After this command is configured, S1 will obtain the fixed IP address 192.168.1.253.

Step 4 Enable the DHCP server function on GE1/0/1 of S2.

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] dhcp select global
[S2-GE1/0/1] quit
```

The **dhcp select global** command enables an interface configured with the DHCP server function to use a global address pool. After receiving a request from a DHCP client, the interface searches all global address pools for the address pool matching the request and then allocates an available address to the DHCP client.

Step 5 Configure DHCP clients.

```
[R1] interface GE 0/0/2
[R1-GE0/0/2] ip address dhcp-alloc
[R1-GE0/0/2] quit
```

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] ip address dhcp-alloc
[S1-GE1/0/1] quit
```

Check the IP address obtained by R1.

```
[R1] display ip interface GE 0/0/2

GE0/0/2 current state : UP
Line protocol current state : UP
The Maximum Transmit Unit : 1500 bytes
input packets : 0, bytes : 0, multicasts : 0
output packets : 0, bytes : 0, multicasts : 0
Directed-broadcast packets:
received packets:          0, sent packets:          0
```

```
forwarded packets:          0, dropped packets:          0
```

```
ARP packet input number:    0
```

```
Request packet:            0
```

```
Reply packet:              0
```

```
Unknown packet:           0
```

Internet Address is allocated by DHCP, 10.0.12.226/24

```
Broadcast address : 0.0.0.0
```

```
TTL being 1 packet number: 0
```

```
TTL invalid packet number: 0
```

```
ICMP packet input number:  0
```

```
Echo reply:                0
```

```
Unreachable:               0
```

```
Source quench:             0
```

```
Routing redirect:          0
```

```
Echo request:              0
```

```
Router advert:             0
```

```
Router solicit:            0
```

```
Time exceed:               0
```

```
IP header bad:             0
```

```
Timestamp request:         0
```

```
Timestamp reply:           0
```

```
Information request:       0
```

```
Information reply:         0
```

```
Netmask request:           0
```

```
Netmask reply:            0
```

```
Unknown type:              0
```

The command output shows that the IP address obtained by R1 is 10.0.12.226/24.

Note: The IP address obtained by R1 is randomly allocated and varies according to the lab environment.

Check the lease information of GE0/0/2 on R1.

```
[R1] display dhcp client interface GE 0/0/2

DHCP client lease information on interface GE0/0/2 :

Current machine state           : Bound

Internet address assigned via : DHCP

Physical address                : fa28-5ddf-0012

IP address                     : 10.0.12.226

Subnet mask                    : 255.255.255.0

Gateway ip address             : 10.0.12.254

DHCP server                    : 10.0.12.254

Lease obtained at              : XXXX-03-06 04:36:00

Lease expires at               : XXXX-03-08 06:36:00

Lease renews at                : XXXX-03-07 05:36:00

Lease rebinds at               : XXXX-03-08 00:21:00

DNS                            : 10.0.12.254

Request option list            : 1 3 6 15 28 33 44 121 184

Class identifier                : huawei AR

Client identifier               : fa28-5ddf-0012
```

The validity period is 2 days and 2 hours, and the DNS address is 10.0.12.254.

Check the routing table of R1.

```
[R1] display ip routing-table

Proto: Protocol      Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

-----

Routing Table : _public_

Destinations : 8      Routes : 8
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	OPR	60	0	D	10.0.12.254	GE0/0/2
10.0.12.0/24	Direct	0	0	D	10.0.12.105	GE0/0/2
10.0.12.105/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

A default route is displayed. The protocol type is **OPR**, the protocol preference is 60, and the next hop is the gateway. The default route delivered by the DHCP server is an open programming route (OPR) with the default preference of 60. You can run the **dhcp client default-route preference** command to change the preference of the default route.

Check the IP address that has been allocated from the address pool **for_r1** on the DHCP server.

[S2] display ip pool name for_r1 used

```

Pool-name       : for_r1
Pool-No         : 0
Lease           : 2 Days 2 Hours 0 Minutes
Domain-name     : -
DNS-server0     : 10.0.12.254
NBNS-server0    : -
Netbios-type    : -
Position        : Local
Status          : Unlocked
Gateway-0       : 10.0.12.254

```

```

Network      : 10.0.12.0
Mask         : 255.255.255.0
VPN instance : --
Logging      : Disable
Conflicted address recycle interval: -

Address Statistic:  Total      :253      Used      :1
                   Idle       :252      Expired   :0
                   Conflict    :0        Disabled  :0
  
```

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
10.0.12.1	10.0.12.254	253	1	252(0)	0	0

Client-ID format as follows:

DHCP : mac-address

IPSec : user-id/portnumber/vrf

SSL-VPN : user-id/session-id

Index	IP	Client-ID	Type	Left	Status
225	10.0.12.226	fa28-5ddf-0012	DHCP	179263	Used

The command output shows that the IP address allocated by the DHCP server is the IP address obtained by R1.

Check the IP address obtained by S1.

```
[S1] display ip interface GE 1/0/1
```


GE1/0/1 current state : UP

Line protocol current state : DOWN

The Maximum Transmit Unit : 1500 bytes

input packets : 0, bytes : 0, multicasts : 0

output packets : 0, bytes : 0, multicasts : 0

Directed-broadcast packets:

received packets: 0, sent packets: 0

forwarded packets: 0, dropped packets: 0

ARP packet input number: 0

Request packet: 0

Reply packet: 0

Unknown packet: 0

Internet protocol processing : disabled

Broadcast address : 0.0.0.0

TTL being 1 packet number: 0

TTL invalid packet number: 0

ICMP packet input number: 0

Echo reply: 0

Unreachable: 0

Source quench: 0

Routing redirect: 0

Echo request: 0

Router advert: 0

Router solicit: 0

Time exceed: 0

IP header bad: 0

Timestamp request: 0

Timestamp reply: 0

Information request:	0
Information reply:	0
Netmask request:	0
Netmask reply:	0
Unknown type:	0

The command output shows that S1 does not obtain an IP address because there is a Layer 3 network between S1 and the DHCP server. Therefore, a DHCP relay agent needs to be deployed.

Check the IP address that has been allocated from the address pool **for_s1** on the DHCP server.

[S2] display ip pool name for_s1 used

```

Pool-name       : for_s1
Pool-No         : 1
Lease           : 1 Days 0 Hours 0 Minutes
Domain-name     : -
DNS-server0     : -
NBNS-server0    : -
Netbios-type    : -
Position        : Local
Status          : Unlocked
Gateway-0       : 192.168.1.254
Network         : 192.168.1.0
Mask            : 255.255.255.0
VPN instance    : --
Logging         : Disable
Conflicted address recycle interval: -

Address Statistic: Total      :253      Used      :1
                   Idle       :252      Expired   :0

```

Conflict

:0

Disabled

:0

Network section

Start

End

Total

Used

Idle(Expired)

Conflict

Disabled

192.168.1.1

192.168.1.254

253

1

252(0)

0

0

Client-ID format as follows:

DHCP

: mac-address

IPSec

: user-id/portnumber/vrf

SSL-VPN

: user-id/session-id

Index

IP

Client-ID

Type

Left

Status

252

192.168.1.253

fa28-5ddf-0022

DHCP

-

Static-bind

The command output shows that the IP address 192.168.1.253 is bound to S1, but S1 has not obtained the IP address.

Step 6 Configure a DHCP relay agent.

Configure R1 and specify the IP address of the DHCP server.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 192.168.1.254 24
[R1-GE0/0/1] dhcp select relay
```

The **dhcp select relay** command enables the DHCP relay function.

```
[R1-GE0/0/1] dhcp relay server-ip 10.0.12.254
```

```
[R1-GE0/0/1] quit
```

The **dhcp relay server-ip** command configures the DHCP server address on a DHCP relay interface.

Configure a route from the DHCP server to the DHCP relay agent.

```
[S2] ip route-static 192.168.1.0 24 10.0.12.105
```

Note that the next-hop address is set to the IP address obtained by GE 0/0/2 of R1. The next-hop address may vary according to the lab environment.

Check the IP address obtained by S1.

```
[S1] display ip interface GE 1/0/1
```

GE1/0/1 current state : UP

Line protocol current state : UP

The Maximum Transmit Unit : 1500 bytes

input packets : 0, bytes : 0, multicasts : 0

output packets : 0, bytes : 0, multicasts : 0

Directed-broadcast packets:

received packets:	0, sent packets:	0
-------------------	------------------	---

forwarded packets:	0, dropped packets:	0
--------------------	---------------------	---

ARP packet input number:	1
--------------------------	---

Request packet:	0
-----------------	---

Reply packet:	1
---------------	---

Unknown packet:	0
-----------------	---

Internet Address is allocated by DHCP, 192.168.1.253/24

Broadcast address : 0.0.0.0

TTL being 1 packet number:	0
----------------------------	---

TTL invalid packet number:	0
----------------------------	---

ICMP packet input number:	0
---------------------------	---

Echo reply:	0
-------------	---

Unreachable:	0
Source quench:	0
Routing redirect:	0
Echo request:	0
Router advert:	0
Router solicit:	0
Time exceed:	0
IP header bad:	0
Timestamp request:	0
Timestamp reply:	0
Information request:	0
Information reply:	0
Netmask request:	0
Netmask reply:	0
Unknown type:	0

The command output shows that S1 has obtained the IP address 192.168.1.253/24.

Check the routing table of S1.

```
[S1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

-----

Routing Table : _public_

Destinations : 8          Routes : 8

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface

0.0.0.0/0          OPR      60     0        D    192.168.1.254          GE1/0/1
```

127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	Direct	0	0	D	192.168.1.253	GE1/0/1
192.168.1.253/32	Direct	0	0	D	127.0.0.1	GE1/0/1
192.168.1.255/32	Direct	0	0	D	127.0.0.1	GE1/0/1
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that S1 has a default route to the gateway.

Check the connectivity between S1 and the DHCP server.

```
[S1] ping 10.0.12.254

PING 10.0.12.254: 56 data bytes, press CTRL_C to break

Reply from 10.0.12.254: bytes=56 Sequence=1 ttl=253 time=1 ms

Reply from 10.0.12.254: bytes=56 Sequence=2 ttl=253 time=1 ms

Reply from 10.0.12.254: bytes=56 Sequence=3 ttl=253 time=1 ms

Reply from 10.0.12.254: bytes=56 Sequence=4 ttl=253 time=1 ms

Reply from 10.0.12.254: bytes=56 Sequence=5 ttl=253 time=1 ms

--- 10.0.12.254 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

The command output shows that S1 can access the DHCP server.

7.3 Reference Configuration

7.3.1 S1 Configuration

```
sysname S1
#
dhcp enable
#
interface GE1/0/1
    undo portswitch
    ip address dhcp-alloc
#
```

7.3.2 R1 Configuration

```
sysname R1
#
dhcp enable
#
interface GE0/0/1
    ip address 192.168.1.254 255.255.255.0
    dhcp select relay
    dhcp relay server-ip 10.0.12.254
#
interface GE0/0/2
    ip address dhcp-alloc
#
```

7.3.3 S2 Configuration

```
sysname S2

#
ip pool for_r1
    network 10.0.12.0 mask 255.255.255.0
    dns-list 10.0.12.254
    gateway-list 10.0.12.254
    lease day 2 hour 2 minute 0
#
ip pool for_s1
    network 192.168.1.0 mask 255.255.255.0
    gateway-list 192.168.1.254
    static-bind ip-address 192.168.1.253 mac-address fa28-5ddf-0022
#
dhcp enable
#
interface GE1/0/1
    undo portswitch
    ip address 10.0.12.254 255.255.255.0
    dhcp select global
#
ip route-static 192.168.1.0 255.255.255.0 10.0.12.226
```

7.4 Quiz

1. What are the differences between the application scenarios of a global address pool and those of an interface address pool?

Answer:

An interface address pool applies to the scenario where the current interface allocates IP addresses on the same network segment as the interface to DHCP clients.

A global address pool can be used to allocate IP addresses on the same network segment as the interface or IP addresses on different network segments (in DHCP relay networking) to DHCP clients.

2. If there are multiple global address pools, how does the DHCP server determine the global address pool from which an IP address is allocated to a DHCP client?

Answer:

In a scenario where no relay agent is deployed, the DHCP server searches all global address pools for the address pool on the same network segment as the interface, and allocates an IP address from the address pool based on the parameters configured for the address pool. In a scenario where a relay agent is deployed, the DHCP server searches all global address pools for the address pool with the network segment requested by the relay agent, and allocates an IP address from the address pool based on the parameters configured for the address pool.

8 OSPF Configuration

8.1 Introduction

8.1.1 About This Lab

Open Shortest Path First (OSPF) is a link-state Interior Gateway Protocol (IGP) developed by the Internet Engineering Task Force (IETF). Currently, OSPFv2 (RFC 2328) is used for IPv4. As a link-state protocol, OSPF offers the following advantages:

- Multicast packet exchange, reducing load on network devices that do not run OSPF
- Support for classless inter-domain routing (CIDR)
- Load balancing among equal-cost routes
- Support for packet authentication

Thanks to these advantages, OSPF is widely adopted as an IGP.

In this lab activity, you will learn the basic configurations and principles of OSPF by setting up single-area OSPF.

8.1.2 Objectives

- Learn basic commands of OSPF.
- Learn how to check the OSPF running status.
- Learn how to control OSPF route selection using cost values.
- Understand how default routes are advertised in OSPF.
- Learn how to configure OSPF authentication.

8.1.3 Networking Topology

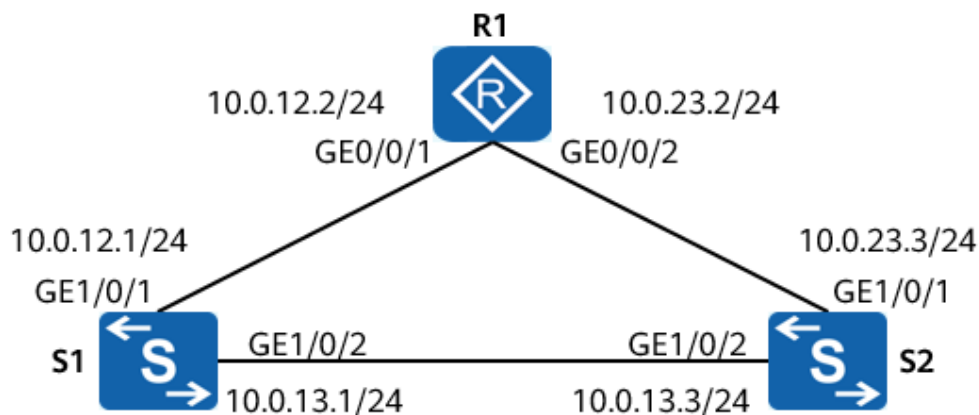


Figure 8-1 Lab topology for OSPF configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

8.1.4 Background

R1, S1, and S2 serve as gateways for their respective networks. You need to configure OSPF to enable connectivity among these networks.

8.2 Task Configuration

8.2.1 Configuration Roadmap

1. Create an OSPF process and enable OSPF on interfaces.
2. Configure OSPF authentication.
3. Configure OSPF to advertise default routes.
4. Control OSPF route selection by modifying cost values.

8.2.2 Configuration Procedure

Step 1 Complete basic device configuration.

Name the devices.

The detailed configuration is not provided here.

Initialize the interface configuration.

```
[S1] interface GE 1/0/1

[S1-GE1/0/1] undo portswitch

[S1-GE1/0/1] quit

[S1] interface GE 1/0/2

[S1-GE1/0/2] undo portswitch

[S1-GE1/0/2] quit
```

```
[S2] interface GE 1/0/1

[S2-GE1/0/1] undo portswitch

[S2-GE1/0/1] quit

[S2] interface GE 1/0/2

[S2-GE1/0/2] undo portswitch

[S2-GE1/0/2] quit
```

Step 2 Configure IP addresses for physical interfaces.

Configure IP addresses for physical interfaces of network devices according to the following table.

Table 8-1 IP addresses of physical interfaces

Network Device	Interface	IP Address/Mask
R1	GE0/0/1	10.0.12.2/24
	GE0/0/2	10.0.23.2/24
S1	GE1/0/1	10.0.12.1/24
	GE1/0/2	10.0.13.1/24
S2	GE1/0/1	10.0.23.3/24
	GE1/0/2	10.0.13.3/24

```
[R1] interface GE0/0/1

[R1-GE0/0/1] ip address 10.0.12.2 24
```

```
[R1-GE0/0/1] quit  
  
[R1] interface GE0/0/2  
  
[R1-GE0/0/2] ip address 10.0.23.2 24  
  
[R1-GE0/0/2] quit
```

```
[S1] interface GE1/0/1  
  
[S1-GE1/0/1] ip address 10.0.12.1 24  
  
[S1-GE1/0/1] quit  
  
[S1] interface GE1/0/2  
  
[S1-GE1/0/2] ip address 10.0.13.1 24  
  
[S1-GE1/0/2] quit
```

```
[S2] interface GE1/0/1  
  
[S2-GE1/0/1] ip address 10.0.23.3 24  
  
[S2-GE1/0/1] quit  
  
[S2] interface GE1/0/2  
  
[S2-GE1/0/2] ip address 10.0.13.3 24  
  
[S2-GE1/0/2] quit
```

Use the ping tool to verify connectivity.

```
[R1] ping 10.0.12.1  
  
PING 10.0.12.1: 56 data bytes, press CTRL_C to break  
  
Reply from 10.0.12.1: bytes=56 Sequence=1 ttl=254 time=11 ms  
  
Reply from 10.0.12.1: bytes=56 Sequence=2 ttl=254 time=9 ms  
  
Reply from 10.0.12.1: bytes=56 Sequence=3 ttl=254 time=11 ms  
  
Reply from 10.0.12.1: bytes=56 Sequence=4 ttl=254 time=9 ms  
  
Reply from 10.0.12.1: bytes=56 Sequence=5 ttl=254 time=9 ms
```

```

--- 10.0.12.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 9/9/11 ms

```

```

[R1] ping 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

    Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=11 ms

    Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=7 ms

    Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=11 ms

    Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=10 ms

    Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=8 ms

--- 10.0.23.3 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 7/9/11 ms

```

Step 3 Create and configure loopback interfaces.

Configure loopback interfaces according to the following table.

Table 8-2 IP addresses of loopback interfaces

Network Device	Interface	IP Address/Mask
R1	LoopBack0	10.0.2.2/32
S1	LoopBack0	10.0.1.1/32
S2	LoopBack0	10.0.3.3/32

Loopback interfaces are logical interfaces that are manually configured and do not physically exist. They can be used to exchange data. Unless manually shut down, the physical and protocol states of a loopback interface remain Up. Typically, a loopback interface uses a 32-bit mask and serves the following purposes:

1. Its IP address is used as the management address of the network device.
2. Its IP address is used as the router ID in dynamic routing protocols.
3. It enhances network reliability.

```
[R1] interface LoopBack0
[R1-LoopBack0] ip address 10.0.2.2 32
[R1-LoopBack0] quit
```

```
[S1] interface LoopBack0
[S1-LoopBack0] ip address 10.0.1.1 32
[S1-LoopBack0] quit
```

```
[S2] interface LoopBack0
[S2-LoopBack0] ip address 10.0.3.3 32
[S2-LoopBack0] quit
```

Verify connectivity between loopback interfaces.

```
[R1] ping -a 10.0.2.2 10.0.1.1

PING 10.0.1.1: 56 data bytes, press CTRL_C to break

Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.1.1 ping statistics ---
```

```
5 packet(s) transmitted
```

```
0 packet(s) received
```

```
100.00% packet loss
```

Because R1 does not have a route to the destination IP address, the ping operation fails.

Step 4 Complete the basic OSPF configuration.

Create an OSPF process.

```
[R1] ospf 1
```

Creating an OSPF process is a prerequisite for configuring OSPF parameters. OSPF multi-process allows multiple OSPF processes to run independently on the same device. Route exchange between different OSPF processes is similar to that between different routing protocols. When creating an OSPF process, specify a process ID. If no process ID is specified, the default process ID 1 is used.

Create an OSPF area and enable OSPF on interfaces.

```
[R1-ospf-1] area 0
```

The **area** command creates an OSPF area and enters its view.

```
[R1-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.255
```

```
[R1-ospf-1-area-0.0.0.0] network 10.0.23.2 0.0.0.255
```

```
[R1-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0
```

```
[R1-ospf-1-area-0.0.0.0] quit
```

```
[R1-ospf-1] quit
```

The **network** *network-address wildcard-mask* command specifies the interface on which OSPF is to be enabled. The *wildcard-mask* is similar to the reversed form of an IP address mask. "1" represents that the corresponding bit in the IP address is ignored, and "0" represents that this bit must be reserved.

In the preceding configuration, OSPF is enabled on three interfaces, all of which are added to area 0.

OSPF can run on an interface only when the following conditions are met:

1. The mask length of the IP address of the interface is greater than or equal to the mask length specified in the **network** command.

Note: If *wildcard-mask* is set to all 0s in the **network** command and the IP address of the interface is the same as that configured in the *network-address* command, the interface also runs OSPF.

2. The primary IP address of the interface must be in the range of the network segment specified in the **network** command.

```
[S1] ospf
```

Warning: If router-id is not specified, the router ID may change after the restart or a router ID conflict may occur.

```
[S1-ospf-1] area 0
```

```
[S1-ospf-1-area-0.0.0.0] network 10.0.12.1 0.0.0.0
```

```
[S1-ospf-1-area-0.0.0.0] network 10.0.13.1 0.0.0.0
```

```
[S1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0
```

```
[S1-ospf-1-area-0.0.0.0] quit
```

```
[S1-ospf-1] quit
```

```
[S2] ospf
```

Warning: If router-id is not specified, the router ID may change after the restart or a router ID conflict may occur.

```
[S2-ospf-1] area 0
```

```
[S2-ospf-1-area-0.0.0.0] network 10.0.13.3 0.0.0.0
```

```
[S2-ospf-1-area-0.0.0.0] network 10.0.23.3 0.0.0.0
```

```
[S2-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0
```

```
[S2-ospf-1-area-0.0.0.0] quit
```

```
[S2-ospf-1] quit
```

Step 5 Check the OSPF status.

Check OSPF neighbor information.

```
[R1] display ospf peer
```

OSPF Process 1 with Router ID **10.0.12.2**

Area 0.0.0.0 interface 10.0.12.2 (GE0/0/1)'s neighbors

```

Router ID: 10.0.12.1      Address : 10.0.12.1
State   : Full           Mode    : Nbr is Slave      Priority: 1
DR      : 10.0.12.2      BDR     : 10.0.12.1      MTU     : 0
Dead timer due (in seconds) : 35
Retrans timer interval      : 5
Neighbor up time           : 00h02m13s
Neighbor up time stamp     : XXXX-XX-XX 11:46:41
Authentication Sequence    : 0
  
```

Area 0.0.0.0 interface 10.0.23.2 (GE0/0/2)'s neighbors

```

Router ID: 10.0.23.3      Address : 10.0.23.3
State   : Full           Mode    : Nbr is Master      Priority: 1
DR      : 10.0.23.2      BDR     : 10.0.23.3      MTU     : 0
Dead timer due (in seconds) : 34
Retrans timer interval      : 5
Neighbor up time           : 00h00m47s
Neighbor up time stamp     : XXXX-XX-XX 11:48:07
Authentication Sequence    : 0
  
```

The **display ospf peer** command displays information about neighbors in each OSPF area. The information includes each neighbor's area, router ID, status, designated router (DR), and backup designated router (BDR).

According to the preceding command output, the router IDs of R1, S1, and S2 are 10.0.12.2, 10.0.12.1, and 10.0.23.3, respectively. Additionally, within the network segment where GE0/0/1 of R1 resides, R1 is the DR and S1 is the BDR.

Note: The DR, BDR, and router ID may vary according to the lab environment. This is because the OSPF router ID can be automatically assigned, and the DR and BDR are elected in non-preemption mode.

Step 6 Change router IDs of OSPF devices.

Change the router ID of each OSPF device to its LoopBack0 IP address, and restart the OSPF process for the new router ID to take effect.

```
[R1] ospf 1 router-id 10.0.2.2
```

```
Warning: The operation will cause reset OSPF. Continue? [Y/N] : y
```

```
[R1-ospf-1] quit
```

```
[S1] ospf 1 router-id 10.0.1.1
```

```
Warning: The operation will cause reset OSPF. Continue? [Y/N] : y
```

```
[S1-ospf-1] quit
```

```
[S2] ospf 1 router-id 10.0.3.3
```

```
Warning: The operation will cause reset OSPF. Continue? [Y/N] : y
```

```
[S2-ospf-1] quit
```

Check OSPF neighbor information.

```
[R1] display ospf peer
```

```
OSPF Process 1 with Router ID 10.0.2.2
```

```
Area 0.0.0.0 interface 10.0.12.2 (GE0/0/1)'s neighbors
```

Router ID: 10.0.1.1	Address : 10.0.12.1	
State : Full	Mode : Nbr is Slave	Priority: 1
DR : 10.0.12.2	BDR : 10.0.12.1	MTU : 0
Dead timer due (in seconds) : 36		
Retrans timer interval : 5		
Neighbor up time : 00h00m38s		
Neighbor up time stamp : XXXX-XX-XX 11:53:11		
Authentication Sequence : 0		

```
Area 0.0.0.0 interface 10.0.23.2 (GE0/0/2)'s neighbors
```

Router ID: 10.0.3.3	Address : 10.0.23.3
----------------------------	---------------------

State	: Full	Mode	: Nbr is Master	Priority:	1
DR	: 10.0.23.2	BDR	: 10.0.23.3	MTU	: 0
Dead timer due (in seconds) : 37					
Retrans timer interval		: 5			
Neighbor up time		: 00h00m09s			
Neighbor up time stamp		: XXXX-XX-XX 11:53:40			
Authentication Sequence		: 0			

The command output shows that each device's router ID is their respective LoopBack0 IP address.

```
# Check the routes learned from OSPF in the IP routing table.
```

```
[R1] display ip routing-table protocol ospf

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

-----

_public_ Routing Table : OSPF

Destinations : 6          Routes : 7

OSPF routing table status : <Active>

Destinations : 3          Routes : 4

Destination/Mask    Proto  Pre  Cost           Flags NextHop          Interface

10.0.1.1/32        OSPF   10   1              D    10.0.12.1             GE0/0/1
10.0.3.3/32        OSPF   10   1              D    10.0.23.3             GE0/0/2
10.0.13.0/24       OSPF   10   2              D    10.0.23.3             GE0/0/2
                   OSPF   10   2              D    10.0.12.1             GE0/0/1

OSPF routing table status : <Inactive>
```

Destinations : 3			Routes : 3			
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.2/32	OSPF	10	0		10.0.2.2	LoopBack0
10.0.12.0/24	OSPF	10	1		10.0.12.2	GE0/0/1
10.0.23.0/24	OSPF	10	1		10.0.23.2	GE0/0/2

The command output shows that three OSPF routes are installed into the IP routing table. There are two equal-cost routes to 10.0.13.0/24, since R1 can reach this destination network through both S1 and S2. The two routes from R1 to S1 and S2 both have a cost value of 1, and their outbound interfaces are directly connected.

Additionally, the three OSPF routes are in **Inactive** state. These routes are the same as the direct routes generated after interface IP addresses are configured. Because direct routes have the highest priority, these OSPF routes are not installed into the IP routing table.

Step 7 Configure OSPF authentication.

Configure interface authentication on R1.

```
[R1] interface GE0/0/1
[R1-GE0/0/1] ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
[R1-GE0/0/1] quit
[R1] interface GE 0/0/2
[R1-GE0/0/2] ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
[R1-GE0/0/2] display this
#
interface GE0/0/2
ip address 10.0.23.2 255.255.255.0
ospf authentication-mode hmac-sha256 1
cipher %+%##!!!!!!!"!!!!!!*!!!!!!y81!>Czw!oj345&&,YP*U%`C,a@@'Gl}TD!!!!!!2jp5!!!!!!=!!!!U^>%Jf#G9!yb:JK
M|W'0jav3[$}ZQ[b>r!!!!!!%+%#
#
return
```

```
[R1-GE0/0/2] quit
```

In the configuration, the password is displayed in ciphertext, since **cipher** denotes encrypted text.

Wait for a while and check the neighbor status.

```
[R1] display ospf peer brief
```

Because authentication is not configured on other network devices, the authentication fails and no neighbor relationship is established.

On S1, configure interface authentication.

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
[S1-GE1/0/1] quit
[S1] interface GE 1/0/2
[S1-GE1/0/2] ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
[S1-GE1/0/2] quit
```

Check the neighbor status of S1.

```
[S1] display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 10.0.1.1
Area 0.0.0.0 interface 10.0.12.1 (GE1/0/1)'s neighbors
  Router ID: 10.0.2.2          Address : 10.0.12.2
  State   : Full              Mode    : Nbr is Master    Priority: 1
  DR      : 10.0.12.2         BDR    : 10.0.12.1    MTU     : 0
  Dead timer due (in seconds) : 35
  Retrans timer interval      : 5
```

```
Neighbor up time      : 00h00m34s
Neighbor up time stamp : XXXX-XX-XX 09:23:33
Authentication Sequence : 1013
```

The command output shows that S1 has established an adjacency with R1.

On S2, configure area authentication.

```
[S2] ospf
[S2-ospf-1] area 0
[S2-ospf-1-area-0.0.0.0] authentication-mode hmac-sha256 1 cipher HCIA-Datcom
[S2-ospf-1-area-0.0.0.0] quit
[S2-ospf-1] quit
```

Check the neighbor status of S2.

```
[S2] display ospf peer brief

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 10.0.3.3

      Peer Statistic Information

Total number of peer(s): 2

Peer(s) in full state: 2

-----
Area Id      Interface      Neighbor id    State
-----
0.0.0.0      GE1/0/1         10.0.2.2      Full
0.0.0.0      GE1/0/2         10.0.1.1      Full
-----
```

The command output shows that S2 has established an adjacency with R1 and S1. (OSPF interface authentication and area authentication both implement OSPF packet authentication on OSPF interfaces.)

Step 8 Assuming R1 serves as the network egress, configure it to advertise default routes.

Configure R1 to advertise default routes.

```
[R1] ospf
[R1-ospf-1] default-route-advertise always
[R1-ospf-1] quit
```

The **default-route-advertise** command advertises default routes to a common OSPF area. If the **always** parameter is not specified, default routes are advertised to other network devices only when the local device has active default routes from other OSPF processes in its routing table. In this example, the local routing table contains no default route. Therefore, the **always** parameter must be specified.

Check the IP routing tables on S1 and S2.

```
[S1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

-----
Routing Table : _public_

          Destinations : 15          Routes : 16

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface

   0.0.0.0/0   O_ASE   150   1             D   10.0.12.2         GE1/0/1
  10.0.1.1/32   Direct     0     0             D   127.0.0.1         LoopBack0
  10.0.2.2/32   OSPF      10     1             D   10.0.12.2         GE1/0/1
  10.0.3.3/32   OSPF      10     1             D   10.0.13.3         GE1/0/2
  10.0.12.0/24  Direct     0     0             D   10.0.12.1         GE1/0/1
  10.0.12.1/32  Direct     0     0             D   127.0.0.1         GE1/0/1
  10.0.12.255/32 Direct     0     0             D   127.0.0.1         GE1/0/1
```


10.0.13.0/24	Direct	0	0	D	10.0.13.1	GE1/0/2
10.0.13.1/32	Direct	0	0	D	127.0.0.1	GE1/0/2
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GE1/0/2
10.0.23.0/24	OSPF	10	2	D	10.0.13.3	GE1/0/2
	OSPF	10	2	D	10.0.12.2	GE1/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[S2] display ip routing-table

Proto: Protocol Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

Routing Table : _public_

Destinations : 15 Routes : 16

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	O_ASE	150	1	D	10.0.23.2	GE1/0/1
10.0.1.1/32	OSPF	10	1	D	10.0.13.1	GE1/0/2
10.0.2.2/32	OSPF	10	1	D	10.0.23.2	GE1/0/1
10.0.3.3/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.12.0/24	OSPF	10	2	D	10.0.13.1	GE1/0/10
	OSPF	10	2	D	10.0.23.2	GE1/0/1
10.0.13.0/24	Direct	0	0	D	10.0.13.3	GE1/0/2
10.0.13.3/32	Direct	0	0	D	127.0.0.1	GE1/0/2
10.0.13.255/32	Direct	0	0	D	127.0.0.1	GE1/0/2

10.0.23.0/24	Direct	0	0	D	10.0.23.3	GE1/0/1
10.0.23.3/32	Direct	0	0	D	127.0.0.1	GE1/0/1
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE1/0/1
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that S1 and S2 have learned the default route.

Step 9 Change the cost value of the corresponding interface on R1 to enable R1's LoopBack0 to access S2's LoopBack0 through the R1 -> S1 -> S2 path.

According to R1's routing table, the route cost for R1 to access S2's LoopBack0 via the R1 -> S2 path is 1. As such, increase the cost value of this route to a value greater than 2.

```
[R1] interface GE 0/0/2
[R1-GE0/0/2] ospf cost 10
[R1-GE0/0/2] quit
```

Check the routing table of R1.

```
[R1] display ip routing-table

Proto: Protocol          Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : _public_

          Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre    Cost           Flags NextHop           Interface

10.0.1.1/32    OSPF      10     1              D    10.0.12.1           GE0/0/1
10.0.2.2/32    Direct    0       0              D    127.0.0.1           LoopBack0
```

10.0.3.3/32	OSPF	10	2	D	10.0.12.1	GE0/0/1
10.0.12.0/24	Direct	0	0	D	10.0.12.2	GE0/0/1
10.0.12.2/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.12.255/32	Direct	0	0	D	127.0.0.1	GE0/0/1
10.0.13.0/24	OSPF	10	2	D	10.0.12.1	GE0/0/1
10.0.23.0/24	Direct	0	0	D	10.0.23.2	GE0/0/2
10.0.23.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
10.0.23.255/32	Direct	0	0	D	127.0.0.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the next hop for R1 to access S2's LoopBack0 is S1.

Verify the result by running the **tracert** command.

```
[R1] tracert -a 10.0.2.2 10.0.3.3

tracert to 10.0.3.3(10.0.3.3), max hops: 64, packet length: 40, press CTRL_C to break

 1 10.0.12.1 11 ms 10 ms 10 ms
 2 10.0.3.3 10 ms 12 ms 10 ms
```

8.3 Verification

Verify the following on your own:

1. Use the **ping** command to verify connectivity between interfaces.
2. Shut down an interface to simulate a link fault and check the changes in routing tables.

8.4 Reference Configuration

8.4.1 R1 Configuration

```
sysname R1

#
interface GE0/0/1

 ip address 10.0.12.2 255.255.255.0

 ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom

#
interface GE0/0/2

 ip address 10.0.23.2 255.255.255.0

 ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom

 ospf cost 10

#
interface LoopBack0

 ip address 10.0.2.2 255.255.255.255

#
ospf 1 router-id 10.0.2.2

 default-route-advertise always

 area 0.0.0.0

  network 10.0.2.2 0.0.0.0

  network 10.0.12.0 0.0.0.255

  network 10.0.23.0 0.0.0.255

#
```

8.4.2 S1 Configuration

```
sysname S1

#
```

```
interface GE1/0/1

undo portswitch

ip address 10.0.12.1 255.255.255.0

ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
#
interface GE1/0/2

undo portswitch

ip address 10.0.13.1 255.255.255.0

ospf authentication-mode hmac-sha256 1 cipher HCIA-Datcom
#
interface LoopBack0

ip address 10.0.1.1 255.255.255.255
#
ospf 1 router-id 10.0.1.1

area 0.0.0.0

network 10.0.1.1 0.0.0.0

network 10.0.12.1 0.0.0.0

network 10.0.13.1 0.0.0.0
#
```

8.4.3 S2 Configuration

```
sysname S2

#
interface GE1/0/1

undo portswitch

ip address 10.0.23.3 255.255.255.0
#
```

```
interface GE1/0/2

undo portswitch

ip address 10.0.13.3 255.255.255.0

#

interface LoopBack0

ip address 10.0.3.3 255.255.255.255

#

ospf 1 router-id 10.0.3.3

area 0.0.0.0

authentication-mode hmac-sha256 1 cipher HCIA-Datcom

network 10.0.3.3 0.0.0.0

network 10.0.13.3 0.0.0.0

network 10.0.23.3 0.0.0.0

#
```

8.5 Quiz

1. In Step 9, what is the path for S2 to reply to the ICMP packets from R1? Explain the reason.

Answer:

The path for S2 to reply to R1 is S2 -> R1. OSPF selects routes based on the sum of outbound interface costs, and in this case the direct path has the lowest cumulative cost.

9 IPv6 Address Configuration

9.1 Introduction

9.1.1 About This Lab

Internet Protocol version 6 (IPv6) is also called IP Next Generation (IPng). It is a set of specifications designed by the Internet Engineering Task Force (IETF) and is the next generation of IPv4.

Compared with IPv4, IPv6 has the following advantages:

- Nearly infinite address space
- Hierarchical address structure
- Plug and play
- Simplified packet header
- Security features
- Mobility
- Enhanced QoS features

This chapter describes how to build an IPv6 network to help you understand the fundamentals and address configuration of IPv6.

9.1.2 Objectives

- Learn how to configure static IPv6 addresses.
- Learn how to configure IPv6 stateless address autoconfiguration.
- Learn how to configure IPv6 static routes.
- Learn how to check IPv6 information.

9.1.3 Networking Topology

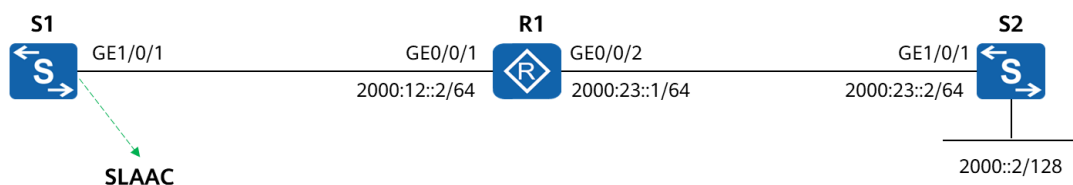


Figure 9-1 Lab topology for IPv6 address configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

1. The two interfaces on R1 are configured with static IPv6 addresses.
2. GE1/0/1 on S1 is configured with an IPv6 address using stateless address autoconfiguration.
3. After obtaining an IPv6 address, S1 can communicate with LoopBack0 on S2.

9.1.4 Background

An enterprise network is deployed with IPv6 for communication. The network devices running on the enterprise network need to be configured.

9.2 Task Configuration

9.2.1 Configuration Roadmap

1. Configure static IPv6 addresses.
2. Configure IPv6 stateless address autoconfiguration.
3. Configure IPv6 static routes.

9.2.2 Configuration Procedure

Step 1 Configure the IPv6 function on interfaces.

Enable the IPv6 function on interfaces.

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] undo portswitch
[S1-GE1/0/1] ipv6 enable
[S1-GE1/0/1] quit
```

The **ipv6 enable** command enables the IPv6 function on an interface.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ipv6 enable
[R1-GE0/0/1] quit
```



```
[R1] interface GE 0/0/2  
[R1-GE0/0/2] ipv6 enable  
[R1-GE0/0/2] quit
```

```
[S2] interface GE 1/0/1  
[S2-GE1/0/1] undo portswitch  
[S2-GE1/0/1] ipv6 enable  
[S2-GE1/0/1] quit
```

Step 2 Configure link-local addresses for interfaces and test the connectivity.

Configure interfaces to automatically generate link-local addresses.

```
[S1] interface GE 1/0/1  
[S1-GE1/0/1] ipv6 address auto link-local  
[S1-GE1/0/1] quit
```

The **ipv6 address auto link-local** command enables an interface to automatically generate a link-local address.

Each interface can have only one link-local address. To prevent link-local address conflicts, automatic generation of link-local addresses is recommended. After an IPv6 global unicast address is configured for an interface, a link-local address is automatically generated.

```
[R1] interface GE 0/0/1  
[R1-GE0/0/1] ipv6 address auto link-local  
[R1-GE0/0/1] quit  
[R1] interface GE 0/0/2  
[R1-GE0/0/2] ipv6 address auto link-local  
[R1-GE0/0/2] quit
```

```
[S2] interface GE 1/0/1  
[S2-GE1/0/1] ipv6 address auto link-local
```

```
[S2-GE1/0/1] quit
```

Check the IPv6 status of interfaces and test the connectivity.

```
[S1] display ipv6 interface GE 1/0/1

GE1/0/1 current state : UP

IPv6 protocol current state : UP

IPv6 is enabled, link-local address is FE80::FA98:FDFF:FEFF:20

No global unicast address configured

Joined group address(es):

    FF02::1

    FF02::2

    FF02::1:FFFF:20

MTU is 1500 bytes

ND DAD is enabled, number of DAD attempts: 1

ND NUD is enabled, number of NUD attempts: 3

ND NUD interval is 5000 milliseconds

ND reachable time is 1200000 milliseconds

ND stale time is 1200 seconds

ND retransmit interval is 1000 milliseconds

ND RAs are halted

ND Proxy is disabled
```

The link-local address of GE1/0/1 on S1 is FE80::FA98:FDFF:FEFF:20.

Note: The link-local address is automatically generated by the device and varies according to the lab environment.

```
[R1] display ipv6 interface GE 0/0/1

GE0/0/1 current state : UP

IPv6 protocol current state : UP

IPv6 is enabled, link-local address is FE80::FA98:FDFF:FEFF:11
```

```
No global unicast address configured

Joined group address(es):

    FF02::1

    FF02::2

    FF02::1:FFFF:11

MTU is 1500 bytes

ND DAD is enabled, number of DAD attempts: 1

ND NUD is enabled, number of NUD attempts: 3

ND NUD interval is 5000 milliseconds

ND reachable time is 1200000 milliseconds

ND stale time is 1200 seconds

ND retransmit interval is 1000 milliseconds

ND RAs are halted
```

The link-local address of GE0/0/1 on R1 is FE80::FA98:FDFF:FEFF:11.

```
[R1] display ipv6 interface GE 0/0/2

GE0/0/2 current state : UP

IPv6 protocol current state : UP

IPv6 is enabled, link-local address is FE80::FA98:FDFF:FEFF:12

No global unicast address configured

Joined group address(es):

    FF02::1

    FF02::2

    FF02::1:FFFF:12

MTU is 1500 bytes

ND DAD is enabled, number of DAD attempts: 1

ND NUD is enabled, number of NUD attempts: 3

ND NUD interval is 5000 milliseconds
```

ND reachable time is 1200000 milliseconds

ND stale time is 1200 seconds

ND retransmit interval is 1000 milliseconds

ND RAs are halted

The link-local address of GE0/0/2 on R1 is FE80::FA98:FDFF:FEFF:12.

```
[S2] display ipv6 interface GE 1/0/1
```

GE1/0/1 current state : UP

IPv6 protocol current state : UP

IPv6 is enabled, link-local address is **FE80::FA98:FDFF:FEFF:30**

No global unicast address configured

Joined group address(es):

FF02::1

FF02::2

FF02::1:FFFF:30

MTU is 1500 bytes

ND DAD is enabled, number of DAD attempts: 1

ND NUD is enabled, number of NUD attempts: 3

ND NUD interval is 5000 milliseconds

ND reachable time is 1200000 milliseconds

ND stale time is 1200 seconds

ND retransmit interval is 1000 milliseconds

ND RAs are halted

ND Proxy is disabled

The link-local address of GE1/0/1 on S2 is FE80::FA98:FDFF:FEFF:30.

Test the connectivity between S1 and R1.

```
[S1] ping ipv6 FE80::FA98:FDFF:FEFF:11 -i GE 1/0/1
```

```
PING FE80::FA98:FDFF:FEFF:11 : 56  data bytes, press CTRL_C to break
```

```
Reply from FE80::FA98:FDFF:FEFF:11
```

```
bytes=56 Sequence=1 hop limit=64 time=16 ms
```

```
Reply from FE80::FA98:FDFF:FEFF:11
```

```
bytes=56 Sequence=2 hop limit=64 time=9 ms
```

```
Reply from FE80::FA98:FDFF:FEFF:11
```

```
bytes=56 Sequence=3 hop limit=64 time=7 ms
```

```
Reply from FE80::FA98:FDFF:FEFF:11
```

```
bytes=56 Sequence=4 hop limit=64 time=10 ms
```

```
Reply from FE80::FA98:FDFF:FEFF:11
```

```
bytes=56 Sequence=5 hop limit=64 time=13 ms
```

```
--- FE80::FA98:FDFF:FEFF:11 ping statistics---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max=7/11/16 ms
```

When the destination IPv6 address of a ping test is a link-local address, you must specify the source interface or source IPv6 address.

Step 3 Configure static IPv6 addresses for interfaces.

```
[R1] interface GE 0/0/1
```

```
[R1-GE0/0/1] ipv6 address 2000:12::2 64
```

```
[R1-GE0/0/1] quit
```

```
[R1] interface GE 0/0/2
```

```
[R1-GE0/0/2] ipv6 address 2000:23::1 64
```

```
[R1-GE0/0/2] quit
```

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] ipv6 address 2000:23::2 64
[S2-GE1/0/1] quit
[S2] interface LoopBack 0
[S2-LoopBack0] ipv6 enable
[S2-LoopBack0] ipv6 address 2000::2 128
[S2-LoopBack0] quit
```

Ping the IP address of an interface on S2 from R1.

```
[R1] ping ipv6 2000:23::2
PING 2000:23::2 : 56 data bytes, press CTRL_C to break
  Reply from 2000:23::2
    bytes=56 Sequence=1 hop limit=64 time=54 ms
  Reply from 2000:23::2
    bytes=56 Sequence=2 hop limit=64 time=56 ms
  Reply from 2000:23::2
    bytes=56 Sequence=3 hop limit=64 time=54 ms
  Reply from 2000:23::2
    bytes=56 Sequence=4 hop limit=64 time=54 ms
  Reply from 2000:23::2
    bytes=56 Sequence=5 hop limit=64 time=55 ms

--- 2000:23::2 ping statistics---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max=54/54/56 ms
```

Check IPv6 neighbor information of GE0/0/2 on R1.

```
[R1] display ipv6 neighbors GE 0/0/2

-----

IPv6 Address : 2000:23::2

Link-layer   : fa98-fdff-0032          State      : REACH
Interface    : GE0/0/2                Age        : 0
VLAN         : -                      CEVLAN     : -
VPN name     : -                      Is Router  : TRUE
Secure FLAG  : UN-SECURE              Nickname   : -
Source IP    : -
Destination IP: -
VNI          : -                      BD         : -
.....
```

The command output shows that the MAC address of an interface on S2 is fa98-fdff-0032. Different from MAC address obtaining through ARP in IPv4, the MAC address of a neighboring device is obtained through NS and NA messages in IPv6.

Step 4 Configure S1 to obtain an IPv6 address in stateless mode.

Configure S1.

```
[S1] interface GE 1/0/1

[S1-GE1/0/1] ipv6 address auto global default

[S1-GE1/0/1] quit
```

The **ipv6 address auto global** command enables the function of automatically generating a global IPv6 address through stateless autoconfiguration.

The **ipv6 address auto global default** command enables the device to use the source IPv6 address in an RA message as the next-hop address of the default route.

Configure R1.

```
[R1] interface GE 0/0/1

[R1-GE0/0/1] ipv6 nd ra halt disable
```

```
[R1-GE0/0/1] ipv6 nd ra prefix 2000:12::/64 3600 1200
[R1-GE0/0/1] quit
```

The **ipv6 nd ra halt disable** command enables the system to send RA messages.

The **ipv6 nd ra prefix** command configures the prefix in RA messages.

valid-lifetime specifies the valid lifetime of the prefix. The valid lifetime is used to determine the on-link status of the prefix. In this example, this parameter is set to 3600 seconds. If this parameter is specified, the prefix carried in a received RA message is valid within the specified lifetime.

preferred-lifetime specifies the preferred lifetime of the address generated based on the prefix through stateless address autoconfiguration. In this example, this parameter is set to 1200 seconds. The preferred lifetime is less than or equal to the valid lifetime.

Check the IPv6 address obtained by S1.

```
[S1] display ipv6 interface GE 1/0/1

GE1/0/1 current state : UP

IPv6 protocol current state : UP

IPv6 is enabled, link-local address is FE80::FA98:FDFF:FEFF:20

Global unicast address(es):

    2000:12::FA98:FDFF:FEFF:20,
    subnet is 2000:12::/64 [SLAAC XXXX-XX-XX 07:39:45 7200S]

Joined group address(es):

    FF02::1

    FF02::2

    FF02::1:FFFF:20

MTU is 1500 bytes

ND DAD is enabled, number of DAD attempts: 1

ND NUD is enabled, number of NUD attempts: 3

ND NUD interval is 5000 milliseconds

ND reachable time is 1200000 milliseconds

ND stale time is 1200 seconds

ND retransmit interval is 1000 milliseconds
```


ND RAs are halted

ND Proxy is disabled

S1 has been automatically configured with the IPv6 address 2000:12::FA98:FDFF:FEFF:20. The first 64 bits of the address are the prefix advertised by R1, and the last 64 bits are the same as the last 64 bits of the link-local address.

Check the IPv6 routing table of S1.

```
[S1] display ipv6 routing-table

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----

Routing Table : _public_

          Destinations : 7          Routes : 7

Destination : ::                                PrefixLength : 0
NextHop      : FE80::FA98:FDFF:FEFF:11          Preference : 64
Cost         : 0                                Protocol    : ND
RelayNextHop : ::                                TunnelID    : 0x0
Interface    : GE1/0/1                          Flags       : D
.....
```

S1 has automatically generated a default route with the next hop being the link-local address of R1. The preference of the route is 64, and the protocol type is ND.

Ping the loopback interface address of S2 from S1.

```
[S1] ping ipv6 2000::2

PING 2000::2 : 56 data bytes, press CTRL_C to break

Request time out
Request time out
Request time out
Request time out
Request time out
```

```
--- 2000::2 ping statistics---  
  5 packet(s) transmitted  
  0 packet(s) received  
100.00% packet loss  
round-trip min/avg/max=0/0/0 ms
```

The ping fails because no route is configured between R1 and S2.

Step 5 Configure IPv6 static routes.

Configure static routes between R1 and S2.

```
[R1] ipv6 route-static 2000::2 128 2000:23::2
```

The **ipv6 route-static** command configures a public network IPv6 static route.

```
[S2] ipv6 route-static 2000:12:: 64 2000:23::1
```

Check the connectivity.

```
[S1] ping ipv6 2000::2  
  
PING 2000::2 : 56 data bytes, press CTRL_C to break  
  
Reply from 2000::2  
bytes=56 Sequence=1 hop limit=62 time=1 ms  
  
Reply from 2000::2  
bytes=56 Sequence=2 hop limit=62 time=1 ms  
  
Reply from 2000::2  
bytes=56 Sequence=3 hop limit=62 time=1 ms  
  
Reply from 2000::2  
bytes=56 Sequence=4 hop limit=62 time=1 ms  
  
Reply from 2000::2  
bytes=56 Sequence=5 hop limit=62 time=1 ms
```

```
--- 2000::2 ping statistics---  
  
  5 packet(s) transmitted  
  
  5 packet(s) received  
  
  0.00% packet loss  
  
round-trip min/avg/max=1/1/1 ms
```

S1 can communicate with the loopback interface of S2.

9.3 Reference Configuration

9.3.1 R1 Configuration

```
sysname R1  
  
#  
  
interface GE0/0/1  
  
  ipv6 enable  
  
  ipv6 address 2000:12::2/64  
  
  ipv6 address auto link-local  
  
  ipv6 nd ra prefix 2000:12::/64 3600 1200  
  
  ipv6 nd ra halt disable  
  
#  
  
interface GE0/0/2  
  
  ipv6 enable  
  
  ipv6 address 2000:23::1/64  
  
  ipv6 address auto link-local  
  
#  
  
ipv6 route-static 2000::2 128 2000:23::2
```

9.3.2 S1 Configuration

```
sysname S1
#
interface GE1/0/1
    undo portswitch
    ipv6 enable
    ipv6 address auto link-local
    ipv6 address auto global default
#
```

9.3.3 S2 Configuration

```
sysname S2
#
interface GE1/0/1
    undo portswitch
    ipv6 enable
    ipv6 address 2000:23::2/64
    ipv6 address auto link-local
#
interface LoopBack0
    ipv6 enable
    ipv6 address 2000::2/128
#
ipv6 route-static 2000:12:: 64 2000:23::1
```

9.4 Quiz

1. Why must the source interface be specified when the connectivity between link-local addresses is checked?

Answer:

A network device has multiple interfaces, and these interfaces belong to the FE80::/10 network segment. When the destination IPv6 address is a link-local address, the outbound interface cannot be determined by querying the routing table. Therefore, the source interface must be specified.

10 ACL Configuration

10.1 Introduction

10.1.1 About This Lab

An access control list (ACL) is a collection of one or more rules. A rule refers to a judgment statement that describes a packet matching condition, which can be a source address, destination address, or port number.

An ACL is essentially a rule-based packet filter. The device filters packets based on given ACL rules and permits or denies the matched packets based on the processing policy of the service module to which the ACL is applied.

10.1.2 Objectives

- Learn how to configure ACLs.
- Learn how to apply an ACL to an interface.
- Understand the basic methods of traffic filtering.

10.1.3 Networking Topology

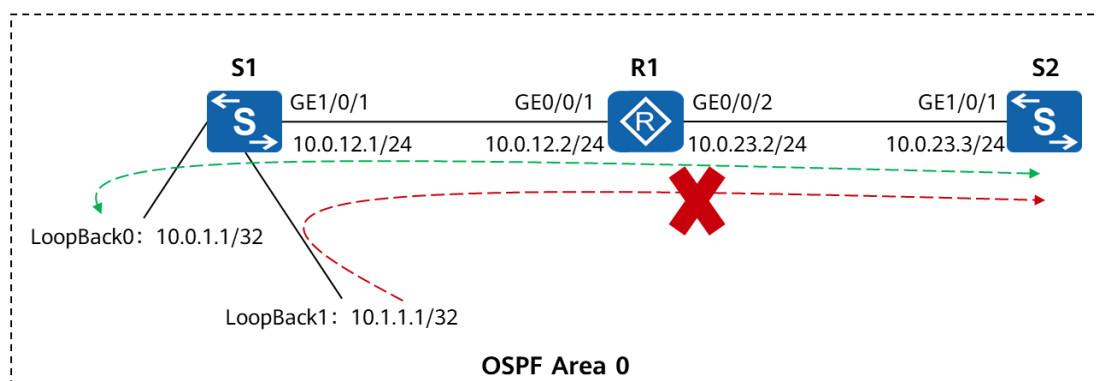


Figure 10-1 Lab topology for ACL configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

10.1.4 Background

As shown in the networking topology, there are reachable routes between S1, R1, and S2. Two logical interfaces LoopBack0 and LoopBack1 are created on S1, with IP addresses 10.0.1.1/32 and 10.1.1.1/32, respectively.

Configure an ACL to allow the LoopBack0 interface on S1 to ping S2's interface IP address, while preventing the LoopBack1 interface on S1 from doing so.

10.2 Task Configuration

10.2.1 Configuration Roadmap

1. Configure IP addresses for devices.
2. Configure OSPF to ensure network connectivity.
3. Configure an ACL to match desired traffic.
4. Configure traffic filtering.

10.2.2 Configuration Procedure

Step 1 Configure IP addresses for devices.

Configure IP addresses for R1, S1, and S2.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 10.0.12.2 24
[R1-GE0/0/1] quit
[R1] interface GE 0/0/2
[R1-GE0/0/2] ip address 10.0.23.2 24
[R1-GE0/0/2] quit
```

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] undo portswitch
[S1-GE1/0/1] ip address 10.0.12.1 24
[S1-GE1/0/1] quit
[S1] interface LoopBack 0
```

```
[S1-LoopBack0] ip address 10.0.1.1 32
[S1-LoopBack0] quit
[S1] interface LoopBack 1
[S1-LoopBack1] ip address 10.1.1.1 32
[S1-LoopBack1] quit
```

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] undo portswitch
[S2-GE1/0/1] ip address 10.0.23.3 24
[S2-GE1/0/1] quit
```

Step 2 Configure OSPF to ensure network connectivity.

Configure OSPF on R1, S1, and S2 and assign them to area 0 to enable connectivity.

```
[R1] ospf
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.0.12.2 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.23.2 0.0.0.0
[R1-ospf-1-area-0.0.0.0] quit
[R1-ospf-1] quit
```

```
[S1] ospf
[S1-ospf-1] area 0
[S1-ospf-1-area-0.0.0.0] network 10.0.12.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0] network 10.1.1.1 0.0.0.0
[S1-ospf-1-area-0.0.0.0] quit
[S1-ospf-1] quit
```



```
[S2] ospf
[S2-ospf-1] area 0
[S2-ospf-1-area-0.0.0.0] network 10.0.23.3 0.0.0.0
[S2-ospf-1-area-0.0.0.0] quit
[S2-ospf-1] quit
```

Run the **ping** command on S1 to test network connectivity.

```
[S1] ping -a 10.0.1.1 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=59 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=90 ms
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=68 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=59 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=83 ms

--- 10.0.23.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 59/71/90 ms
```

```
[S1] ping -a 10.1.1.1 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=59 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=77 ms
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=108 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=95 ms
```

```
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=72 ms
```

```
--- 10.0.23.3 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 59/82/108 ms
```

Step 3 Configure an ACL to match desired traffic.

Configure an ACL on R1 to deny packets with the source IP address 10.1.1.1 and permit other packets.

```
[R1] acl 3000
[R1-acl4-advance-3000] rule deny ip source 10.1.1.1 0.0.0.0
[R1-acl4-advance-3000] rule permit ip
[R1-acl4-advance-3000] quit
```

Configure a traffic policy in modular QoS command line interface (MQC) mode.

```
[R1] traffic classifier test
[R1-classifier-test] if-match acl 3000
[R1-classifier-test] quit
[R1] traffic behavior test
[R1-behavior-test] permit
[R1-behavior-test] quit
[R1] traffic policy test
[R1-trafficpolicy-test] classifier test behavior test
[R1-trafficpolicy-test] quit
```

Filter traffic on GE0/0/2 of R1.

```
[R1] interface GE 0/0/2
[R1-GE0/0/2] traffic-policy test outbound
[R1-GE0/0/2] quit
```

Run the **ping** command on S1 to test network connectivity.

```
[S1] ping -a 10.0.1.1 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.23.3: bytes=56 Sequence=1 ttl=254 time=59 ms
Reply from 10.0.23.3: bytes=56 Sequence=2 ttl=254 time=90 ms
Reply from 10.0.23.3: bytes=56 Sequence=3 ttl=254 time=68 ms
Reply from 10.0.23.3: bytes=56 Sequence=4 ttl=254 time=59 ms
Reply from 10.0.23.3: bytes=56 Sequence=5 ttl=254 time=83 ms

--- 10.0.23.3 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 59/71/90 ms
```

```
[S1] ping -a 10.1.1.1 10.0.23.3

PING 10.0.23.3: 56 data bytes, press CTRL_C to break

Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.0.23.3 ping statistics ---
```

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss

The preceding command output shows that an ACL is applied to the outbound direction of GE0/0/2 on R1 to prevent S1's LoopBack1 interface from accessing S2's physical interface IP address.

10.3 Reference Configuration

10.3.1 R1 Configuration

```
sysname R1

#
acl number 3000
    rule 5 deny ip source 10.1.1.1 0
    rule 10 permit ip
#
traffic classifier test type or
    if-match acl 3000
#
traffic behavior test
#
traffic policy test
    classifier test behavior test precedence 5
#
interface GE0/0/1
    ip address 10.0.12.2 255.255.255.0
#
interface GE0/0/2
```

```
ip address 10.0.23.2 255.255.255.0

traffic-policy test outbound

#

ospf 1

area 0.0.0.0

network 10.0.12.2 0.0.0.0

network 10.0.23.2 0.0.0.0

#
```

10.3.2 S1 Configuration

```
sysname S1

#

interface GE1/0/1

undo portswitch

ip address 10.0.12.1 255.255.255.0

#

interface LoopBack0

ip address 10.0.1.1 255.255.255.255

#

interface LoopBack1

ip address 10.1.1.1 255.255.255.255

#

ospf 1

area 0.0.0.0

network 10.0.1.1 0.0.0.0

network 10.0.12.1 0.0.0.0

network 10.1.1.1 0.0.0.0
```

#

10.3.3 S2 Configuration

```
sysname S2

#

interface GE1/0/1

  undo portswitch

  ip address 10.0.23.3 255.255.255.0

#

ospf 1

  area 0.0.0.0

    network 10.0.23.3 0.0.0.0

#
```

10.4 Quiz

1. In addition to applying a traffic policy to GE0/0/2 of R1, what other methods can be used to filter traffic?

Answer:

Apply a traffic policy to the inbound direction of GE0/0/1 of R1.

11

File Management Configuration

11.1 Introduction

11.1.1 About This Lab

The device supports multiple file management modes. You can select a proper file management mode based on the task and security requirements.

Files can be managed through File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP), and Secure File Transfer Protocol (SFTP).

During file management, a device may work as either a server or a client:

1. If the device works as a server, you can access the device from a client to manage files on the device and transfer files between the client and device.
2. If the device works as a client, you can access another device (the server) from the device to manage and transfer files.

11.1.2 Objectives

- Understand the process of establishing an FTP connection.
- Learn how to configure FTP server parameters.
- Learn how to transfer files to an FTP server.

11.1.3 Networking Topology

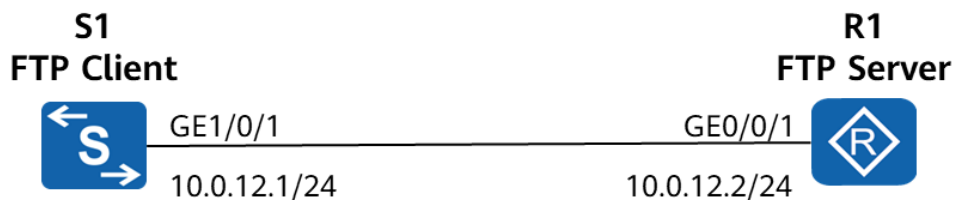


Figure 11-1 Lab topology for file management configuration

The router used in this lab is ENSP-AR, the switch is ENSP-LSW.

S1 functions as an FTP client, and R1 functions as an FTP server.

11.1.4 Background

S1 needs to manage the configuration file of R1.

11.2 Task Configuration

11.2.1 Configuration Roadmap

1. Configure the FTP server function and parameters.
2. Configure local FTP users.
3. Log in to the FTP server from the FTP client.
4. Perform file operations through the FTP client.

11.2.2 Configuration Procedure

Step 1 Perform basic device configurations.

Configure IP addresses for devices.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 10.0.12.2 24
[R1-GE0/0/1] quit
```

```
[S1] vlan 2
[S1-vlan2] quit
[S1] interface GE 1/0/1
[S1-GE1/0/1] port link-type access
[S1-GE1/0/1] port default vlan 2
[S1-GE1/0/1] quit
[S1] interface Vlanif 2
[S1-Vlanif2] ip address 10.0.12.1 24
[S1-Vlanif2] quit
```


Save the configuration file for subsequent verification.

```
<S1> save test1.cfg
```

Warning: Are you sure to save the configuration to flash:/test1.cfg? [Y/N] : **y**

Now saving the current configuration to the slot 1

Info: Save the configuration successfully.

```
<R1> save test2.cfg
```

Warning: Are you sure to save the configuration to flash:/test2.cfg? [Y/N] : **y**

Now saving the current configuration to the slot 0

Info: Save the configuration successfully.

Display the current file list.

```
<S1> dir
```

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time	FileName
0	drwx	-	XX XX XXXX	07:27:19	\$_autobackup
1	dr-x	-	XX XX XXXX	01:00:00	\$_install_mod
2	dr-x	-	XX XX XXXX	07:26:52	\$_license
3	dr-x	-	XX XX XXXX	07:27:17	\$_startup
4	dr-x	-	XX XX XXXX	07:27:38	\$_system
5	dr-x	-	XX XX XXXX	07:27:38	\$_user
6	-rw-	4,995	XX XX XXXX	07:57:05	1.cfg
7	-rw-	3,659	XX XX XXXX	08:00:40	device.sys
8	drwx	-	XX XX XXXX	07:27:51	dhcp
9	lrwx	17	XX XX XXXX	07:26:47	ensp-lsw.cc -> system file
10	-rw-	2,477	XX XX XXXX	07:28:57	env_cfg
11	drwx	-	XX XX XXXX	07:27:20	eva

```

12 drwx      - XX XX XXXX 07:57:37 full_kpi
13 drwx      - XX XX XXXX 07:27:05 hips
14 drwx      - XX XX XXXX 07:27:06 hrp
15 drwx      - XX XX XXXX 07:27:06 idm
16 dr-x      - XX XX XXXX 07:42:37 logfile
17 drwx      - XX XX XXXX 07:27:37 lpustat
18 -rw-      506 XX XX XXXX 08:00:57 mem_env_cfg
19 -rw-      1,488 XX XX XXXX 07:27:11 nvram
20 drwx      - XX XX XXXX 07:27:08 pki
21 drwx      - XX XX XXXX 07:27:08 security
22 -rw-      175 XX XX XXXX 08:00:22 stats_tlv.dat
23 -rw-      4,993 XX XX XXXX 08:00:40 test1.cfg
24 drwx      - XX XX XXXX 07:27:45 webm
25 drwx      - XX XX XXXX 07:27:38 ztp

```

152,579,072 KB total (143,497,908 KB free)

<R1> dir

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time	FileName
0	drwx		- XX XX XXXX	07:27:05	\$_autobackup
1	dr-x		- XX XX XXXX	01:00:00	\$_install_mod
2	dr-x		- XX XX XXXX	07:26:32	\$_license
3	dr-x		- XX XX XXXX	07:27:03	\$_startup
4	dr-x		- XX XX XXXX	07:27:28	\$_system
5	dr-x		- XX XX XXXX	07:27:28	\$_user
6	-rw-	6,409	XX XX XXXX	07:57:25	1.cfg

7	-rw-	2,311,364	XX XX XXXX 07:27:46	application.txt
8	-rw-	2,311,364	XX XX XXXX 07:27:46	application_touch.txt
9	drwx	-	XX XX XXXX 07:27:35	default-sdb
10	-rw-	3,219	XX XX XXXX 08:00:52	device.sys
11	drwx	-	XX XX XXXX 07:27:51	dhcp
12	lrwx	16	XX XX XXXX 07:26:23	ensp-ar.cc -> system file
13	-rw-	2,570	XX XX XXXX 07:29:35	env_cfg
14	drwx	-	XX XX XXXX 07:27:28	full_kpi
15	drwx	-	XX XX XXXX 07:26:46	hips
16	drwx	-	XX XX XXXX 07:26:47	hrp
17	drwx	-	XX XX XXXX 07:27:35	idm
18	drwx	-	XX XX XXXX 07:29:51	idname
19	drwx	-	XX XX XXXX 07:26:47	loc
20	dr-x	-	XX XX XXXX 07:57:28	logfile
21	-rw-	296	XX XX XXXX 08:02:38	mem_env_cfg
22	drwx	-	XX XX XXXX 07:26:50	nlog
23	drwx	-	XX XX XXXX 07:26:48	pki
24	drwx	-	XX XX XXXX 07:26:48	security
25	-rw-	175	XX XX XXXX 07:59:44	stats_tlv.dat
26	-rw-	14,155	XX XX XXXX 07:27:46	subcategory.txt
27	-rw-	6,409	XX XX XXXX 08:00:52	test2.cfg
28	drwx	-	XX XX XXXX 07:27:35	update
29	drwx	-	XX XX XXXX 07:27:36	webm
30	drwx	-	XX XX XXXX 07:27:28	ztp

152,579,072 KB total (143,502,460 KB free)

The configuration files of the two devices have been saved successfully.

Note: The number of files varies according to the device. You do not need to pay attention to other files.

Step 2 Configure the FTP server function and parameters on R1.

```
<R1> install feature-software WEAKEA
```

Info: This operation will install weak algorithm software, and weak algorithms will be allowed to be used.

Info: Operating, please wait for a moment.....done.

Info: Succeeded in installing the software.

```
<R1> system-view
```

Enter system view, return user view with return command.

```
[R1] ftp server enable
```

Warning: FTP is not a secure protocol, and it is recommended to use SFTP.

```
[R1] ftp server source all-interface
```

Warning: FTP server source configuration will take effect in the next login. Continue? [Y/N]: **y**

Warning: It expands the range of accessed IP.

The **ftp server enable** command enables the FTP server function on the device and allows FTP users to log in to the device. By default, the FTP function is disabled.

The **ftp server source all-interface** command specifies the source interface of the FTP server. Other optional parameters include the FTP server port number and FTP connection idle time.

Step 3 Configure a local FTP user.

```
[R1] aaa
```

```
[R1-aaa] local-user ftp-client password irreversible-cipher Huawei@123
```

Info: The initial password of ftp-client must be changed during login.

```
[R1-aaa] local-user ftp-client service-type ftp
```

Warning: If the service type of local users is configured multiple times, the new configuration overwrites the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: **y**

Warning: The user access modes include Telnet, FTP, or HTTP, so security risks exist.

Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

```
[R1-aaa] local-user ftp-client privilege level 3
```

Warning: This operation may affect online users and will change the user privilege level, Continue?

```
[Y/N]: y
```

Set the privilege level of the FTP user.

```
[R1-aaa] local-user ftp-client ftp-directory flash:/
```

Info: The configured FTP path must be an absolute path, for example, flash:/ftp.

Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

```
[R1-aaa] quit
```

Configure the authorized directory of the FTP user to ensure that the FTP user can log in to the system.

Step 4 Log in to the FTP server from the FTP client.

Log in to the FTP server from the FTP client.

```
<S1> install feature-software WEAKEA
```

Info: This operation will install weak algorithm software, and weak algorithms will be allowed to be used.

Info: Operating, please wait for a moment.....done.

Info: Succeeded in installing the software.

```
<S1> ftp 10.0.12.2
```

Warning: FTP is not secure. Using SFTP is recommended.

Trying 10.0.12.2 ...

Press CTRL + K to abort

Connected to 10.0.12.2.

220 FTP service ready.

User(10.0.12.2:(none)): **ftp-client**

331 Password required for ftp-client.

Enter password: **Huawei@123**

```
230 User logged in.
```

You have logged in to the file system of R1.

Step 5 Perform operations on the file system of R1.

Set the transfer mode.

```
[ftp] ascii
```

```
200 Type set to A.
```

The device supports file transfer in ASCII and binary modes.

ASCII mode is used to transfer text files, while binary mode is used to transfer application files, such as system software, images, video files, compressed files, and database files. The configuration file to be downloaded is a text file. Therefore, you need to set the file transfer mode to ASCII. By default, the file transfer mode is ASCII.

Download a configuration file.

```
[ftp] get test2.cfg
```

```
Warning: The file may not transfer correctly in ASCII mode.
```

```
213 6409
```

```
200 Port command okay.
```

```
150 Opening ASCII mode data connection for /test2.cfg.
```

```
/ 100% [*****]
```

```
226 Transfer complete.
```

```
FTP: 6409 byte(s) received in 0.060 second(s) 104.300Kbyte(s)/sec.
```

Delete a configuration file.

```
[ftp] delete test2.cfg
```

```
Warning: File test2.cfg will be deleted. Continue? [Y/N]: y
```

```
250 DELE command successful.
```

Upload a configuration file.

```
[ftp] put test1.cfg
Warning: The file may not transfer correctly in ASCII mode.
200 Port command okay.
150 Opening ASCII mode data connection for /test1.cfg.

/      100% [*****]

226 Transfer complete.

FTP: 4993 byte(s) send in 0.020 second(s) 243.750Kbyte(s)/sec.
```

Close the FTP connection.

```
[ftp] bye
221 Server closing.
<S1>
```

11.3 Verification

Check the file directories of S1 and R1.

```
<S1> dir | include test
```

Idx	Attr	Size(Byte)	Date	Time	FileName
23	-rw-	4,993	XX XX XXXX	08:00:40	test1.cfg
24	-rw-	6,409	XX XX XXXX	08:12:04	test2.cfg

The preceding command output shows that S1 has downloaded the configuration file **test2.cfg** of R1. The **include test** command indicates that the output information must contain the keyword **test**.

```
<R1> dir | include test
```

Idx	Attr	Size(Byte)	Date	Time	FileName
27	-rw-	4,993	XX XX XXXX	08:13:13	test1.cfg

The preceding command output shows that the configuration file **test2.cfg** of R1 does not exist and the configuration file **test1.cfg** of S1 has been obtained.

11.4 Reference Configuration

11.4.1 S1 Configuration

```
sysname S1

#
vlan batch 2

#
interface Vlanif2

 ip address 10.0.12.1 255.255.255.0

#
interface GE1/0/1

 port link-type access

 port default vlan 2

#
```

11.4.2 R1 Configuration

```
sysname R1

#
```



```
ftp server enable

ftp server source all-interface

#

aaa

local-user ftp-client password irreversible-cipher Huawei@123

local-user ftp-client privilege level 3

local-user ftp-client ftp-directory flash:/ read execute write

local-user ftp-client service-type ftp

#

interface GE0/0/1

ip address 10.0.12.2 255.255.255.0

#
```

11.5 Quiz

1. Does FTP work in active or passive mode by default?

Answer:

FTP works in active mode by default.

12 Remote Login Configuration

12.1 Introduction

12.1.1 About This Lab

Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.

AAA provides the following functions:

- Authentication: verifies whether users are allowed to access the network.
- Authorization: authorizes users to access specific services.
- Accounting: records the network resources used by users for charging purposes.

Users can use one or more security services provided by AAA. For example, if a company wants to authenticate employees that access certain network resources, the network administrator only needs to configure an authentication server. If the company also wants to record the operations performed by employees on the network, an accounting server is needed.

In summary, AAA authorizes users to access specific resources and records user operations. AAA is widely used because it features good scalability and facilitates centralized user information management. AAA can be implemented using multiple protocols. RADIUS is most commonly used in actual scenarios.

In this lab, you will configure local AAA to manage and control resources for remote Telnet users.

12.1.2 Objectives

- Learn how to configure local AAA.
- Learn how to create a domain.
- Learn how to create a local user.
- Understand domain-based user management.

12.1.3 Networking Topology

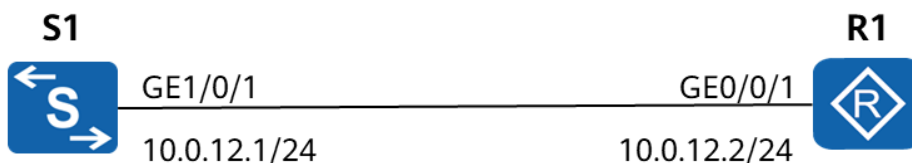


Figure 12-1 Lab topology for remote login configuration

The router used in this lab is ENSP-AR, and the switch is ENSP-LSW.

12.1.4 Background

S1 simulates a client. Resources accessible to users who manage R1 need to be controlled on R1 so that only authenticated users are allowed to access specific resources. As such, you need to configure local AAA authentication on S1 and R1 and manage users based on domains.

12.2 Task Configuration

12.2.1 Configuration Roadmap

1. Configure AAA schemes.
2. Create a domain and apply the AAA schemes to the domain.
3. Configure local users.

12.2.2 Configuration Procedure

Step 1 Perform basic device configurations.

Configure IP addresses for interconnection between R1 and S1.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 10.0.12.2 24
[R1-GE0/0/1] quit
```

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] undo portswitch
```

```
[S1-GE1/0/1] ip address 10.0.12.1 24
[S1-GE1/0/1] quit
```

Step 2 Configure AAA schemes.

Configure authentication and authorization schemes.

```
[R1] aaa
```

Enter the AAA view.

```
[R1-aaa] authentication-scheme datacom
```

Create an authentication scheme named **datacom**.

```
[R1-aaa-authen-datacom] authentication-mode local
```

Set the authentication mode to local authentication.

```
[R1-aaa-authen-datacom] quit
[R1-aaa] authorization-scheme datacom
```

Create an authorization scheme named **datacom**.

```
[R1-aaa-author-datacom] authorization-mode local
```

Set the authorization mode to local authorization.

```
[R1-aaa-author-datacom] quit
[R1-aaa] quit
```

A device functioning as an AAA server is called a local AAA server, which performs user authentication and authorization, but not user accounting.

Similar to the remote AAA server, the local AAA server needs to have the user names, passwords, and permission information of local users configured. A local AAA server performs authentication and authorization faster than a remote AAA server, which reduces operation costs. However, the information storage capacity of a local AAA server is limited by the available space on the device hardware.

Step 3 Create a domain and apply the AAA schemes to the domain.

```
[R1] aaa
[R1-aaa] domain datacom
```

The device manages users based on domains. A domain is a group of users and each user belongs to a domain. A user uses the AAA configuration in the domain to which the user belongs. Create a domain named **datacom**.

```
[R1-aaa-domain-datacom] authentication-scheme datacom
```

Specify the authentication scheme **datacom** for users in the domain.

```
[R1-aaa-domain-datacom] authorization-scheme datacom
[R1-aaa-domain-datacom] quit
```

Specify the authorization scheme **datacom** for users in the domain.

Step 4 Configure a local user.

Create a local user and password.

```
[R1-aaa] local-user hcia@datacom password irreversible-cipher Huawei@123
```

Info: The initial password of hcia@datacom must be changed during login.

If the user name contains a delimiter of an at sign (@), the character string before the at sign is the user name and that following the at sign is the domain name. If the user name does not contain @, the entire character string is the user name and the domain name is the default one.

Configure parameters for the local user, such as access type and privilege level.

```
[R1-aaa] local-user hcia@datacom service-type telnet
```

Warning: If the service type of local users is configured multiple times, the new configuration overwrites the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: **y**

Warning: The user access modes include Telnet, FTP, or HTTP, so security risks exist.

Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

The **local-user service-type** command configures the access type for a local user. After you specify the access type of a user, the user can successfully log in only when the configured access type is used. If the access type is set to **telnet**, the user cannot access the device through a web page. Multiple access types can be configured for a user.

Table 12-1 Description of the service-type parameter

Parameter	Description
none	Indicates that no access type is specified.
http	Indicates an HTTP user.
ftp	Indicates an FTP user.
ssh	Indicates an SSH user.
telnet	Indicates a Telnet user, which is typically a network administrator.
terminal	Indicates a terminal user, which is typically a user connected using a console port.

```
[R1-aaa] local-user hcia@datacom privilege level 3
```

Warning: This operation may affect online users and will change the user privilege level, Continue?

```
[Y/N] : y
```

```
[R1-aaa] return
```

Specify a privilege level for the local user. After logging in to the device, a user can run only the commands of the same privilege level or lower privilege levels.

Step 5 Enable the Telnet function on R1.

Install the weak security algorithm/protocol feature package WEAKEA on R1.

```
<R1> install feature-software WEAKEA
```

Info: Checking, please wait for a moment...done.

Info: Start to install the package WEAKEA.

Info: Operating, please wait for a moment..This operation will install weak algorithm software, and weak algorithms will be allowed to be used.

.....done.

Info: Succeeded in installing the software.

Due to variations in security performance between protocols or algorithms, some protocols or algorithms may pose security risks. For security purposes, you are advised not to use

insecure protocols or algorithms. If you need to use them, install the weak security algorithm or protocol feature package WEAKEA. By default, the device provides the weak security algorithm/protocol feature package WEAKEA.

```
<R1> system-view
```

```
[R1] telnet server enable
```

Enable the Telnet server function on the device. This function is enabled by default on some devices.

```
[R1] telnet server-source all-interface
```

Warning: Telnet server source configuration will take effect in the next login. Continue? [Y/N] : **y**

Warning: It expands the range of accessed IP.

Configure any interface having an IP address configured as the source interface of a Telnet server.

```
[R1] user-interface vty 0 4
```

The **user-interface vty** command displays the VTY user interface view. In this view, you can set Telnet and SSH parameters. The value of **vtty** ranges from 0 to 20.

By default, when the device starts with factory settings, the number of VTY interfaces is 5, the authentication mode is AAA, and the protocol is SSH.

```
[R1-ui-vty0-4] authentication-mode aaa
```

The **authentication-mode** command sets an authentication mode for accessing the user interface. By default, the authentication mode for accessing a user interface is not configured. An authentication mode must be configured for accessing the user interface. Otherwise, users cannot log in to the device.

```
[R1-ui-vty0-4] protocol inbound telnet
```

Warning: TELNET is not a secure protocol. SSH protocol is recommended.

```
[R1-ui-vty0-4] quit
```

The **protocol inbound** command specifies the protocols that the current user interface supports.

Step 6 Verify the configuration.

Install the weak security algorithm/protocol feature package WEAKEA on S1.

```
<S1> install feature-software WEAKEA
```

```
Info: Checking, please wait for a moment...done.
```

```
Info: Start to install the package WEAKEA.
```

```
Info: Operating, please wait for a moment..This operation will install weak algorithm software, and weak algorithms will be allowed to be used.
```

```
.....done.
```

```
Info: Succeeded in installing the software.
```

Remotely access R1 from S1 through Telnet.

```
<S1> telnet 10.0.12.2
```

```
Trying 10.0.12.2 ...
```

```
Press CTRL+K to abort
```

```
Connected to 10.0.12.2 ...
```

```
Warning: Telnet is not a secure protocol, and it is recommended to use Stelnet.
```

```
Username: hcia@datacom
```

```
Password: Huawei@123
```

```
Warning: The initial password poses security risks.The password needs to be changed, Continue? [Y/N] :y
```

```
Please enter old password: Huawei@123
```

```
Please enter new password: Huawei@1234
```

```
Please confirm new password: Huawei@1234
```

```
The password has been changed successfully.
```

```
Info: The max number of VTY users is 21, the number of current VTY users online is 1, and total number of terminal users online is 2.
```

```
The current login time is XXXX-XX-XX 00:04:42.
```

```
<R1>
```


By default, you must change the password upon the first login. After the password is changed, you can log in to R1.

Display online users on R1.

```
<R1> display users
```

NOTE:

User-Intf: The absolute number and the relative number of user interface

Authen: Whether the authentication passes

Author: Command line authorization flag

User-Intf		Delay	Type	Network Address	Authen	Author	Username

0	CON 0	00:01:50	--	0	pass	no	Unspecified
* 34	VTY 0	00:00:00	TEL	10.0.12.1	pass	no	hcia@datacom

The preceding command output shows that S1 has logged in to R1 through Telnet, and the VTY user interface number is 0.

Step 7 Enable STelnet on R1.

Telnet is insecure, while STelnet is secure and recommended on the live network.

Create an SSH user.

```
[R1] ssh user sshclient@datacom
```

```
[R1] ssh user sshclient@datacom authentication-type password
```

```
[R1] ssh user sshclient@datacom service-type stelnet
```

When the password, password-rsa, password-x509v3-rsa, password-dsa, password-ecc, password-sm2, password-x509v3-ecdsa-sha2, or password-sm2-sm3 mode is used to log in to the device, you need to create a local user with the same user name as the SSH user in the AAA view.

Create a local AAA user.

```
[R1] aaa
```

```
[R1-aaa] local-user sshclient@datacom password irreversible-cipher Huawei@123

[R1-aaa] local-user sshclient@datacom privilege level 3

Warning: This operation may affect online users and will change the user privilege level, Continue?
[Y/N] : y

[R1-aaa] local-user sshclient@datacom service-type ssh

Warning: If the service type of local users is configured multiple times, the new configuration overwrites
the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N] : y

Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory)
of a local user, the rights of users already online do not change. The change takes effect to users who
go online after the change.

[R1-aaa] quit
```

Configure the SSH server.

```
[R1] ssh server-source all-interface

Warning: SSH server source configuration will take effect in the next login. Continue? [Y/N] : y

Warning: It expands the range of accessed IP.

[R1] stelnet server enable

Info: Succeeded in starting the STelnet server.

Warning: After configuring the source interface or source address, the IPV4/IPV6 listening socket will be
created.
```

Configure the VTY user interface to support SSH.

```
[R1] user-interface vty 5 10

[R1-ui-vty5-10] authentication-mode aaa

[R1-ui-vty5-10] protocol inbound ssh

[R1-ui-vty5-10] quit
```

Step 8 Remotely access R1 from S1 through STelnet.

Enable first login for the SSH client.

```
[S1] ssh client first-time enable
```

Warning: After this command is run, the SSH client unconditionally trusts the connected SSH server.

The **ssh client first-time enable** command enables first login for the SSH client.

When an STelnet/SFTP client attempts to log in to an SSH server for the first time, it does not check the validity of the SSH server public key because it has not saved the public key.

Log in to R1 from S1 through STelnet.

```
[S1] stelnet 10.0.12.2
Trying 10.0.12.2 ...
Press CTRL + K to abort
Connected to 10.0.12.2 ...
The server's key fingerprint is ssh-rsa 3072 CFokN9ZDH/07wxl9QXw8A5Seh8BfGEKmq9WBelWtEjg.
The server is not authenticated. Continue to access it? [Y/N] : y
Save the server's public key? [Y/N] : y
The server's public key will be saved with the name 10.0.12.2. Please wait...
Please input the username: sshclient@datacom
Enter password: Huawei@123
Warning: The initial password poses security risks.
The password needs to be changed, Continue? [Y/N] : y
Please enter old password: Huawei@123
Please enter new password: Huawei@1234
Please confirm new password: Huawei@1234
The password has been changed successfully.

Info: The max number of VTY users is 21, the number of current VTY users online is 1, and total number
of terminal users online is 2.

The current login time is XXXX-XX-XX 11:19:58.
<R1>
```

During the login, S1 has the public key of R1 automatically allocated and saves the public key.

Display the users who have logged in to R1.

```
<R1> display users
```

NOTE:

User-Intf: The absolute number and the relative number of user interface

Authen: Whether the authentication passes

Author: Command line authorization flag

User-Intf		Delay	Type	Network Address	Authen	Author	Username

0	CON 0	00:01:12	--	0	pass	no	Unspecified
* 39	VTY 5	00:00:00	SSH	10.0.12.1	pass	no	sshclient@datacom

The command output shows that the login type of the user **sshclient@datacom** is SSH and the VTY user interface number is 5.

Step 9 Perform local AAA user management.

Display the AAA summary.

```
[R1] display aaa configuration
```

```

Domain Name Delimiter      : @
Domainname parse direction : Left to right
Domainname location        : After-delimiter
Administrator user default domain: default_admin
Normal user default domain : default
Domain                     : total: 256    used: 3
Authentication-scheme      : total: 32     used: 2
Accounting-scheme          : total: 32     used: 1
Authorization-scheme        : total: 32     used: 2
Service-scheme             : total: 256    used: 0

```

Recording-scheme	: total: 32	used: 0
Local-user	: total: 512	used: 2
Local-user block retry-interval	: 5 Min(s)	
Local-user block retry-time	: 3	
Local-user block time	: 5 Min(s)	
Remote-admin-user block retry-interval	: 5 Min(s)	
Remote-admin-user block retry-time	: 30	
Remote-admin-user block time	: 5 Min(s)	
Session timeout invalid enable	: No	

The **display aaa configuration** command displays the AAA summary, such as the domain, authentication scheme, authorization scheme, and accounting scheme.

The preceding command output shows the following information:

1. The domain name delimiter is at sign (@).
2. A maximum of 512 local users can be created, and two local users have been created.
3. The authentication retry interval of a local account is 5 minutes.
4. The maximum number of consecutive authentication failures of a local account is 3.
5. The lockout duration of a local account is 5 minutes.

Display all the IP addresses that fail authentication.

```
[R1] display vty ip-block all
[R1] display ssh server ip-block all
```

The **display vty ip-block all** command displays all the IP addresses that fail authentication.

The **display ssh server ip-block all** command displays the IP addresses of all the clients that fail SSH authentication.

Currently, no IP address is locked due to authentication failures.

If an IP address is locked, run the following commands to unlock it:

1. Run the **activate ssh server ip-block ip-address** command to unlock the IP address of a user that fails the authentication for SSH connections.
2. Run the **activate vty ip-block ip-address** command to unlock the IP address of a user that fails the authentication for Telnet connections.

If you do not want to enable the IP address locking function, run the following commands:

1. Run the **telnet server ip-block disable** command to disable the function of locking IP addresses in Telnet access scenarios.
2. Run the **ssh server ip-block disable** command to disable the client IP address locking function on an SSH server.

Display local user information.

```
[R1] display local-user username hcia@datacom
```

The contents of local user(s):

Password : *****

State : active

Service-type-mask : S

Privilege level : 3

Ftp-directory : -

Access-limit : Yes

Access-limit-max : 4294967295

Accessed-num : 0

Idle-timeout : -

User-group : -

Password-force-change: Yes

Original-password : No

Password-set-time : XXXX-XX-XX 10:32:34

Password-expired : No

Password-expire-time : XXXX-XX-XX 10:32:34

Account-expire-time : -

Last login ip : 10.0.12.1

Last login time : XXXX-XX-XX 10:32:19

Login fail count : 0

The **display local-user username** *user-name* command displays the attributes of a local administrator.

The **display local-user state { active | block }** command displays information about local administrators in the active or blocked state. Users in the active state can process authentication requests, and users in the blocked users reject authentication requests.

If a user is locked, you can run the **local-user *user-name* state active** command to set the local user to the active state.

12.3 Reference Configuration

12.3.1 R1 Configuration

```
sysname R1

#
telnet server enable
telnet server-source all-interface
#
aaa
 authentication-scheme datacom
   authentication-mode local
 authorization-scheme datacom
   authorization-mode local
 domain datacom
   authentication-scheme datacom
   accounting-scheme default
   authorization-scheme datacom
 local-user hcia@datacom password irreversible-cipher Huawei@123
 local-user hcia@datacom privilege level 3
 local-user hcia@datacom service-type telnet
 local-user sshclient@datacom password irreversible-cipher Huawei@123
 local-user sshclient@datacom privilege level 3
 local-user sshclient@datacom service-type ssh
```

```
#  
interface GE0/0/1  
    ip address 10.0.12.2 255.255.255.0  
#  
stelnet server enable  
ssh user sshclient@datacom  
ssh user sshclient@datacom authentication-type password  
ssh user sshclient@datacom service-type stelnet  
ssh server-source all-interface  
#  
user-interface vty 0 4  
    authentication-mode aaa  
    protocol inbound telnet  
#  
user-interface vty 5 20  
    authentication-mode aaa  
    protocol inbound ssh  
#
```

12.3.2 S1 Configuration

```
sysname S1  
#  
rsa peer-public-key 10.0.12.2  
    public-key-code begin  
    3082018A  
    02820181  
    00C7CD46 93D06C46 836FC539 DF9B3B40 63C4AE66 AD4AB5FF 5E333806 BB7AC5FE
```



```
D60936B1 4C404CBB 8B9402D2 82075518 0CF755AF B64F31A0 77427356 4B7C028E
F27DE79A BAC9ADFC B5A7EF1D EF0397D5 4C9E7A7F 6D81F95F E31B6D55 53091B99
4472F713 F10F21A5 F984AE33 B3FA315C 173A33FB 0927BFBF 16C73F77 7A0040E8
1CDF54DA 4CAB9010 C4BBF15F 1A64961D F95EE0E3 E8383EFF F5A257F6 888024B8
3D2289E2 138E1DA4 790F1C73 F46754FB 55F0D0A0 131315BB 101D1C42 7926CC6C
7CE3FC43 7B31581B B49A8392 DE303467 22EF0D84 E1EB1FDE 0869D679 DB188EFF
53E5F166 46BBFCA1 D466CA23 1E853A7A 7A3797A0 24DD941C 52556147 329D3CE6
3E8C6B33 02DD9EE5 4A077D7E 5D163BEE C4E4983B BCCECB44 81BD2BF1 D0AEB438
C5C2524B 95371BDD AAB9ACF7 FA00F0A5 65B9FBAA 918168F2 B4B867F8 3F2B22AA
550B5C50 08D71D83 B9D5B59D BC1DD579 D07910E5 6F8A1344 896797BB 3E9B3B57
EB034801 D8DD5AC8 7F7D255A 27D1E357 D221644B ADCA0CEA C79CCB91 ADD3AC4A
91
0203
010001
public-key-code end
peer-public-key end
#
interface GE1/0/1
undo portswitch
ip address 10.0.12.1 255.255.255.0
#
ssh client first-time enable
ssh client peer 10.0.12.2 assign rsa-key 10.0.12.2
#
```

12.4 Quiz

1. Why is STelnet more secure than Telnet?

Answer:

STelnet is implemented based on SSH, supports encrypted transmission, and provides key and password authentication. Telnet sends accounts, passwords, and instructions in plaintext, which is prone to eavesdropping and man-in-the-middle attacks. Therefore, STelnet is more secure than Telnet.

13 NAT Configuration

13.1 Introduction

13.1.1 About This Lab

Network address translation (NAT) translates IP addresses in IP datagram headers as packets are forwarded between private and public networks. As a transitional solution, NAT enables address reuse to mitigate IPv4 address exhaustion. In addition to addressing the IP address shortage, NAT offers the following benefits:

- Protects private networks from external attacks.
- Enables and regulates communication between private and public networks.

In this lab activity, you will understand the fundamentals of NAT by configuring it in various scenarios.

13.1.2 Objectives

- Learn how to configure Network Address Port Translation (NAPT).
- Learn how to configure NAT in Easy IP mode.
- Learn how to configure NAT Server.

13.1.3 Networking Topology

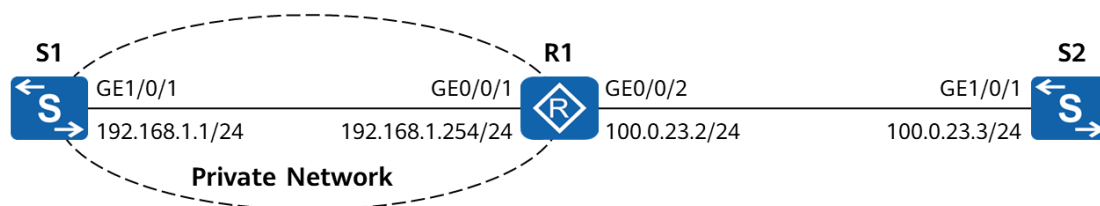


Figure 13-1 Lab topology for NAT configuration

The router used in this lab is ENSP-AR, and the switches are ENSP-LSW.

1. The network between R1 and S1 is an enterprise intranet and uses private IPv4 addresses.
2. S1 simulates an intranet client. R1 acts as both the gateway for S1, as well as the egress router that connects the intranet to the public network.

3. S2 simulates the public network.

13.1.4 Background

Due to the shortage of IPv4 addresses, enterprises typically use private IPv4 addresses on their internal networks. However, enterprise users often need to access public networks, and enterprises also need to provide services for external users. NAT is required to meet these needs.

13.2 Task Configuration

13.2.1 Configuration Roadmap

1. Configure NAPT.
2. Configure NAT in Easy IP mode.
3. Configure NAT Server.

13.2.2 Configuration Procedure

Step 1 Complete basic configurations.

Configure IP addresses for interfaces and routes.

```
[S1] interface GE 1/0/1
[S1-GE1/0/1] undo portswitch
[S1-GE1/0/1] ip address 192.168.1.1 24
[S1-GE1/0/1] quit
[S1] ip route-static 0.0.0.0 0 192.168.1.254
[S1] quit
```

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 192.168.1.254 24
[R1-GE0/0/1] quit
[R1] interface GE 0/0/2
[R1-GE0/0/2] ip address 100.0.23.2 24
```

```
[R1-GE0/0/2] quit
[R1] ip route-static 0.0.0.0 0 100.0.23.3
```

```
[S2] interface GE 1/0/1
[S2-GE1/0/1] undo portswitch
[S2-GE1/0/1] ip address 100.0.23.3 24
[S2-GE1/0/1] quit
```

Configure the Telnet function on S1 (for subsequent verification).

```
<S1> install feature-software WEAKEA
Info: Operating, please wait for a moment.....done.
Info: Succeeded in installing the software.
<S1> system-view
[S1] telnet server enable
[S1] telnet server-source all-interface
Warning: Telnet server source configuration will take effect in the next login. Continue? [Y/N] : y
[S1] aaa
[S1-aaa] local-user hcia-datcom password irreversible-cipher Huawei@123
[S1-aaa] local-user hcia-datcom service-type telnet
Warning: If the service type of local users is configured multiple times, the new configuration overwrites the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: y
Warning: The user access modes include Telnet, FTP, or HTTP, so security risks exist.
Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.
[S1-aaa] local-user hcia-datcom privilege level 3
Warning: This operation may affect online users and will change the user privilege level, Continue? [Y/N] : y
[S1-aaa] quit
[S1] user-interface vty 0 4
```

```
[S1-ui-vty0-4] authentication-mode aaa
[S1-ui-vty0-4] protocol inbound telnet
[S1-ui-vty0-4] quit
```

Step 2 Configure NAPT.

Assume that the enterprise has obtained public IP addresses ranging from 100.1.23.1 to 100.1.23.254.

Configure a NAT address pool.

```
[R1] nat address-group test 1
[R1-address-group-test] section 1 100.1.23.1 100.1.23.254
[R1-address-group-test] mode pat
[R1-address-group-test] quit
```

The **nat address-group** command configures a NAT address pool.

The **section** command configures an address segment.

The **mode pat** command sets the address pool mode to PAT. By default, the NAT mode of an address pool is PAT (that is, NAPT).

Configure a NAT policy.

```
[R1] nat-policy
[R1-policy-nat] rule name test
[R1-policy-nat-rule-test] source-address range 192.168.1.1 192.168.1.254
[R1-policy-nat-rule-test] action source-nat address-group test
[R1-policy-nat-rule-test] quit
[R1-policy-nat] quit
```

The **source-address range** command defines a matching condition in a NAT policy rule.

The **action** command defines the action in the NAT policy rule. In this example, the **source-nat** action is configured to perform source NAT on data flows matching the NAT policy rule.

Enable NAT on GE0/0/2 of R1.

```
[R1] interface GE 0/0/2
```

```
[R1-GE0/0/2] nat enable
[R1-GE0/0/2] quit
```

On S2, configure a static route to IP addresses in the NAT address pool.

```
[S2] ip route-static 100.1.23.0 24 100.0.23.2
```

Test the connectivity on S1.

```
[S1] ping -c 1000 100.0.23.3

PING 100.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 100.0.23.3: bytes=56 Sequence=1 ttl=254 time=11 ms

Reply from 100.0.23.3: bytes=56 Sequence=2 ttl=254 time=14 ms

Reply from 100.0.23.3: bytes=56 Sequence=3 ttl=254 time=8 ms

Reply from 100.0.23.3: bytes=56 Sequence=4 ttl=254 time=9 ms

Reply from 100.0.23.3: bytes=56 Sequence=5 ttl=254 time=10 ms

Reply from 100.0.23.3: bytes=56 Sequence=6 ttl=254 time=9 ms

Reply from 100.0.23.3: bytes=56 Sequence=7 ttl=254 time=3 ms
```

Configure S1 to send 1000 consecutive ping packets.

Check the NAT session table on R1.

```
[R1] display session all verbose

Session Table Information:

Protocol                : 1 (ICMP)
SrcAddr VPN             : 192.168.1.1
DestAddr VPN            : 100.0.23.3
Type Code IcmpId       : 8 0 5999
Time To Live            : 19 s
NAT Info
New SrcAddr             : 100.1.23.200
```

New DestAddr	: -
New IcmpId	: 2048
Total : 1	

The NAT session table shows that the private address of S1 is translated into 100.1.23.200.

Step 3 Configure NAT in Easy IP mode.

Modify the NAT policy. Specifically, change the action in the NAT policy rule to **easy-ip**.

```
[R1] nat-policy
[R1-policy-nat] rule name test
[R1-policy-nat-rule-test] action source-nat easy-ip
[R1-policy-nat-rule-test] quit
[R1-policy-nat] quit
```

Test the connectivity on S1.

```
[S1] ping -c 1000 100.0.23.3

PING 100.0.23.3: 56 data bytes, press CTRL_C to break

Reply from 100.0.23.3: bytes=56 Sequence=1 ttl=254 time=11 ms
Reply from 100.0.23.3: bytes=56 Sequence=2 ttl=254 time=14 ms
Reply from 100.0.23.3: bytes=56 Sequence=3 ttl=254 time=8 ms
Reply from 100.0.23.3: bytes=56 Sequence=4 ttl=254 time=9 ms
Reply from 100.0.23.3: bytes=56 Sequence=5 ttl=254 time=10 ms
Reply from 100.0.23.3: bytes=56 Sequence=6 ttl=254 time=9 ms
Reply from 100.0.23.3: bytes=56 Sequence=7 ttl=254 time=3 ms
```

Configure S1 to send 1000 consecutive ping packets.

Check the NAT session table.

```
[R1] display session all verbose
```


Session Table Information:	
Protocol	: 1 (ICMP)
SrcAddr VPN	: 192.168.1.1
DestAddr VPN	: 100.0.23.3
Type Code IcmpId	: 8 0 8882
Time To Live	: 20 s
NAT Info	
New SrcAddr	: 100.0.23.2
New DestAddr	: -
New IcmpId	: 2048
Total : 1	

The NAT session table shows that the private address of S1 is translated into 100.0.23.2, which is the IP address of GE0/0/2 on R1.

Step 4 Configure NAT Server.

Assume that S1 needs to provide the Telnet service for the public network. Because S1 does not have a public IP address, configure NAT Server on the outbound interface of R1.

Configure NAT Server on R1.

```
[R1] nat server for_telnet protocol tcp global 100.0.23.2 telnet inside 192.168.1.1 telnet
```

The **nat server** command defines a mapping table of internal servers so that external users can access the servers through address and port translation.

Log in to S1 from S2 through Telnet.

```
<S2> install feature-software WEAKEA
<S2> telnet 100.0.23.2
Trying 100.0.23.2 ...
Press CTRL+K to abort
Connected to 100.0.23.2 ...
Warning: Telnet is not a secure protocol, and it is recommended to use Stelnet.
```

Username: **hcia-datacom**

Password: **Huawei@123**

Warning: The initial password poses security risks.The password needs to be changed, Continue? [Y/N] :
y

Please enter old password: **Huawei@123**

Please enter new password: **Huawei@1234**

Please confirm new password: **Huawei@1234**

The password has been changed successfully.

Info: The max number of VTY users is 5, the number of current VTY users online is 1, and total number of terminal users online is 2.

The current login time is XXXX-XX-XX XX:XX:XX.

<S1>

Check the NAT session table on R1.

[R1] display session all verbose

Session Table Information:

Protocol : 6 (TCP)

SrcAddr Port VPN : **100.0.23.3 55108**

DestAddr Port VPN : **100.0.23.2 23**

Time To Live : 596 s

NAT Info

New SrcAddr : -

New SrcPort : -

New DestAddr : **192.168.1.1**

New DestPort : **23**

Total : 1

The NAT session table shows that the destination address 100.0.23.2 and destination port number 23, which are requested by S2, are translated into 192.168.1.1 and 23, respectively.

13.3 Reference Configuration

13.3.1 S1 Configuration

```
sysname S1

#

telnet server enable

telnet server-source all-interface

#

aaa

 local-user hcia-datcom password irreversible-cipher Huawei@123

 local-user hcia-datcom privilege level 3

 local-user hcia-datcom service-type telnet

#

interface GE1/0/1

 undo portswitch

 ip address 192.168.1.1 255.255.255.0

#

 ip route-static 0.0.0.0 0.0.0.0 192.168.1.254

#

user-interface vty 0 4

 authentication-mode aaa

 protocol inbound telnet

#
```

13.3.2 R1 Configuration

```
sysname R1

#

interface GE0/0/1

 ip address 192.168.1.254 255.255.255.0

#

interface GE0/0/2

 ip address 100.0.23.2 255.255.255.0

 nat enable

#

 ip route-static 0.0.0.0 0.0.0.0 100.0.23.3

#

 nat address-group test 1

 mode pat

 section 1 100.1.23.1 100.1.23.254

#

 nat server for_telnet protocol tcp global 100.0.23.2 telnet inside 192.168.1.1 telnet

#

 nat-policy

 rule name test

 source-address range 192.168.1.1 192.168.1.254

 action source-nat easy-ip

#
```

13.3.3 S2 Configuration

```
sysname S2

#
```

```
interface GE1/0/1

undo portswitch

ip address 100.0.23.3 255.255.255.0

#

ip route-static 100.1.23.0 255.255.255.0 100.0.23.2

#
```

13.4 Quiz

1. When configuring NAT Server, should the destination port numbers before translation be the same as those after translation?

Answer: No.

14

WAC + Fit AP Networking Configuration

14.1 Introduction

14.1.1 About This Lab

Wired LANs use wired cables or optical fibers as transmission media, which are expensive and have fixed locations. As further emphasis was placed on network mobility, wired LANs became unable to meet users' requirements. This led to the development of WLAN, which has become the most cost-effective and convenient network access mode. WLAN technology allows users to easily access a wireless network and move around within the coverage area of the wireless network.

In this lab activity, you will learn the overall process and configuration for constructing a basic WLAN in the typical WAC + Fit AP networking.

14.1.2 Objectives

- Understand how to configure a Fit AP to go online.
- Understand commands for configuring various WLAN profiles.
- Understand the basic process for WLAN configuration.

14.1.3 Networking Topology

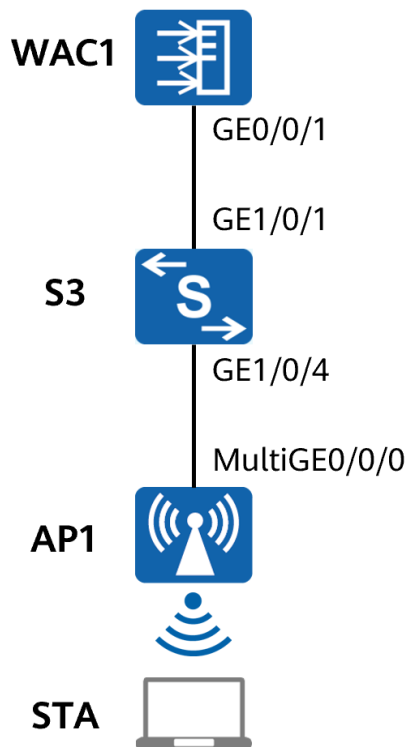


Figure 14-1 Lab topology for the WAC + Fit AP networking configuration

The switch used in this lab is ENSP-LSW, the wireless controller is ENSP-AC, and the wireless access point is ENSP-AP.

1. WAC1 and AP1 are on the same Layer 2 network.
2. WAC1 functions as a DHCP server to assign IP addresses to AP1 and STAs.
3. The direct data forwarding mode is used.

14.1.4 Background

An enterprise wants to enable users to access the network through a WLAN, meeting the basic mobile office requirements.

14.1.5 Data Planning

Table 14-1 WLAN data planning

Item	Setting
Management VLAN for AP1	VLAN 100
Service VLAN for STAs	VLAN 101

Item	Setting
DHCP server	WAC1 functions as a DHCP server to assign IP addresses to AP1.
	WAC1 functions as a DHCP server to assign IP addresses to STAs.
IP address pool for AP1	192.168.100.1–192.168.100.253/24
IP address pool for STAs	192.168.101.1–192.168.101.253/24
IP address of WAC1's source interface	VLANIF 100: 192.168.100.254/24
AP group	Name: ap-group1
	Referenced profiles: VAP profile HCIA-WLAN and regulatory domain profile default
Regulatory domain profile	Name: default
	Country code: CN
SSID profile	Name: HCIA-WLAN
	SSID name: HCIA-WLAN
Security profile	Name: HCIA-WLAN
	Security policy: open
	Password: HCIA-Datcom
VAP profile	Name: HCIA-WLAN
	Forwarding mode: direct forwarding
	Service VLAN: VLAN 101
	Referenced profiles: SSID profile HCIA-WLAN and security profile HCIA-WLAN
Radio parameters	Radio: radio 0
	Frequency band: 2.4G
	Channel: disables automatic channel selection

14.2 Task Configuration

14.2.1 Configuration Roadmap

1. Configure the connectivity of the wired network.
2. Configure AP1 to go online.
 - (a) Create an AP group to which APs requiring the same configuration are added, implementing unified configuration.
 - (b) Configure WAC1 system parameters, including the country code and WAC1 source interface for communication with AP1.
 - (c) Configure the AP authentication mode and import AP1 offline so that AP1 can go online.
3. Configure STAs to go online. Specifically, configure WLAN service parameters and deliver them to AP1 so that STAs can access the network.

14.2.2 Configuration Procedure

Step 1 Complete basic device configurations.

Name the devices.

The detailed configuration is not provided here.

Step 2 Configure the wired network.

Configure VLANs.

```
[WAC1] vlan batch 100 101

[WAC1] interface GE 0/0/1

[WAC1-GE0/0/1] port link-type trunk

[WAC1-GE0/0/1] port trunk allow-pass vlan 100

[WAC1-GE0/0/1] quit
```

```
[S3] vlan batch 100 101

[S3] interface GE 1/0/1

[S3-GE1/0/1] port link-type trunk

[S3-GE1/0/1] port trunk allow-pass vlan 100

[S3-GE1/0/1] quit
```

```
[S3] interface GE 1/0/4  
[S3-GE1/0/4] port link-type trunk  
[S3-GE1/0/4] port trunk allow-pass vlan 100 101  
[S3-GE1/0/4] port trunk pvid vlan 100  
[S3-GE1/0/4] quit
```

Configure IP addresses for interfaces.

```
[WAC1] interface Vlanif 100  
[WAC1-Vlanif100] ip address 192.168.100.254 24  
[WAC1-Vlanif100] quit
```

```
[S3] interface Vlanif 100  
[S3-Vlanif100] ip address 192.168.100.1 24  
[S3-Vlanif100] quit  
[S3] interface Vlanif 101  
[S3-Vlanif101] ip address 192.168.101.254 24  
[S3-Vlanif101] quit
```

Configure DHCP.

```
[S3] dhcp enable  
[S3] interface Vlanif 100  
[S3-Vlanif100] dhcp select interface  
[S3-Vlanif100] dhcp server excluded-ip-address 192.168.100.254  
[S3-Vlanif100] quit  
[S3] interface Vlanif 101  
[S3-Vlanif101] dhcp select interface  
[S3-Vlanif101] quit
```

Step 3 Configure AP1 to go online.

Create the AP group **ap-group1**.

```
[WAC1] wlan  
[WAC1-wlan] ap-group name ap-group1  
[WAC1-wlan-ap-group-ap-group1] quit
```

Create a regulatory domain profile and configure the country code of WAC1 in the profile.

```
[WAC1-wlan] regulatory-domain-profile name default
```

A regulatory domain profile provides configurations such as the country code, calibration channel set, and calibration bandwidth for APs.

By default, the system provides the regulatory domain profile **default**. This step enters the view of the regulatory domain profile **default**.

```
[WAC1-wlan-regulate-domain-default] country-code cn  
Info: The current country code is same with the input country code.  
[WAC1-wlan-regulate-domain-default] quit
```

A country code identifies the country or region in which APs are deployed. Different countries or regions have their own regulations on AP radio attributes, such as the transmit power and supported channels. The correct country code configuration ensures that AP radio attributes comply with local laws and regulations. By default, the country code of a device is **CN**.

Bind the regulatory domain profile to the AP group.

```
[WAC1-wlan] ap-group name ap-group1  
[WAC1-wlan-ap-group-ap-group1] regulatory-domain-profile default  
Warning: This configuration change will clear the channel and power configurations of radios, and may restart APs. Continue? [Y/N]: y  
[WAC1-wlan-ap-group-ap-group1] quit  
[WAC1-wlan] quit
```

The **regulatory-domain-profile** command in the AP group view binds a regulatory domain profile to the AP group. By default, the regulatory domain profile **default** is bound to an AP group. In the regulatory domain profile **default**, the country code is **CN**, 2.4 GHz

calibration channels include channels 1, 6, and 11, and 5 GHz calibration channels include channels 149, 153, 157, 161, and 165.

On WAC1, configure the source interface of CAPWAP tunnels.

```
[WAC1] capwap source interface Vlanif 100
```

Set the DTLS PSK(contains 8-32 plain-text characters, or 128 or 148 cipher-text characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm PSK: **Huawei@123**

Info: Deliver DTLS PSK to devices using CAPWAP connections. It may take a few minutes.

Set the user name for FIT APs(The value is a string of 4 to 31 characters, which can contain letters, underscores, and digits, and must start with a letter): **admin**

Set the password for FIT APs(plain-text password of 8-128 characters or cipher-text password of 128-268 characters that must be a combination of at least three of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm password: **Huawei@123**

Set the PSK of the global offline management VAP(plain-text password of 8-63 characters or cipher-text password of 128-188 characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm PSK: **Huawei@123**

Warning: Ensure that the management VLAN and service VLAN are different. Otherwise, services may be interrupted.

Info: Ensure that the management VLAN for devices is within the VLAN range mapping the source interface. Otherwise, the devices cannot go online.

The **capwap source interface** command configures the interface used by a WAC to establish CAPWAP tunnels with APs for communication.

The CAPWAP source interface can be successfully configured only when all of the following security-related configurations exist: PSK for DTLS encryption, user name and password for logging in to an AP, and password for logging in to the global offline management VAP. If any of these configurations does not exist, the system prompts you to supplement related configurations first.

Import AP1 offline on WAC1 and add AP1 to the AP group **ap-group1**.

APs can be added on a WAC in the following ways:

1. Manual import: The MAC addresses and SNs of APs are added on a WAC before they go online. The WAC automatically starts to set up a connection with an AP if the AP's MAC address and SN match the preconfigured ones.

2. Automatic discovery: When the AP authentication mode is set to non-authentication, or the AP authentication mode is set to MAC or SN authentication and the MAC address or SN of an AP is whitelisted, the WAC can automatically discover the AP and allow it to go online.

3. Manual confirmation: The AP authentication mode is set to MAC address or SN authentication, and the corresponding AP whitelist is configured on the WAC. When an AP out of the whitelist connects to the WAC, the WAC adds the AP to the unauthenticated AP list. An AP can go online only after its identity is manually confirmed.

```
[WAC1] wlan
[WAC1-wlan] ap auth-mode mac-auth
```

The **ap auth-mode** command configures the AP authentication mode. Only authenticated APs can go online. The authentication modes include MAC address authentication, SN authentication, and non-authentication. The default AP authentication mode is MAC address authentication.

Check the MAC address of an AP using LLDP.

```
[S3] display lldp neighbor interface GE 1/0/4

GE1/0/4 has 1 neighbor(s):

Neighbor index          :1
Chassis type             :MAC Address
Chassis ID               : fa27-e678-0030
Port ID subtype          :Interface Name
Port ID                  :MultiGE0/0/0
Port description         :--
System name              :HUAWEI
System description       :Huawei AP
Huawei YunShan OS
Version 1.24.0.1 (AirEngine 5700 V600R024C00SPC100)
Copyright (C) 20XX-20XX Huawei Technologies Co., Ltd.
HUAWEI AirEngine5773-21
System capabilities supported :wlanAccessPoint
```

```
System capabilities enabled      :wlanAccessPoint
Management address type        :IPv4
```

Note: For the MAC address and SN of an AP, check the MAC address label and SN label in the packing box.

Add an AP based on the AP's MAC address.

```
[WAC1-wlan] ap-id 1 ap-mac fa27-e678-0030
```

The **ap-id** command adds an offline AP or displays the AP view.

When adding an AP, configure the parameter **ap-mac** if MAC address authentication is used, and configure the parameter **ap-sn** if SN authentication is used.

To enter the AP view, you only need to enter the AP ID.

```
[WAC1-wlan-ap-1] ap-name AP1
```

The **ap-name** command configures the name of an AP. AP names must be unique. If the AP name is not configured, the AP's MAC address is used as the AP name by default after the AP goes online.

```
[WAC1-wlan-ap-1] ap-group ap-group1
```

Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N] : **y**

```
[WAC1-wlan-ap-1] quit
```

```
[WAC1-wlan] quit
```

The **ap-group** command adds the AP to an AP group. Then, WAC1 delivers the configurations in the AP group to the AP. For example, after AP1 is added to **ap-group1**, the regulatory domain profile, radio profile, and VAP profile configurations in **ap-group1** are delivered to AP1. By default, an AP is not added to any group. After the AP group is changed for an AP, WAC1 delivers the new configuration to the AP. Then the AP restarts and joins the new AP group.

Wait for several minutes and check AP information.

```
[WAC1] display ap all
```

Total AP information:

```
nor      : normal      [1]
```

ExtralInfo : Extra information									
Total: 1									

ID	MAC	Name	Group	IP	Type	State	STA	Uptime	ExtralInfo

1	fa27-e678-0030	AP1	ap-group1	192.168.100.23	AirEngine5773-21	nor	0	4M:36S	-

The **display ap** command displays AP information, including the IP address, model (**AirEngine5773-21**), state (**nor**), and online duration of the AP.

You can also specify **by-state state** or **by-ssid ssid** in this command to filter APs by state or SSID.

The command output shows that AP1 is working properly.

Table 14-2 AP state description

AP State	Description
commit-failed (cmtfa)	WLAN service configurations fail to be delivered to an AP after the AP goes online on a WAC.
committing (cmt)	WLAN service configurations are being delivered to an AP after the AP goes online on a WAC.
config (cfg)	WLAN service configurations are being delivered to an AP when the AP is going online on a WAC.
config-failed (cfgfa)	WLAN service configurations fail to be delivered to an AP when the AP is going online on a WAC.
download (dload)	An AP is being upgraded.
fault	An AP fails to go online.
idle	It is the initialization state of an AP before it establishes a link with the WAC for the first time.
normal (nor)	An AP is working properly.
standby (stdby)	An AP is in normal state on the standby WAC.
ver-mismatch (vmiss)	The version of an AP does not match that of a WAC on which the AP is to go online.
countryCode-mismatch (cmiss)	The country code of an AP does not match that of a WAC on which the AP is about to go online.

AP State	Description
type-mismatch (tmiss)	The AP type does not match that configured on the WAC.
unauth	The AP is not authenticated.

Step 4 Configure WLAN service parameters.

Create the security profile **HCIA-WLAN** and configure a security policy in the profile.

```
[WAC1] wlan
[WAC1-wlan] security-profile name HCIA-WLAN
[WAC1-wlan-sec-prof-HCIA-WLAN] security open
[WAC1-wlan-sec-prof-HCIA-WLAN] quit
```

The **security psk** command configures WPA/WPA2 pre-shared key (PSK) authentication and encryption.

The **security open** command configures open authentication, which means that no authentication or encryption is performed and any user can access the network without authentication.

Create the SSID profile **HCIA-WLAN** and set the SSID name to **HCIA-WLAN**.

```
[WAC1-wlan] ssid-profile name HCIA-WLAN
```

The SSID profile **HCIA-WLAN** is created.

```
[WAC1-wlan-ssid-prof-HCIA-WLAN] ssid HCIA-WLAN
[WAC1-wlan-ssid-prof-HCIA-WLAN] quit
```

The SSID name is set to **HCIA-WLAN**.

Create the VAP profile **HCIA-WLAN**, set the data forwarding mode and service VLAN, and bind the security profile and SSID profile to the VAP profile.

```
[WAC1-wlan] vap-profile name HCIA-WLAN
```

The **vap-profile** command creates a VAP profile.

You can configure the data forwarding mode in a VAP profile and bind the SSID profile, security profile, and traffic profile to the VAP profile.


```
[WAC1-wlan-vap-prof-HCIA-WLAN] forward-mode direct-forward
```

The **forward-mode** command configures the data forwarding mode in a VAP profile. By default, the data forwarding mode is direct forwarding.

```
[WAC1-wlan-vap-prof-HCIA-WLAN] service-vlan vlan-id 101
```

The **service-vlan** command configures a service VLAN for a VAP. After STAs access the WLAN, the user data forwarded by APs carry the service VLAN tag.

```
[WAC1-wlan-vap-prof-HCIA-WLAN] security-profile HCIA-WLAN
```

The security profile **HCIA-WLAN** is bound to the VAP profile.

```
[WAC1-wlan-vap-prof-HCIA-WLAN] ssid-profile HCIA-WLAN
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] quit
```

The SSID profile **HCIA-WLAN** is bound to the VAP profile.

Bind the VAP profile **HCIA-WLAN** to the AP group and apply the VAP profile configurations to radio 0 of APs.

```
[WAC1-wlan] ap-group name ap-group1
```

```
[WAC1-wlan-ap-group-ap-group1] vap-profile HCIA-WLAN wlan 1 radio 0
```

```
[WAC1-wlan-ap-group-ap-group1] radio 0
```

```
[WAC1-wlan-ap-group-ap-group1-radio-0] calibrate auto-channel-select disable
```

```
[WAC1-wlan-ap-group-ap-group1-radio-0] quit
```

```
[WAC1-wlan-ap-group-ap-group1] quit
```

```
[WAC1-wlan] quit
```

The **vap-profile** command binds a VAP profile to radios. After this command is executed, all configurations in the VAP profile, including the configurations in the profiles bound to the VAP profile, are delivered to AP radios.

```
[WAC1] display vap all
```

WID : WLAN ID

Total: 1									
AP ID	AP name	Rfid	WID	BSSID	Status	Auth type	STA	SSID	Radio Type
1	AP1	0	1	FA27-E678-0030	ON	Open	0	HCIA-WLAN	11be

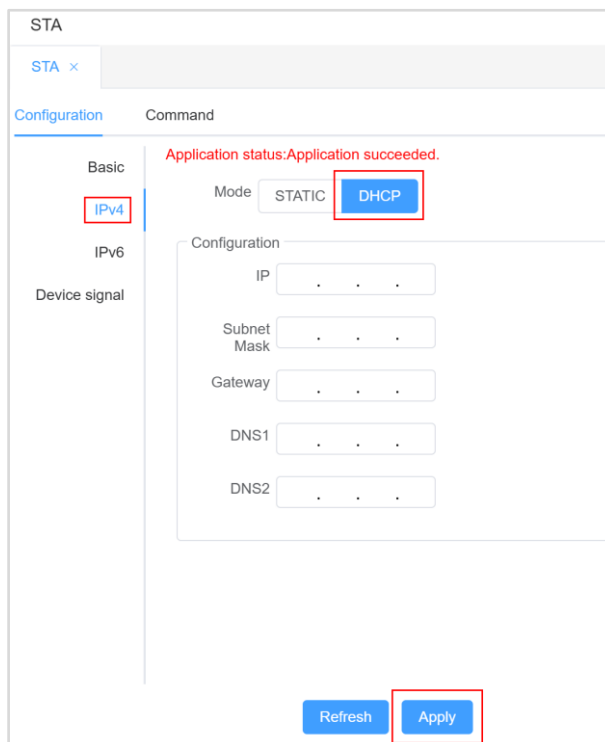
The **display vap** command displays information about service VAPs. The command output shows that the SSID name is **HCIA-WLAN**, the authentication mode is **open**, and the number of access STAs is **0**.

14.3 Verification

Use a STA to perform the following verification:

1. Use a STA to connect to the WLAN with the SSID of **HCIA-WLAN**. Check the IP address obtained by the STA and ping 192.168.101.254 from the STA.

Double-click the activated STA. In the dialog box that is displayed, select **IPv4**, select **DHCP**, and click **Apply**.



STA

STA x

Configuration Command

Basic

IPv4

IPv6

Device signal

Application status: Application succeeded.

Mode

STATIC

DHCP

Configuration

IP

Subnet Mask

Gateway

DNS1

DNS2

Refresh

Apply

Drag the STA to the wireless signal coverage area of AP1. On the CLI of the STA, run the following commands to search for and associate with the SSID:

```
PC> wnmcli device wifi list --rescan yes
```

SSID	Status
HCIA-WLAN	down

```
PC> wnmcli device wifi connect HCIA-WLAN
```

```
Connect is done
```

wnmcli device wifi list --rescan yes is used to list available wireless signals.

wnmcli device wifi connect ~~XX~~ is used to connect to a specific wireless signal.

Check the IP address obtained by the STA.

```
PC> ifconfig
```

Local IPv6 address.....:	fe80::6b8:99ff:fe6c:213f
IPv6 address.....:	::
IPv6 gateway.....:	::
IPv4 address.....:	192.168.101.118
Subnet mask.....:	255.255.255.0
Gateway.....:	192.168.101.254
Physical address.....:	04:b8:99:6c:21:3f
DNS server.....:	

Ping the gateway on the CLI of the STA.

```
PC> ping 192.168.101.254
```

```
PING 192.168.101.254 (192.168.101.254) 56(84) bytes of data.
```

```
64 bytes from 192.168.101.254: icmp_seq=1 ttl=255 time=108 ms
```

```
64 bytes from 192.168.101.254: icmp_seq=2 ttl=255 time=9.16 ms
```

```
64 bytes from 192.168.101.254: icmp_seq=3 ttl=255 time=7.01 ms
```

```
64 bytes from 192.168.101.254: icmp_seq=4 ttl=255 time=9.64 ms
```

```
64 bytes from 192.168.101.254: icmp_seq=5 ttl=255 time=5.90 ms

--- 192.168.101.254 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 5.900/27.924/107.908/40.015 ms
```

2. After the STA connects to the SSID, run the **display station all** command on WAC1 to check STA information.

```
[WAC1] display station all

Rf/WLAN: Radio ID/WLAN ID

Rx/Tx: link receive rate/link transmit rate(Mbps)

Total: 1 2.4G: 1 5G: 0 6G: 0

-----
STA MAC          AP ID  Ap name Rf/WLAN Band Type Rx/Tx RSSI VLAN IP address      SSID
-----
04b8-996c-213f  1      AP1     0/1      2.4G 11g 0/0   -   101  192.168.101.118  HCIA-WLAN
-----
```

14.4 Reference Configuration

14.4.1 WAC1 Configuration

```
sysname WAC1

#

wlan

security-profile name HCIA-WLAN

security open

ssid-profile name HCIA-WLAN

ssid HCIA-WLAN
```

```
vap-profile name HCIA-WLAN

service-vlan vlan-id 101

ssid-profile HCIA-WLAN

security-profile HCIA-WLAN

ap-group name default

ap-group name ap-group1

radio 0

    vap-profile HCIA-WLAN wlan 1

    calibrate auto-channel-select disable

ap-id 1 type-id 220 ap-mac fa27-e678-0030 ap-sn 21500868341234500058

ap-name AP1

ap-group ap-group1

provision-ap

#

capwap source interface vlanif 100

#

vlan batch 100 to 101

#

interface Vlanif100

    ip address 192.168.100.254 255.255.255.0

#

interface GE0/0/1

    port link-type trunk

    port trunk allow-pass vlan 100

#
```

14.4.2 S3 Configuration

```
sysname S3

#
dhcp enable

#
vlan batch 100 to 101

#
interface Vlanif100

 ip address 192.168.100.1 255.255.255.0

 dhcp select interface

 dhcp server excluded-ip-address 192.168.100.254

#
interface Vlanif101

 ip address 192.168.101.254 255.255.255.0

 dhcp select interface

#
interface GE1/0/1

 port link-type trunk

 port trunk allow-pass vlan 100

#
interface GE1/0/4

 port link-type trunk

 port trunk pvid vlan 100

 port trunk allow-pass vlan 100 to 101

#
```

14.5 Quiz

1. What configuration is required on WAC1 to enable AP1 to broadcast a signal for guest access?

Answer:

First, create an SSID profile and set the SSID name as required. Then, create a VAP profile, associate the created SSID profile with the VAP profile, and bind the VAP profile to the AP group.

15

Network Programming and Automation (Omitted)

Currently, network programming and automation lab cannot be completed using the eNSP Pro simulator. For this lab, please refer to the HCIA-Datacom V2.0 lab guide (real device version).

16

Network Troubleshooting

16.1 Introduction

16.1.1 About This Lab

Network engineers are expected not only to plan and deploy network services, but also to identify and resolve faults by following defined procedures.

On a wired network, common faults include terminal address obtaining failures, network loops, interface flapping, and connectivity issues. On a WLAN, common faults include AP online failures and STA online failures.

This lab introduces several typical faults to help you master the basic troubleshooting process and common troubleshooting commands.

16.1.2 Objectives

- Understand the basic troubleshooting process of campus networks.
- Master the commands for troubleshooting campus network faults.

16.1.3 Networking Topology

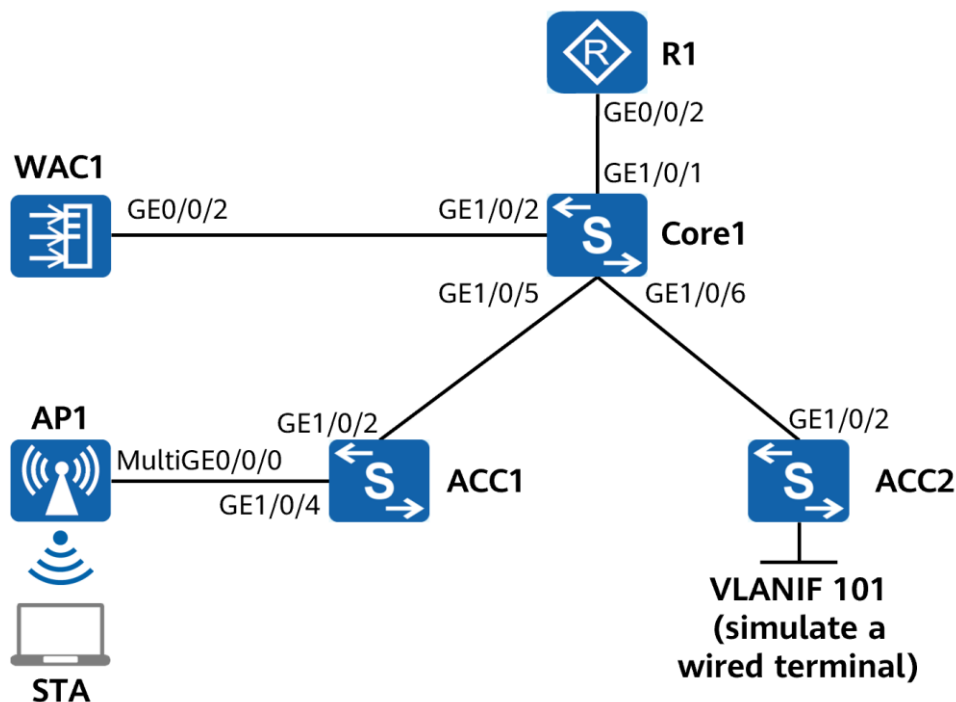


Figure 16-1 Lab topology for network troubleshooting

The router used in this lab is ENSP-AR, the switches are ENSP-LSW, the wireless controller is ENSP-AC, and the wireless access point is ENSP-AP.

1. Core1 is the core switch of the campus network, and its downstream devices are Layer 2 devices.
2. OSPF is configured on Core1, WAC1, and R1 to implement communication between them. Core1 functions as a Layer 3 service gateway and also a DHCP server to assign IP addresses to wireless and wired terminals. VLANIF 101 on ACC2 is used to simulate a wired terminal.
3. WAC1 and AP1 are located on a Layer 3 network. Core1 functions as a DHCP server to assign an IP address to AP1. Service data is forwarded in direct mode.

16.1.4 Background

A small enterprise plans to build a campus network for employees to access. After the administrator completes network deployment based on the service plan, it is found that terminals fail to obtain IP addresses, the network is disconnected, APs fail to go online, and STAs cannot detect wireless signals.

The administrator needs to troubleshoot these network faults to restore services.

16.1.5 Data Planning

Table 16-1 VLAN planning

VLAN ID	VLAN Description
3	Management VLAN for APs
10	VLAN for interconnection between Core1 and R1
11	VLAN for interconnection between Core1 and WAC1
100	Service VLAN for STAs
101	Service VLAN for wired terminals

Table 16-2 IP address planning

Network Device	Interface	IP Address/Mask
Core1	VLANIF 3	192.168.3.254/24
	VLANIF 10	192.168.10.1/30
	VLANIF 11	192.168.11.1/30
	VLANIF 100	192.168.100.254/24
	VLANIF 101	192.168.101.254/24
	LoopBack 0	10.0.1.1/32
R1	GE0/0/2	192.168.10.2/30
	LoopBack 0	10.0.2.2/32
WAC1	VLANIF 11	192.168.11.2/30
	LoopBack 0	10.0.3.3/32

Table 16-3 WLAN data planning

Item	Setting
Management VLAN for APs	VLAN 3
Service VLAN for STAs	VLAN 100
DHCP server	Core1 functions as a DHCP server to assign a management IP address to AP1.
	Core1 functions as a DHCP server to assign a service IP address to the STA.

Item	Setting
IP address pool for APs	192.168.3.1-192.168.3.253/24
IP address pool for STAs	192.168.100.1-192.168.100.253/24
IP address of the WAC's source interface	VLANIF 11: 192.168.11.2/30
AP group	Name: ap-group1
	Referenced profiles: VAP profile HCIA-WLAN and regulatory domain profile default
Regulatory domain profile	Name: default
	Country code: CN
SSID profile	Name: HCIA-WLAN
	SSID: HCIA-Guest
Security profile	Name: HCIA-WLAN
	Security policy: open
VAP profile	Name: HCIA-WLAN
	Forwarding mode: direct forwarding
	Service VLAN: VLAN 100
	Referenced profiles: SSID profile HCIA-WLAN and security profile HCIA-WLAN
Radio parameters	Radio: radio 0
	Frequency band: 2.4G
	Channel: disable automatic channel selection

16.2 Task Configuration

16.2.1 Configuration Roadmap

1. Import the pre-configuration to complete network service configuration.

2. Troubleshoot the wired terminal's failure to obtain an IP address.
3. Troubleshoot a Layer 3 network connectivity issue.
4. Troubleshoot an AP's failure to go online on the WAC.
5. Troubleshoot a STA's failure to detect wireless signals.

16.2.2 Configuration Procedure

Step 1 Import the pre-configuration.

Import the pre-configuration of R1 in the system view of R1.

```
#
sysname R1

#
interface GE0/0/2
ip address 192.168.10.2 255.255.255.252

#
interface LoopBack0
ip address 10.0.2.2 255.255.255.255

#
ospf 1 router-id 10.0.2.2
area 0.0.0.0
network 192.168.10.2 0.0.0.0

#
lldp enable

#
```

Import the pre-configuration of WAC1 in the system view of WAC1.

```
#
sysname WAC1

#
vlan batch 11 100
```

```
#
interface Vlanif11
 ip address 192.168.11.2 255.255.255.252
#
interface GE0/0/2
 port link-type access
 port default vlan 11
#
interface LoopBack0
 ip address 10.0.3.3 255.255.255.255
#
ospf 1 router-id 10.0.3.3
 area 0.0.0.0
  network 10.0.3.3 0.0.0.0
  network 192.168.11.2 0.0.0.0
#
wlan
 security-profile name HCIA-WLAN
 security open
 ssid-profile name HCIA-WLAN
 ssid HCIA-Guest
 vap-profile name HCIA-WLAN
 service-vlan vlan-id 100
 security-profile HCIA-WLAN
 regulatory-domain-profile name default
 ap-group name ap-group1
 radio 0
 vap-profile HCIA-WLAN wlan 1
```

```
calibrate auto-channel-select disable

#
capwap source interface vlanif 11

Huawei@123
Huawei@123
admin
Huawei@123
Huawei@123
Huawei@123
Huawei@123
#
```

Import the pre-configuration of Core1 in the system view of Core1.

```
#
sysname Core1

#
dhcp enable

#
vlan batch 3 10 to 11 100 to 101

#
interface Vlanif3

ip address 192.168.3.254 255.255.255.0

dhcp select interface

dhcp server gateway-list 192.168.3.254

dhcp server option 43 sub-option 2 ip-address 192.168.11.2

#
interface Vlanif10

ip address 192.168.10.1 255.255.255.252
```

```
#  
  
interface Vlanif11  
  
ip address 192.168.11.1 255.255.255.252  
  
#  
  
interface Vlanif100  
  
ip address 192.168.100.254 255.255.255.0  
  
dhcp select interface  
  
dhcp server gateway-list 192.168.100.254  
  
#  
  
interface Vlanif101  
  
shutdown  
  
ip address 192.168.101.254 255.255.255.0  
  
dhcp select interface  
  
dhcp server gateway-list 192.168.101.254  
  
#  
  
interface GE1/0/1  
  
port link-type access  
  
port default vlan 10  
  
#  
  
interface GE1/0/2  
  
port link-type access  
  
port default vlan 11  
  
#  
  
interface GE1/0/5  
  
port link-type trunk  
  
undo port trunk allow-pass vlan 1  
  
port trunk allow-pass vlan 3 100  
  
#
```



```
interface GE1/0/6

port link-type trunk

port trunk allow-pass vlan 101

#

interface LoopBack0

ip address 10.0.1.1 255.255.255.255

#

ospf 1 router-id 10.0.1.1

area 0.0.0.0

network 10.0.1.1 0.0.0.0

network 192.168.3.0 0.0.0.255

network 192.168.10.1 0.0.0.0

network 192.168.11.1 0.0.0.0

network 192.168.100.0 0.0.0.255

network 192.168.101.0 0.0.0.255

#
```

Import the pre-configuration of ACC1 in the system view of ACC1.

```
#

sysname ACC1

#

dhcp enable

#

dhcp snooping enable

#

vlan batch 3 100

#

interface GE1/0/2
```

```
port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 3 100

dhcp snooping enable

dhcp snooping trusted

#

interface GE1/0/4

port link-type trunk

port trunk pvid vlan 3

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 3 100

dhcp snooping enable

#
```

Import the pre-configuration of ACC2 in the system view of ACC2.

```
#

sysname ACC2

#

vlan batch 101

#

interface Vlanif101

ip address dhcp-alloc

#

interface GE1/0/2

port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 101

#
```

Step 2 Troubleshoot the wired terminal's failure to obtain an IP address.

Check whether a wired terminal fails to obtain an IP address.

On ACC2, ping the service gateway address of the wired terminal.

```
[ACC2] ping 192.168.101.254

PING 192.168.101.254: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

--- 192.168.101.254 ping statistics ---

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss
```

The command output shows that the ping test fails.

On ACC2, check whether VLANIF 101 obtains an IP address.

```
[ACC2] display ip interface Vlanif 101

Vlanif101 current state :      UP

Line protocol current state :  DOWN

The Maximum Transmit Unit : 1500 bytes

input packets : 0,      bytes : 0,      multicasts : 0

output packets : 0,      bytes : 0,      multicasts : 0

Directed-broadcast packets:

received packets:      0,      sent packets:      0

forwarded packets:      0,      dropped packets:      0
```

Internet protocol processing : disabled

Broadcast address : 0.0.0.0

TTL being 1 packet number: 0

TTL invalid packet number: 0

ICMP packet input number: 0

Echo reply: 0

Unreachable: 0

Source quench: 0

Routing redirect: 0

Echo request: 0

Router advert: 0

Router solicit: 0

Time exceed: 0

IP header bad: 0

Timestamp request: 0

Timestamp reply: 0

Information request: 0

Information reply: 0

Netmask request: 0

Netmask reply: 0

Unknown type: 0

The command output shows that VLANIF 101 on ACC2 does not obtain an IP address from the wired service network segment 192.168.101.0/24.

Check whether the status of the interconnection interface is normal. Specifically, check whether the link between the DHCP client and server is normal.

Check the interface status on ACC2.

[ACC2] display interface brief | include up

PHY: Physical

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

(b): BFD down

(e): ETHOAM down

(d): Dampening Suppressed

(p): port alarm down

(dl): DLDP down

(c): CFM down

(sd): STP instance discarding

(ms): MACsec down

(ed): error down

(lb): LBDT block

InUti/OutUti: input utility rate/output utility rate

Interface	PHY	Protocol	InUti	OutUti	inErrors	outErrors
GE1/0/2	up	up	0.01%	0.01%	0	0
Vlanif101	up	down	--	--	0	0

The command output shows that the status of the directly connected physical interface between ACC2 and Core1 is normal, and the physical status of VLANIF 101 on ACC2 is normal but the protocol status is **down**. This indicates that the physical link between the DHCP client and the server is normal.

Check whether the service VLAN is allowed by the interface.

[ACC2] display vlan

The total number of vlans is : 2

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning;				STAT: Statistic;			
BC: Broadcast;		MC: Multicast;		UC: Unknown-unicast;			
FWD: Forward;		DSD: Discard;					

VID		Ports					

1		UT:					
101		TG:GE1/0/2(U)					

VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description

1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
101	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0101

[Core1] display vlan			
The total number of vlans is : 6			

U: Up;	D: Down;	TG: Tagged;	UT: Untagged;
MP: Vlan-mapping;		ST: Vlan-stacking;	
#: ProtocolTransparent-vlan;		*: Management-vlan;	
MAC-LRN: MAC-address learning;		STAT: Statistic;	
BC: Broadcast;	MC: Multicast;	UC: Unknown-unicast;	
FWD: Forward;	DSD: Discard;		

VID	Ports		

1	UT:		
3	TG:GE1/0/5(U)		

10	UT:GE1/0/1(U)
11	UT:GE1/0/2(U)
100	TG:GE1/0/5(U)
101	TG:GE1/0/6(U)

VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description

1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
3	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0003
10	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0010
11	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0011
100	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0100
101	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0101

The command output indicates that the interconnection interface between ACC2 and Core1 allows packets from service VLAN 101 to pass through.

Check whether the DHCP client configuration is correct.

On ACC2, check the configuration of VLANIF 101.

```
[ACC2] interface Vlanif 101
[ACC2-Vlanif101] display this
#
interface Vlanif101
  ip address dhcp-alloc
#
return
[ACC2-Vlanif101] quit
```

You can see that VLANIF 101 on ACC2 has been configured to dynamically obtain an IP address through DHCP.

Check the DHCP server configuration.

On Core1, check the configuration of the interface address pool.

```
[Core1] display ip pool interface Vlanif101
```

Pool-name	:	Vlanif101
Pool-No	:	2
Lease	:	1 Days 0 Hours 0 Minutes
Domain-name	:	-
DNS-server0	:	-
NBNS-server0	:	-
Netbios-type	:	-
Position	:	Interface
Status	:	Unlocked
Gateway-0	:	192.168.101.254
Network	:	192.168.101.0
Mask	:	255.255.255.0
VPN instance	:	--
Logging	:	Disable

Conflicted address recycle interval: -

Address Statistic: Total	:253	Used	:0
Idle	:253	Expired	:0
Conflict	:0	Disabled	:0

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
-------	-----	-------	------	---------------	----------	----------

192.168.101.1	192.168.101.254	253	0	253(0)	0	0
---------------	-----------------	-----	---	--------	---	---

The command output shows that the service IP address pool range of the wired terminal is correct, but no IP address is allocated.

Based on the preceding analysis, the interface address pool on Core1 is correctly configured but no address is allocated. As such, you need to check the interface status of the interface address pool.

Check the status of VLANIF 101 on Core1.

```
[Core1] display interface Vlanif 101

Vlanif101 current state : Administratively DOWN (ifindex: 40)

Line protocol current state : DOWN

Description:

Route Port,The Maximum Transmit Unit is 1500

Internet Address is 192.168.101.254/24

IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 20df-73f5-4575

Physical is VLANIF

Current system time: XXXX-XX-XX 09:23:21

    Input bandwidth utilization  : --

    Output bandwidth utilization : --
```

The command output shows that the status of VLANIF 101 on Core1 is **Administratively DOWN**, indicating that the **shutdown** command is run on the interface by mistake.

Check the configuration of VLANIF 101 on Core1.

```
[Core1] interface Vlanif 101

[Core1-Vlanif101] display this

#

interface Vlanif101

shutdown

ip address 192.168.101.254 255.255.255.0

dhcp select interface

dhcp server gateway-list 192.168.101.254

#
```

```
return
```

The command output shows that VLANIF 101 is in the **shutdown** state and you can run the **undo shutdown** command to recover the interface status.

```
[Core1-Vlanif101] undo shutdown
```

```
[Core1-Vlanif101] quit
```

Check whether VLANIF 101 on ACC2 obtains an IP address.

```
[ACC2] display ip interface Vlanif 101
```

```
Vlanif101 current state :      UP
```

```
Line protocol current state :  UP
```

```
The Maximum Transmit Unit : 1500 bytes
```

```
input packets : 0,  bytes : 0,  multicasts : 0
```

```
output packets : 0,  bytes : 0,  multicasts : 0
```

```
Directed-broadcast packets:
```

```
received packets:0,      sent packets:      0
```

```
forwarded packets: 0,    dropped packets:  0
```

```
Internet Address is allocated by DHCP, 192.168.101.43/24
```

```
Broadcast address : 0.0.0.0
```

```
TTL being 1 packet number:      0
```

```
TTL invalid packet number:      0
```

```
ICMP packet input number:      0
```

```
Echo reply:      0
```

```
Unreachable:      0
```

```
Source quench:      0
```

```
Routing redirect:      0
```

```
Echo request:      0
```

```
Router advert:      0
```

```
Router solicit:      0
```

Time exceed:	0
IP header bad:	0
Timestamp request:	0
Timestamp reply:	0
Information request:	0
Information reply:	0
Netmask request:	0
Netmask reply:	0
Unknown type:	0

You can see that VLANIF 101 on ACC2 has obtained a service IP address on the wired service network segment 192.168.101.0/24 through DHCP.

Ping the wired service gateway address on ACC2 again.

```
[ACC2] ping -a 192.168.101.43 192.168.101.254

PING 192.168.101.254: 56 data bytes, press CTRL_C to break

Reply from 192.168.101.254: bytes=56 Sequence=1 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 192.168.101.254 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

The command output shows that ACC2 can successfully ping the gateway address of the wired service network segment on Core1.

In this way, the wired terminal's failure to obtain an IP address has been resolved.

Step 3 Troubleshoot a Layer 3 network connectivity issue.

Check whether a Layer 3 network connectivity issue occurs.

The network service requires that the loopback interfaces of Layer 3 devices (Core1, R1, and WAC1) can ping each other.

On Core1, ping the loopback interface address (10.0.2.2) of R1.

```
[Core1] ping 10.0.2.2

PING 10.0.2.2: 56 data bytes, press CTRL_C to break

Request time out

Request time out

Request time out

Request time out

Request time out

--- 10.0.2.2 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

The command output shows that the ping test fails.

Check whether the status of the interconnection interface is normal.

```
[Core1] display interface brief | include up

PHY: Physical

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

(b): BFD down

(e): ETHOAM down
```

(d): Dampening Suppressed

(p): port alarm down

(dl): DLDP down

(c): CFM down

(sd): STP instance discarding

(ms): MACsec down

(ed): error down

(lb): LBDT block

InUti/OutUti: input utility rate/output utility rate

Interface	PHY	Protocol	InUti	OutUti	inErrors	outErrors
GE1/0/1	up	up	0.01%	0.01%	0	0
GE1/0/2	up	up	0.01%	0.01%	0	0
GE1/0/5	up	up	0.01%	0.01%	0	0
GE1/0/6	up	up	0.01%	0.01%	0	0
LoopBack0	up	up(s)	--	--	0	0
Vlanif3	up	up	--	--	0	0
Vlanif10	up	up	--	--	0	0
Vlanif11	up	up	--	--	0	0
Vlanif100	up	up	--	--	0	0
Vlanif101	up	up	--	--	0	0

The command output shows that the status of the interface connecting Core1 to R1 is normal.

Check whether the service VLAN is allowed by the interface.

[Core1] display vlan

The total number of vlans is : 6

U: Up;

D: Down;

TG: Tagged;

UT: Untagged;

MP: Vlan-mapping;

ST: Vlan-stacking;

#: ProtocolTransparent-vlan;				*: Management-vlan;					
MAC-LRN: MAC-address learning;				STAT: Statistic;					
BC: Broadcast;		MC: Multicast;		UC: Unknown-unicast;					
FWD: Forward;		DSD: Discard;							

VID		Ports							

1		UT:							
3		TG:GE1/0/5(U)							
10		UT:GE1/0/1(U)							
11		UT:GE1/0/2(U)							
100		TG:GE1/0/5(U)							
101		TG:GE1/0/6(U)							
VID	Type	Status	Property	MAC-LRN	STAT	BC	MC	UC	Description

1	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0001
3	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0003
10	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0010
11	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0011
100	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0100
101	common	enable	default	enable	disable	FWD	FWD	FWD	VLAN 0101

The command output indicates that the interconnection interface between Core1 and R1 allows packets from service VLAN 10 to pass through.

Check the interface IP addresses.

```
[Core1] display ip interface brief
```

*down: administratively down

!down: FIB overload down

^down: standby

(l): loopback

(s): spoofing

(d): Dampening Suppressed

(ed): error down

The number of interface that is UP in Physical is 7

The number of interface that is DOWN in Physical is 1

The number of interface that is UP in Protocol is 7

The number of interface that is DOWN in Protocol is 1

Interface	IP Address/Mas	Physical	Protocol	VPN
LoopBack0	10.0.1.1/32	up	up(s)	--
Vlanif3	192.168.3.254/24	up	up	--
Vlanif10	192.168.10.1/30	up	up	--
Vlanif11	192.168.11.1/30	up	up	--
Vlanif100	192.168.100.254/24	up	up	--
Vlanif101	192.168.101.254/24	up	up	--

[R1] display ip interface brief

*down: administratively down

!down: FIB overload down

^down: standby

(l): loopback

(s): spoofing

(d): Dampening Suppressed

(ed): error down

The number of interface that is UP in Physical is 4

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 4

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol	VPN
GE0/0/2	192.168.10.2/30	up	up	--
LoopBack0	10.0.2.2/32	up	up(s)	--

The command output shows that the IP address of the interface connecting Core1 to R1 and the loopback interface are correct and the interfaces are working properly.

Check whether routes are normal.

[Core1] display ip routing-table

Proto: Protocol Pre: Preference

Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route

Routing Table : _public_

Destinations : 21		Routes : 21				
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	OSPF	10	1	D	192.168.11.2	Vlanif11
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.3.0/24	Direct	0	0	D	192.168.3.254	Vlanif3
192.168.3.254/32	Direct	0	0	D	127.0.0.1	Vlanif3
192.168.3.255/32	Direct	0	0	D	127.0.0.1	Vlanif3
192.168.10.0/30	Direct	0	0	D	192.168.10.1	Vlanif10
192.168.10.1/32	Direct	0	0	D	127.0.0.1	Vlanif10
192.168.10.3/32	Direct	0	0	D	127.0.0.1	Vlanif10
192.168.11.0/30	Direct	0	0	D	192.168.11.1	Vlanif11
192.168.11.1/32	Direct	0	0	D	127.0.0.1	Vlanif11
192.168.11.3/32	Direct	0	0	D	127.0.0.1	Vlanif11

192.168.100.0/24	Direct	0	0	D	192.168.100.254	Vlanif100
192.168.100.254/32	Direct	0	0	D	127.0.0.1	Vlanif100
192.168.100.255/32	Direct	0	0	D	127.0.0.1	Vlanif100
192.168.101.0/24	Direct	0	0	D	192.168.101.254	Vlanif101
192.168.101.254/32	Direct	0	0	D	127.0.0.1	Vlanif101
192.168.101.255/32	Direct	0	0	D	127.0.0.1	Vlanif101
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the IP routing table of Core1 contains a direct route to the subnet where R1 resides, but does not contain the OSPF route to the loopback interface address of R1.

```
[R1] display ip routing-table
```

```
Proto: Protocol      Pre: Preference
```

```
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
```

```
-----
```

```
Routing Table : _public_
```

```
Destinations : 14      Routes : 14
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	OSPF	10	1	D	192.168.10.1	GE0/0/2
10.0.2.2/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.3.3/32	OSPF	10	2	D	192.168.10.1	GE0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.3.0/24	OSPF	10	2	D	192.168.10.1	GE0/0/2
192.168.10.0/30	Direct	0	0	D	192.168.10.2	GE0/0/2
192.168.10.2/32	Direct	0	0	D	127.0.0.1	GE0/0/2
192.168.10.3/32	Direct	0	0	D	127.0.0.1	GE0/0/2

192.168.11.0/30	OSPF	10	2	D	192.168.10.1	GE0/0/2
192.168.100.0/24	OSPF	10	2	D	192.168.10.1	GE0/0/2
192.168.101.0/24	OSPF	10	2	D	192.168.10.1	GE0/0/2
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that the IP routing table of R1 contains a direct route to the subnet where Core1 resides, the OSPF route to Core1's loopback interface address, and the direct route to R1's loopback interface address.

Based on the preceding analysis, the OSPF neighbor relationship between Core1 and R1 is normal, but R1 does not advertise the routes associated with the address of LoopBack0 to OSPF. As a result, Core1 cannot learn the route.

Check the OSPF configuration on R1.

```
[R1] ospf 1
[R1-ospf-1] display this
#
ospf 1 router-id 10.0.2.2
area 0.0.0.0
network 192.168.10.2 0.0.0.0
#
return
```

The command output shows that OSPF is not enabled on LoopBack0 of R1.

Modify the OSPF configuration of R1 and enable OSPF on LoopBack0.

```
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0
[R1-ospf-1-area-0.0.0.0] quit
[R1-ospf-1] quit
```

Check the IP routing table of Core1 again.

```
[Core1] display ip routing-table
```

Proto: Protocol		Pre: Preference				
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route						

Routing Table : _public_						
Destinations : 22		Routes : 22				
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
10.0.2.2/32	OSPF	10	1	D	192.168.10.2	Vlanif10
10.0.3.3/32	OSPF	10	1	D	192.168.11.2	Vlanif11
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.3.0/24	Direct	0	0	D	192.168.3.254	Vlanif3
192.168.3.254/32	Direct	0	0	D	127.0.0.1	Vlanif3
192.168.3.255/32	Direct	0	0	D	127.0.0.1	Vlanif3
192.168.10.0/30	Direct	0	0	D	192.168.10.1	Vlanif10
192.168.10.1/32	Direct	0	0	D	127.0.0.1	Vlanif10
192.168.10.3/32	Direct	0	0	D	127.0.0.1	Vlanif10
192.168.11.0/30	Direct	0	0	D	192.168.11.1	Vlanif11
192.168.11.1/32	Direct	0	0	D	127.0.0.1	Vlanif11
192.168.11.3/32	Direct	0	0	D	127.0.0.1	Vlanif11
192.168.100.0/24	Direct	0	0	D	192.168.100.254	Vlanif100
192.168.100.254/32	Direct	0	0	D	127.0.0.1	Vlanif100
192.168.100.255/32	Direct	0	0	D	127.0.0.1	Vlanif100
192.168.101.0/24	Direct	0	0	D	192.168.101.254	Vlanif101
192.168.101.254/32	Direct	0	0	D	127.0.0.1	Vlanif101
192.168.101.255/32	Direct	0	0	D	127.0.0.1	Vlanif101
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that Core1 has learned the OSPF route to the loopback interface address of R1.

Check the connectivity between Core1 and the loopback interface of R1 again.

```
[Core1] ping 10.0.2.2

PING 10.0.2.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=255 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=255 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=3 ttl=255 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=4 ttl=255 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=5 ttl=255 time=1 ms

--- 10.0.2.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

The command output shows that Core1 successfully pings the loopback interface address of R1.

In this way, the Layer 3 network connectivity issue has been resolved.

Step 4 Troubleshoot AP1's failure to go online on WAC1.

Check whether AP1 fails to go online.

On WAC1, check the online status of AP1.

```
[WAC1] display ap all

Total AP information:

unauth: unauth          [1]

ExtraInfo : Extra information

Total: 1

-----
```

ID	MAC	Name	Group	IP Type	State	STA	Uptime	ExtraInfo
-	fa53-fa95-0060	-	-	-	AirEngine5773-21 unauth	-	-	-

The command output shows that AP1 is not online and is in **unauth** state.

Check the reason for AP1's failure to go online.

[WAC1] display ap online-fail-record all		
Total APs: 1 Total records: 1		
MAC	Last fail time	Reason
fa53-fa95-0060	20XX-11-26/09:43:43	The AP is not in the MAC whitelist

The command output shows that the reason for AP1's failure to go online is **The AP is not in the MAC whitelist**.

Manually add AP1 to the MAC address authentication list.

On WAC1, add AP1 and configure its name and group.

[WAC1] wlan
[WAC1-wlan] ap-id 1 ap-mac fa53-fa95-0060
[WAC1-wlan-ap-1] ap-name AP1
[WAC1-wlan-ap-1] ap-group ap-group1
Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]: y
[WAC1-wlan-ap-1] quit
[WAC1-wlan] quit

You can run the **display ap all** command on WAC1 to obtain the MAC address of AP1.

After a period of time, check the online status of AP1 again.

```
[WAC1] display ap all
```

Total AP information:

nor : normal [1]

ExtralInfo : Extra information

Total: 1

ID	MAC	Name	Group	IP	Type	State	STA	Uptime	ExtralInfo
1	fa53-fa95-0060	AP1	ap-group1	192.168.3.188	AirEngine5773-21	nor	0	22S	-

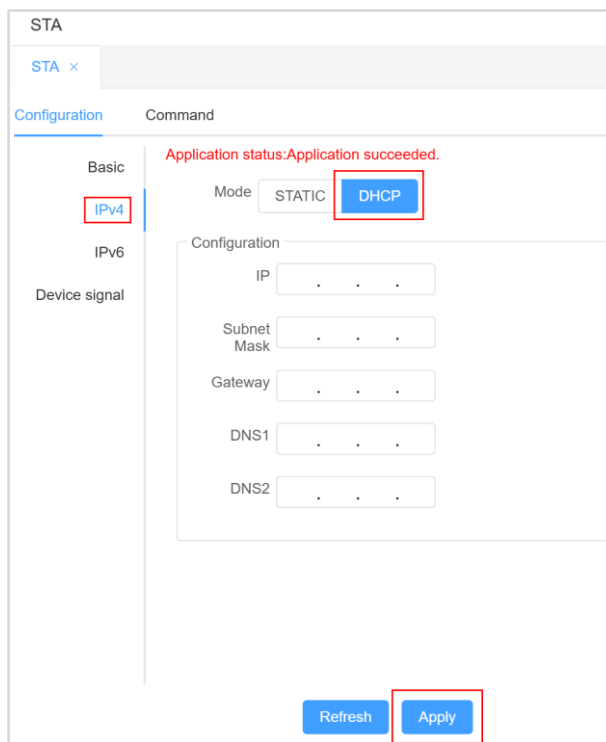
The command output shows that AP1 has successfully gone online on WAC1 and is in **nor** state (indicating that AP1 is normal).

In this way, AP1's failure to go online on WAC1 has been resolved.

Step 5 Troubleshoot a STA's failure to detect wireless signals.

Check whether a STA fails to detect wireless signals.

After AP1 goes online successfully, double-click the activated STA. In the dialog box that is displayed, select **IPv4**, select **DHCP**, and click **Apply**.



STA

STA x

Configuration Command

Basic

IPv4

IPv6

Device signal

Mode

STATIC

DHCP

Configuration

IP

Subnet Mask

Gateway

DNS1

DNS2

Refresh

Apply

Application status: Application succeeded.

Drag the STA into the wireless signal coverage of AP1, and then run the following command on the CLI of the STA to search for the SSID.

```
PC> wnmcli device wifi list --rescan yes
```

SSID	Status
HUAWEI-WLAN	down

The SSID **HCIA-Guest** cannot be found on the STA, and only **HUAWEI-WLAN** is displayed.

Log in to WAC1 and check the VAP status.

```
[WAC1] display vap all
```

```
WID : WLAN ID
```

```
Total: q
```

AP ID	AP name	Rfid	WID	BSSID	Status	Auth type	STA	SSID	Radio Type
1	AP1	0	1	FA53-FA95-0060	ON	Open	0	HUAWEI-WLAN	11be

The command output shows that the SSID name associated with the VAP is **HUAWEI-WLAN**.

Check the WLAN service configuration.

On WAC1, check the VAP and SSID profile configurations.

```
[WAC1] wlan
```

```
[WAC1-wlan] display this
```

```
#
```

```
wlan
```

```
.....
```

```
security-profile name HCIA-WLAN
```

```
security open
```

```
ssid-profile name HCIA-WLAN
```

```

ssid HCIA-Guest

vap-profile name HCIA-WLAN

service-vlan vlan-id 100

security-profile HCIA-WLAN

regulatory-domain-profile name default

ap-group name ap-group1

radio 0

    vap-profile HCIA-WLAN wlan 1

    calibrate auto-channel-select disable

ap-id 1 type-id 220 ap-mac fa53-fa95-0060 ap-sn 21500868341234500106

ap-name AP1

ap-group ap-group1

.....

#

return

```

The command output shows that the SSID profile is correctly configured on WAC1 but is not bound to the VAP profile.

Modify the VAP profile configuration of the WLAN.

On WAC1, bind the SSID profile to the VAP profile.

```

[WAC1-wlan] vap-profile name HCIA-WLAN

[WAC1-wlan-vap-prof-HCIA-WLAN] ssid-profile HCIA-WLAN

Warning: This action may cause service interruption. Continue? [Y/N]: y

[WAC1-wlan-vap-prof-HCIA-WLAN] quit

[WAC1-wlan] quit

```

Check the VAP status again.

```

[WAC1] display vap all

WID : WLAN ID

```


Total: 1

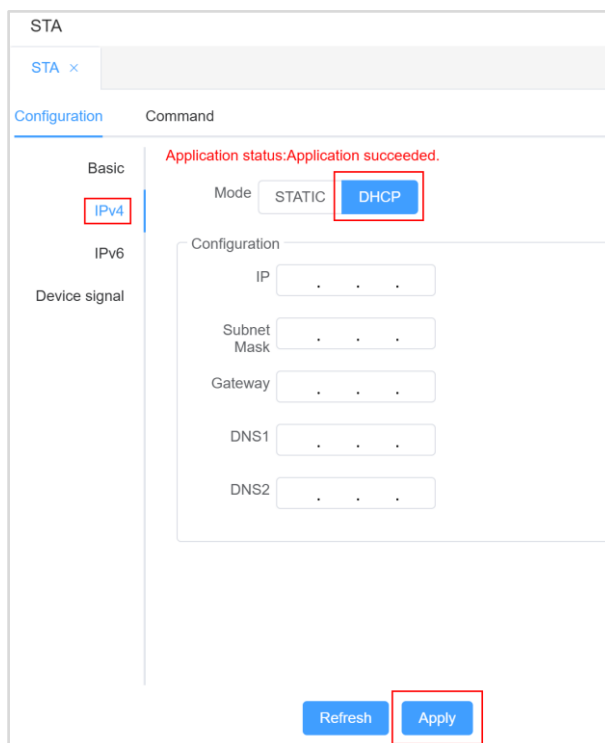
AP ID	AP name	Rfid WID	BSSID	Status	Auth type	STA	SSID	Radio Type
1	AP1	0	1	FA53-FA95-0060	ON	Open	0	HCIA-Guest 11be

The command output shows that the SSID has been changed to **HCIA-Guest**.
In this way, the STA's failure to detect wireless signals has been resolved.

16.3 Verification

Use a STA to perform the following verification:

1. Double-click the activated STA. In the dialog box that is displayed, select **IPv4**, select **DHCP**, and click **Apply**.



2. Use the STA to search for and connect to the wireless signal with the SSID **HCIA-Guest**.

```
PC> wnmcli device wifi list --rescan yes
```

SSID	Status
------	--------

HCIA-Guest down

```
PC> wnmcli device wifi connect HCIA-Guest
```

```
Connect is done
```

wnmcli device wifi list --rescan yes is used to list available wireless signals.

wnmcli device wifi connect ~~XX~~ is used to connect to a specific wireless signal.

3. Check the IP address obtained by the STA on the CLI of the terminal, and run the **ping 192.168.100.254** command on the STA.

```
PC> ifconfig
```

```
Local IPv6 address..... : fe80::a2e:cbff:fec4:6d69
```

```
IPv6 address..... : ::
```

```
IPv6 gateway..... : ::
```

```
IPv4 address..... : 192.168.100.207
```

```
Subnet mask..... : 255.255.255.0
```

```
Gateway..... : 192.168.100.254
```

```
Physical address..... : 08:2e:cb:c4:6d:69
```

```
DNS server..... :
```

```
PC> ping 192.168.100.254
```

```
PING 192.168.100.254 (192.168.100.254) 56(84) bytes of data.
```

```
64 bytes from 192.168.100.254: icmp_seq=1 ttl=255 time=41.6 ms
```

```
64 bytes from 192.168.100.254: icmp_seq=2 ttl=255 time=43.0 ms
```

```
64 bytes from 192.168.100.254: icmp_seq=3 ttl=255 time=18.3 ms
```

```
64 bytes from 192.168.100.254: icmp_seq=4 ttl=255 time=39.3 ms
```

```
64 bytes from 192.168.100.254: icmp_seq=5 ttl=255 time=16.6 ms
```

```
--- 192.168.100.254 ping statistics ---
```

```
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 16.632/31.774/43.015/11.759 ms
```

4. After the STA connects to the SSID, run the **display station all** command on WAC1 to check STA information.

```
[WAC1] display station all

Rf/WLAN: Radio ID/WLAN ID

Rx/Tx: link receive rate/link transmit rate(Mbps)

Total: 1 2.4G: 1 5G: 0 6G: 0

-----
STA MAC      AP ID Ap name Rf/WLAN Band Type Rx/Tx RSSI  VLAN IP address  SSID
-----
082e-cbc4-6d69 1   AP1      0/1    2.4G  11g  0/0   -    100 192.168.100.207 HClA-Guest
-----
```

5. After the STA obtains the IP address, run the **display ip pool interface Vlanif100 used** command on Core1 to check the address allocation in the interface address pool.

```
[Core1] display ip pool interface Vlanif100 used

Pool-name      : Vlanif100
Pool-No        : 1
Lease           : 1 Days 0 Hours 0 Minutes
Domain-name    : -
DNS-server0    : -
NBNS-server0   : -
Netbios-type    : -
Position       : Interface
Status         : Unlocked
Gateway-0      : 192.168.100.254
```

Network : 192.168.100.0

Mask : 255.255.255.0

VPN instance : --

Logging : Disable

Conflicted address recycle interval: -

Address Statistic:	Total	:253	Used	:1
	Idle	:252	Expired	:0
	Conflict	:0	Disabled	:0
	Ping	:0	Decline	:0

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
192.168.100.1	192.168.100.254	253	1	252(0)	0	0

Client-ID format as follows:

DHCP : mac-address

IPSec : user-id/portnumber/vrf

SSL-VPN : user-id/session-id

Index	IP	Client-ID	Type	Left	Status
206	192.168.100.207	082e-cbc4-6d69	DHCP	86130	Used

6. After the STA obtains the IP address, run the **ping 10.0.3.3** command to test the Layer 3 connectivity between the STA and WAC1.

```
PC> ping 10.0.3.3

PING 10.0.3.3 (10.0.3.3) 56(84) bytes of data:

64 bytes from 10.0.3.3: icmp_seq=1 ttl=254 time=54.7 ms
64 bytes from 10.0.3.3: icmp_seq=2 ttl=254 time=20.9 ms
64 bytes from 10.0.3.3: icmp_seq=3 ttl=254 time=36.1 ms
64 bytes from 10.0.3.3: icmp_seq=4 ttl=254 time=24.1 ms
64 bytes from 10.0.3.3: icmp_seq=5 ttl=254 time=36.4 ms

--- 10.0.3.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 20.861/34.439/54.681/11.889 ms
```

Use a wired terminal (VLANIF 101 of ACC2) to perform the following verification:

1. Run the **display ip interface Vlanif 101** command on ACC2 to check the IP address dynamically obtained by VLANIF 101, and run the **ping -a *dynamically obtained IP address* 192.168.101.254** command on ACC2.

```
[ACC2] display ip interface Vlanif 101

Vlanif101 current state : UP

Line protocol current state : UP

The Maximum Transmit Unit : 1500 bytes

input packets : 0, bytes : 0, multicasts : 0

output packets : 0, bytes : 0, multicasts : 0

Directed-broadcast packets:

received packets:          0,    sent packets:          0

forwarded packets:        0,    dropped packets:        0

Internet Address is allocated by DHCP, 192.168.101.43/24

Broadcast address : 0.0.0.0

TTL being 1 packet number:          0

TTL invalid packet number:          0
```

ICMP packet input number:	5
Echo reply:	5
Unreachable:	0
Source quench:	0
Routing redirect:	0
Echo request:	0
Router advert:	0
Router solicit:	0
Time exceed:	0
IP header bad:	0
Timestamp request:	0
Timestamp reply:	0
Information request:	0
Information reply:	0
Netmask request:	0
Netmask reply:	0
Unknown type:	0

```
[ACC2] ping -a 192.168.101.43 192.168.101.254
```

```
PING 192.168.101.254: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.101.254: bytes=56 Sequence=1 ttl=255 time=11 ms
```

```
Reply from 192.168.101.254: bytes=56 Sequence=2 ttl=255 time=14 ms
```

```
Reply from 192.168.101.254: bytes=56 Sequence=3 ttl=255 time=11 ms
```

```
Reply from 192.168.101.254: bytes=56 Sequence=4 ttl=255 time=12 ms
```

```
Reply from 192.168.101.254: bytes=56 Sequence=5 ttl=255 time=10 ms
```

```
--- 192.168.101.254 ping statistics ---
```

```
5 packet(s) transmitted
```

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 10/11/14 ms

2. After VLANIF 101 obtains the IP address, run the **display ip pool interface Vlanif101 used** command on Core1 to check the address allocation in the interface address pool.

```
[Core1] display ip pool interface Vlanif101 used
```

Pool-name : Vlanif101

Pool-No : 2

Lease : 1 Days 0 Hours 0 Minutes

Domain-name : -

DNS-server0 : -

```
NBNS-server0      : -
```

Netbios-type : -

Position : Interface

Status : Unlocked

Gateway-0 : 192.168.101.254

Network : 192.168.101.0

Mask : 255.255.255.0

VPN instance : --

Logging : Disable

Conflicted address recycle interval: -

Address Statistic: Total :253 Used :1

Idle :252 Expired :0

Conflict	:0	Disabled	:0
----------	----	----------	----

Ping	:0	Decline	:0
------	----	---------	----

Network section						
Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
192.168.101.1	192.168.101.254	253	1	252(0)	0	0
Client-ID format as follows:						
DHCP : mac-address						
IPSec : user-id/portnumber/vrf						
SSL-VPN : user-id/session-id						
Index	IP	Client-ID	Type	Left	Status	
42	192.168.101.43	fa67-91b3-0045	DHCP	82784	Used	

3. After VLANIF 101 obtains the IP address, run the **ping 10.0.2.2** command to test the Layer 3 connectivity with R1.

```
[ACC2] ping 10.0.2.2
```

```
PING 10.0.2.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=254 time=38 ms
```

```
Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=254 time=17 ms
```

```
Reply from 10.0.2.2: bytes=56 Sequence=3 ttl=254 time=55 ms
```

```
Reply from 10.0.2.2: bytes=56 Sequence=4 ttl=254 time=42 ms
```

```
Reply from 10.0.2.2: bytes=56 Sequence=5 ttl=254 time=24 ms
```

```
--- 10.0.2.2 ping statistics ---
```

```
5 packet(s) transmitted
```



```
5 packet(s) received  
0.00% packet loss  
round-trip min/avg/max = 17/35/55 ms
```

16.4 Reference Configuration

16.4.1 R1 Configuration

```
#  
sysname R1  
  
#  
interface GE0/0/2  
  
ip address 192.168.10.2 255.255.255.252  
  
#  
interface LoopBack0  
  
ip address 10.0.2.2 255.255.255.255  
  
#  
ospf 1 router-id 10.0.2.2  
  
area 0.0.0.0  
  
network 10.0.2.2 0.0.0.0  
  
network 192.168.10.2 0.0.0.0  
  
#  
lldp enable  
  
#  
return
```

16.4.2 WAC1 Configuration

```
#
```

```
sysname WAC1

#
wlan

security-profile name HCIA-WLAN

security open

ssid-profile name HCIA-WLAN

ssid HCIA-Guest

vap-profile name HCIA-WLAN

service-vlan vlan-id 100

ssid-profile HCIA-WLAN

security-profile HCIA-WLAN

regulatory-domain-profile name default

ap-group name ap-group1

radio 0

vap-profile HCIA-WLAN wlan 1

calibrate auto-channel-select disable

ap-id 1 type-id 220 ap-mac fa53-fa95-0060 ap-sn 21500868341234500106

ap-group ap-group1

provision-ap

#

capwap source interface vlanif 11

#

vlan batch 11 100

#

interface Vlanif11

ip address 192.168.11.2 255.255.255.252

#

interface GE0/0/2
```

```
port link-type access
port default vlan 11
#
interface LoopBack0
ip address 10.0.3.3 255.255.255.255
#
ospf 1 router-id 10.0.3.3
area 0.0.0.0
network 10.0.3.3 0.0.0.0
network 192.168.11.2 0.0.0.0
#
return
```

16.4.3 Core1 Configuration

```
#
sysname Core1
#
dhcp enable
#
vlan batch 3 10 to 11 100 to 101
#
interface Vlanif3
ip address 192.168.3.254 255.255.255.0
dhcp select interface
dhcp server gateway-list 192.168.3.254
dhcp server option 43 sub-option 2 ip-address 192.168.11.2
#
```

```
interface Vlanif10

ip address 192.168.10.1 255.255.255.252

#

interface Vlanif11

ip address 192.168.11.1 255.255.255.252

#

interface Vlanif100

ip address 192.168.100.254 255.255.255.0

dhcp select interface

dhcp server gateway-list 192.168.100.254

#

interface Vlanif101

ip address 192.168.101.254 255.255.255.0

dhcp select interface

dhcp server gateway-list 192.168.101.254

#

interface GE1/0/1

port link-type access

port default vlan 10

#

interface GE1/0/2

port link-type access

port default vlan 11

#

interface GE1/0/5

port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 3 100
```

```
#  
interface GE1/0/6  
    port link-type trunk  
    port trunk allow-pass vlan 101  
#  
interface LoopBack0  
    ip address 10.0.1.1 255.255.255.255  
#  
ospf 1 router-id 10.0.1.1  
    area 0.0.0.0  
        network 10.0.1.1 0.0.0.0  
        network 192.168.3.0 0.0.0.255  
        network 192.168.10.1 0.0.0.0  
        network 192.168.11.1 0.0.0.0  
        network 192.168.100.0 0.0.0.255  
        network 192.168.101.0 0.0.0.255  
#  
return
```

16.4.4 ACC1 Configuration

```
#  
sysname ACC1  
#  
dhcp enable  
#  
dhcp snooping enable  
#
```

```
vlan batch 3 100

#
interface GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3 100
dhcp snooping enable
dhcp snooping trusted
#
interface GE1/0/4
port link-type trunk
port trunk pvid vlan 3
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3 100
dhcp snooping enable
#
return
```

16.4.5 ACC2 Configuration

```
#
sysname ACC2
#
vlan batch 101
#
interface Vlanif101
ip address dhcp-alloc
#
```

```
interface GE1/0/2

port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 101

#

return
```

16.5 Quiz

1. When detecting connectivity issues between networks, what testing tools are generally used?

Answer:

The ping command is used to check network connectivity and host reachability.

The tracert command is used to check the device nodes that the packet passes through from the source to the destination and the network reachability.

2. What are the common causes of the failure for AP1 to go online on WAC1?

Answer:

Hardware issues: The physical device, connection component, or power module is faulty.

Software issues: Network configurations are incorrect (for example, incorrect DHCP configuration, network interworking configuration, AP authentication configuration, or blacklist and whitelist configuration), the software version does not match, or WAC licenses are insufficient.

17

Campus Network Deployment

17.1 Introduction

17.1.1 About This Lab

Communication networks are ubiquitous in the information society, and campus networks are always a core part. In a city, except roads and homes, everything is a campus, including factories, government agencies, shopping malls, office buildings, and school campuses. According to statistics, 90% of urban residents work and live in campuses, 80% of gross domestic product (GDP) is created in campuses, and each person stays in campuses for 18 hours every day. Campus networks are the key infrastructure that connects campuses to the digital world, making them an indispensable part of campus construction. They are playing an increasingly important role in daily work, R&D, production, and operation management.

In this lab activity, you will have a good understanding on the application of common campus network technologies by deploying a typical campus network.

17.1.2 Objectives

- Understand how to configure a Layer 2 campus network.
- Understand how to configure a Layer 3 campus network.
- Understand how to configure a wireless campus network.
- Understand how to configure campus network security and services.

17.1.3 Networking Topology

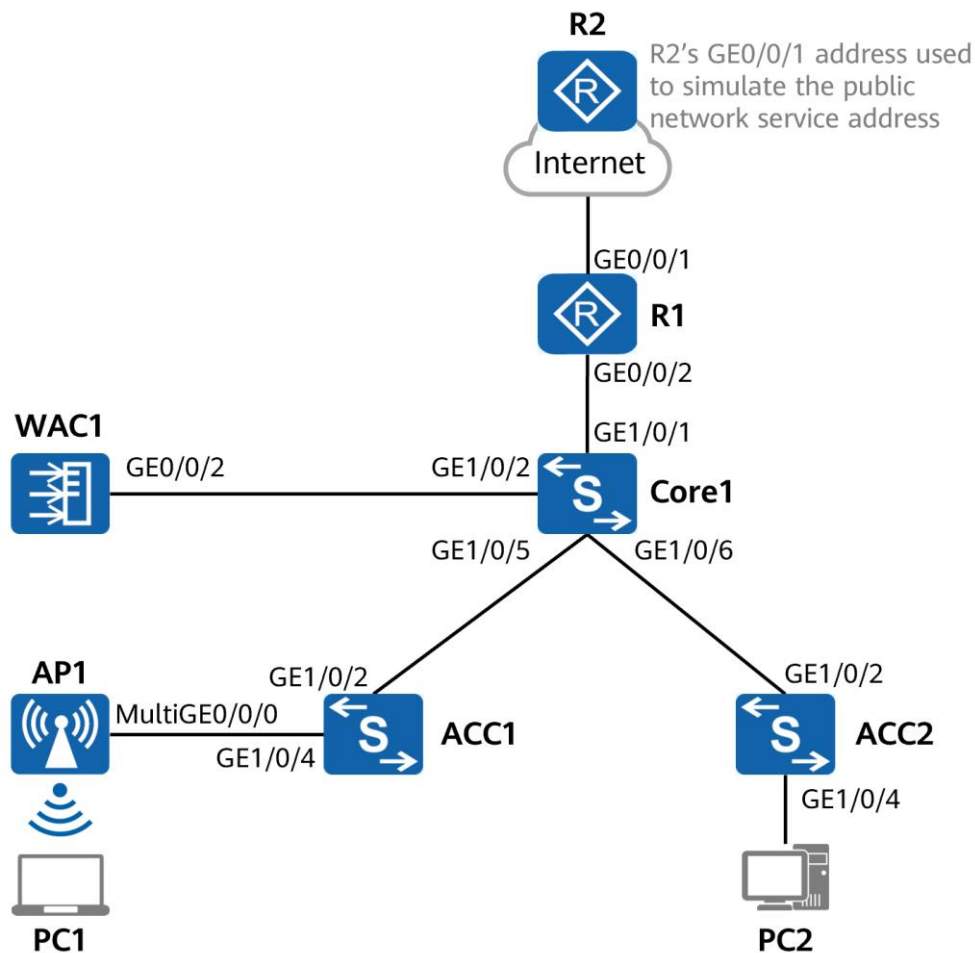


Figure 17-1 Lab topology for campus network deployment

The router used in this lab is ENSP-AR, the switches are ENSP-LSW, the wireless controller is ENSP-AC, and the wireless access point is ENSP-AP.

1. Core1 functions as the root bridge, and its downstream devices are Layer 2 devices.
2. OSPF is configured on Core1, WAC1, and R1 to implement communication between them. Core1 functions as a Layer 3 service gateway and also a DHCP server to assign IP addresses to PC1 and PC2.
3. WAC1 and AP1 are located on a Layer 3 network. Core1 functions as a DHCP server to assign an IP address to AP1. Service data is forwarded in direct mode.
4. A NAT policy is configured to allow only wired intranet users to access the public network.
5. The administrator can remotely log in to all devices through STelnet for O&M and commissioning.
6. The configuration file of Core1 (core switch) needs to be backed up to the FTP server (WAC1).

17.1.4 Background

A small enterprise plans to build a campus network for employees, on which network devices need to be managed and monitored in a centralized manner. The campus has about 50 wired and wireless terminals. After accessing the network, terminals can dynamically obtain service IP addresses. In addition, wired users need to access the external network.

17.2 Task Configuration

17.2.1 Configuration Roadmap

1. Configure interfaces and VLANs for the switches, WAC, and router, and configure Layer 3 interface addresses and routes to ensure network connectivity.
2. Configure the DHCP function on the core switch so that the switch can function as a DHCP server to assign IP addresses to APs and wireless/wired users.
3. Configure wireless services on the WAC so that APs and STAs can go online.
4. Configure the outbound NAT function to enable wired users to access public network services.
5. Configure SSH and FTP functions to enable the core switch to remotely manage access devices and back up configuration files to the FTP server.

17.2.2 Configuration Procedure

Step 1 Complete basic device configuration.

Name the devices.

The detailed configuration is not provided here.

Enable LLDP on R1.

```
[R1] lldp enable
```

Info: Global LLDP is enabled successfully.

Check the LLDP neighbor relationship of Core1.

```
[Core1] display lldp neighbor brief
```

Local Interface	Exptime(s)	Neighbor	Interface	Neighbor Device
-----------------	------------	----------	-----------	-----------------

GE1/0/1	116	GE0/0/2	R1
GE1/0/2	109	GE0/0/2	WAC1
GE1/0/5	100	GE1/0/2	ACC1
GE1/0/6	109	GE1/0/2	ACC2

Step 2 Configure interface-based VLAN assignment.

Configure VLANs for Core1, ACC1, ACC2, and WAC1 according to the following table.

Table 17-1 VLAN planning

VLAN ID	VLAN Description
2	Management VLAN for switches
3	Management VLAN for APs
10	VLAN for interconnection between Core1 and R1
11	VLAN for interconnection between Core1 and WAC1
100	Service VLAN for STAs
101	Service VLAN for wired terminals

On Core1, create VLANs 2, 3, 10, 11, 100, and 101, and add the corresponding interfaces to the VLANs.

```
[Core1] vlan batch 2 to 3 10 to 11 100 to 101

[Core1] interface GE 1/0/1
[Core1-GE1/0/1] port link-type access
[Core1-GE1/0/1] port default vlan 10
[Core1-GE1/0/1] quit

[Core1] interface GE 1/0/2
[Core1-GE1/0/2] port link-type access
[Core1-GE1/0/2] port default vlan 11
[Core1-GE1/0/2] quit

[Core1] interface GE 1/0/5
[Core1-GE1/0/5] port link-type trunk
```

```
[Core1-GE1/0/5] port trunk allow-pass vlan 2 3 100
[Core1-GE1/0/5] undo port trunk allow-pass vlan 1
[Core1-GE1/0/5] quit
[Core1] interface GE 1/0/6
[Core1-GE1/0/6] port link-type trunk
[Core1-GE1/0/6] port trunk allow-pass vlan 2 101
[Core1-GE1/0/6] undo port trunk allow-pass vlan 1
[Core1-GE1/0/6] quit
```

On ACC1, create VLANs 2, 3, and 100, and add the corresponding interfaces to the VLANs.

```
[ACC1] vlan batch 2 to 3 100
[ACC1] interface GE 1/0/2
[ACC1-GE1/0/2] port link-type trunk
[ACC1-GE1/0/2] port trunk allow-pass vlan 2 3 100
[ACC1-GE1/0/2] undo port trunk allow-pass vlan 1
[ACC1-GE1/0/2] quit
[ACC1] interface GE 1/0/4
[ACC1-GE1/0/4] port link-type trunk
[ACC1-GE1/0/4] port trunk allow-pass vlan 3 100
[ACC1-GE1/0/4] port trunk pvid vlan 3
[ACC1-GE1/0/4] undo port trunk allow-pass vlan 1
[ACC1-GE1/0/4] quit
```

On ACC2, create VLANs 2 and 101, and add the corresponding interfaces to the VLANs.

```
[ACC2] vlan batch 2 101
[ACC2] interface GE 1/0/2
[ACC2-GE1/0/2] port link-type trunk
[ACC2-GE1/0/2] port trunk allow-pass vlan 2 101
```

```
[ACC2-GE1/0/2] undo port trunk allow-pass vlan 1
[ACC2-GE1/0/2] quit
[ACC2] interface GE 1/0/4
[ACC2-GE1/0/4] port link-type access
[ACC2-GE1/0/4] port default vlan 101
[ACC2-GE1/0/4] quit
```

On WAC1, create VLANs 11 and 100, and add the corresponding interface to the VLAN.

```
[WAC1] vlan batch 11 100
[WAC1] interface GE 0/0/2
[WAC1-GE0/0/2] port link-type access
[WAC1-GE0/0/2] port default vlan 11
[WAC1-GE0/0/2] quit
```

Note: VLAN 100 is a wireless service VLAN. Because wireless service traffic is directly forwarded and WAC1 is deployed in off-path mode, the interfaces of WAC1 do not need to be configured to allow traffic from VLAN 100.

Check the created VLANs.

Note: The following information displays only the VLANs created in this step.

```
[Core1] display vlan

The total number of vlans is : 7

-----

U: Up;           D: Down;           TG: Tagged;      UT: Untagged;
MP: Vlan-mapping;      ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
MAC-LRN: MAC-address learning;  STAT: Statistic;
BC: Broadcast;      MC: Multicast;      UC: Unknown-unicast;
FWD: Forward;      DSD: Discard;

-----
```

VID	Ports
1	UT:
2	TG:GE1/0/5(U) GE1/0/6(U)
3	TG:GE1/0/5(U)
10	UT:GE1/0/1(U)
11	UT:GE1/0/2(U)
100	TG:GE1/0/5(U)
101	TG:GE1/0/6(U)
...	

[ACC1] display vlan

The total number of vlans is : 4

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning; STAT: Statistic;

BC: Broadcast; MC: Multicast; UC: Unknown-unicast;

FWD: Forward; DSD: Discard;

VID	Ports
1	UT:
2	TG:GE1/0/2(U)
3	UT:GE1/0/4(U)
	TG:GE1/0/2(U)
100	TG:GE1/0/2(U) GE1/0/4(U)

...

[ACC2] display vlan

The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning; STAT: Statistic;

BC: Broadcast; MC: Multicast; UC: Unknown-unicast;

FWD: Forward; DSD: Discard;

VID	Ports
1	UT:
2	TG:GE1/0/2(U)
101	UT:GE1/0/4(U) TG:GE1/0/2(U)

...

[WAC1] display vlan

The total number of vlans is : 3

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

MAC-LRN: MAC-address learning; STAT: Statistic;

BC: Broadcast; MC: Multicast; UC: Unknown-unicast;

FWD: Forward; DSD: Discard;

VID Ports

VID	Ports
1	UT:
11	UT:GE0/0/2(U)
100	
...	

Step 3 Enable STP.

Enable STP globally on the Layer 2 network and configure Core1 as the root bridge to prevent network reconvergence caused by incorrect cable connections or addition of new devices.

Enable STP globally on Core1, ACC1, and ACC2.

```
[Core1] stp enable
```

```
[ACC1] stp enable
```

```
[ACC2] stp enable
```

Change the spanning tree working mode to STP.

```
[Core1] stp mode stp
```

```
[ACC1] stp mode stp
```

```
[ACC2] stp mode stp
```


Configure Core1 as the root bridge.

```
[Core1] stp root primary
```

Check the spanning tree interface status on Core1.

```
[Core1] display stp brief
```

MSTID	Port	Role	STP State	Protection	Cost	Edged
0	GE1/0/1	DESI	forwarding	none	20000	disable
0	GE1/0/2	DESI	forwarding	none	20000	disable
0	GE1/0/5	DESI	forwarding	none	20000	disable
0	GE1/0/6	DESI	forwarding	none	20000	disable

Step 4 Configure IP addresses for devices.

Configure IP addresses for Core1, ACC1, ACC2, R1, and WAC1 according to the following table.

Table 17-2 IP address planning

Network Device	Interface	IP Address/Mask
Core1	VLANIF 2	192.168.2.254/24
	VLANIF 3	192.168.3.254/24
	VLANIF 10	192.168.10.1/30
	VLANIF 11	192.168.11.1/30
	VLANIF 100	192.168.100.254/24
	VLANIF 101	192.168.101.254/24
	LoopBack 0	10.0.1.1/32
ACC1	VLANIF 2	192.168.2.1/24
ACC2	VLANIF 2	192.168.2.2/24
R1	GE0/0/2	192.168.10.2/30
	LoopBack 0	10.0.2.2/32
WAC1	VLANIF 11	192.168.11.2/30
	LoopBack 0	10.0.3.3/32

Configure IP addresses for the management interfaces of Core1, ACC1, and ACC2.

```
[Core1] interface Vlanif 2
[Core1-Vlanif2] ip address 192.168.2.254 24
[Core1-Vlanif2] quit
[Core1] interface Vlanif 3
[Core1-Vlanif3] ip address 192.168.3.254 24
[Core1-Vlanif3] quit
```

```
[ACC1] interface Vlanif 2
[ACC1-Vlanif2] ip address 192.168.2.1 24
[ACC1-Vlanif2] quit
```

```
[ACC2] interface Vlanif 2
[ACC2-Vlanif2] ip address 192.168.2.2 24
[ACC2-Vlanif2] quit
```

Configure IP addresses for the interconnection interfaces on Core1, R1, and WAC1.

```
[Core1] interface Vlanif 10
[Core1-Vlanif10] ip address 192.168.10.1 30
[Core1-Vlanif10] quit
[Core1] interface Vlanif 11
[Core1-Vlanif11] ip address 192.168.11.1 30
[Core1-Vlanif11] quit
```

```
[R1] interface GE0/0/2
[R1-GE0/0/2] ip address 192.168.10.2 30
```

```
[R1-GE0/0/2] quit
```

```
[WAC1] interface Vlanif 11  
[WAC1-Vlanif11] ip address 192.168.11.2 30  
[WAC1-Vlanif11] quit
```

Configure IP addresses for service interfaces on Core1.

```
[Core1] interface Vlanif 100  
[Core1-Vlanif100] ip address 192.168.100.254 24  
[Core1-Vlanif100] quit  
[Core1] interface Vlanif 101  
[Core1-Vlanif101] ip address 192.168.101.254 24  
[Core1-Vlanif101] quit
```

Configure IP addresses for loopback interfaces on Core1, R1, and WAC1.

```
[Core1] interface LoopBack 0  
[Core1-LoopBack0] ip address 10.0.1.1 32  
[Core1-LoopBack0] quit
```

```
[R1] interface LoopBack 0  
[R1-LoopBack0] ip address 10.0.2.2 32  
[R1-LoopBack0] quit
```

```
[WAC1] interface LoopBack 0  
[WAC1-LoopBack0] ip address 10.0.3.3 32  
[WAC1-LoopBack0] quit
```

Use the ping tool to test the connectivity of the direct links between devices.

```
[Core1] ping 192.168.10.2
```

```
PING 192.168.10.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.10.2: bytes=56 Sequence=1 ttl=255 time=1 ms
```

```
Reply from 192.168.10.2: bytes=56 Sequence=2 ttl=255 time=1 ms
```

```
Reply from 192.168.10.2: bytes=56 Sequence=3 ttl=255 time=1 ms
```

```
Reply from 192.168.10.2: bytes=56 Sequence=4 ttl=255 time=1 ms
```

```
Reply from 192.168.10.2: bytes=56 Sequence=5 ttl=255 time=1 ms
```

```
--- 192.168.10.2 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
[Core1] ping 192.168.11.2
```

```
PING 192.168.11.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.11.2: bytes=56 Sequence=1 ttl=254 time=1 ms
```

```
Reply from 192.168.11.2: bytes=56 Sequence=2 ttl=254 time=1 ms
```

```
Reply from 192.168.11.2: bytes=56 Sequence=3 ttl=254 time=1 ms
```

```
Reply from 192.168.11.2: bytes=56 Sequence=4 ttl=254 time=1 ms
```

```
Reply from 192.168.11.2: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
--- 192.168.11.2 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
[Core1] ping 192.168.2.1
```

```
PING 192.168.2.1: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.2.1: bytes=56 Sequence=1 ttl=254 time=1 ms
```

```
Reply from 192.168.2.1: bytes=56 Sequence=2 ttl=254 time=1 ms
```

```
Reply from 192.168.2.1: bytes=56 Sequence=3 ttl=254 time=1 ms
```

```
Reply from 192.168.2.1: bytes=56 Sequence=4 ttl=254 time=1 ms
```

```
Reply from 192.168.2.1: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
--- 192.168.2.1 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
[Core1] ping 192.168.2.2
```

```
PING 192.168.2.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 192.168.2.2: bytes=56 Sequence=1 ttl=254 time=1 ms
```

```
Reply from 192.168.2.2: bytes=56 Sequence=2 ttl=254 time=1 ms
```

```
Reply from 192.168.2.2: bytes=56 Sequence=3 ttl=254 time=1 ms
```

```
Reply from 192.168.2.2: bytes=56 Sequence=4 ttl=254 time=1 ms
```

```
Reply from 192.168.2.2: bytes=56 Sequence=5 ttl=254 time=1 ms
```

```
--- 192.168.2.2 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

Step 5 Configure DHCP.

Core1 functions as the DHCP server to assign management IP addresses to APs on the campus network and service IP addresses to wireless and wired terminals connected to the network.

Enable the DHCP function on Core1.

```
[Core1] dhcp enable
```

Configure an interface address pool to assign an IP address to AP1 and specify the IP address used by WAC1 to establish a CAPWAP tunnel.

```
[Core1] interface Vlanif 3
[Core1-Vlanif3] dhcp select interface
[Core1-Vlanif3] dhcp server gateway-list 192.168.3.254
[Core1-Vlanif3] dhcp server option 43 sub-option 2 ip-address 192.168.11.2
[Core1-Vlanif3] quit
```

Configure an interface address pool to assign IP addresses to wireless and wired terminals.

```
[Core1] interface Vlanif 100
[Core1-Vlanif100] dhcp select interface
[Core1-Vlanif100] dhcp server gateway-list 192.168.100.254
[Core1-Vlanif100] quit
[Core1] interface Vlanif 101
[Core1-Vlanif101] dhcp select interface
[Core1-Vlanif101] dhcp server gateway-list 192.168.101.254
[Core1-Vlanif101] quit
```

Enable DHCP snooping on ACC1 and ACC2.

```
[ACC1] dhcp enable
[ACC1] dhcp snooping enable
```

```
[ACC1] interface GE 1/0/2
[ACC1-GE1/0/2] dhcp snooping enable
[ACC1-GE1/0/2] dhcp snooping trusted
[ACC1-GE1/0/2] quit
[ACC1] interface GE 1/0/4
[ACC1-GE1/0/4] dhcp snooping enable
[ACC1-GE1/0/4] quit
```

```
[ACC2] dhcp enable
[ACC2] dhcp snooping enable
[ACC2] interface GE 1/0/2
[ACC2-GE1/0/2] dhcp snooping enable
[ACC2-GE1/0/2] dhcp snooping trusted
[ACC2-GE1/0/2] quit
[ACC2] interface GE 1/0/4
[ACC2-GE1/0/4] dhcp snooping enable
[ACC2-GE1/0/4] quit
```

Note: In the current version, eNSP Pro does not support transparent transmission between the DHCP client and DHCP server through a switch. If this is the case, you need to configure DHCP snooping on the transparent transmission switch.

Check the IP address obtained by AP1.

```
[ACC1] display lldp neighbor interface GE 1/0/4
GE1/0/4 has 1 neighbor(s):

Neighbor index          :    1
Chassis type            :    MAC Address
Chassis ID              :    fa53-fa95-0060
Port ID subtype         :    Interface Name
```

```

Port ID                :    MultiGE0/0/0
Port description       :    --
System name            :    HUAWEI
System description     :    Huawei AP
Huawei YunShan OS
Version 1.24.0.1 (AirEngine 5700 V600R024C10SPC100)
Copyright (C) 20XX-20XX Huawei Technologies Co., Ltd.
HUAWEI AirEngine5773-21
Current Work Mode: FIT
System capabilities supported      :    wlanAccessPoint
System capabilities enabled        :    wlanAccessPoint
Management address type           :    IPv4
Management address                :    192.168.3.203
Expired time                      :    105s

```

Note: The preceding command output contains only key information. You can see that AP1 has successfully obtained a management IP address on the management network segment 192.168.3.0/24.

Check the IP address obtained by the wired terminal.

On ACC2, create VLANIF 101 to simulate a wired terminal, and configure VLANIF 101 to dynamically obtain an IP address through DHCP.

```

[ACC2] interface Vlanif 101
[ACC2-Vlanif101] ip address dhcp-alloc
[ACC2-Vlanif101] quit

```

Check the IP address obtained by VLANIF 101 on ACC2.

```

[ACC2] display ip interface Vlanif 101
Vlanif101 current state :    UP

```


Line protocol current state : UP

The Maximum Transmit Unit : 1500 bytes

input packets : 0, bytes : 0, multicasts : 0

output packets : 0, bytes : 0, multicasts : 0

Directed-broadcast packets:

received packets: 0, sent packets: 0

forwarded packets: 0, dropped packets: 0

Internet Address is allocated by DHCP, 192.168.101.192/24

Broadcast address : 0.0.0.0

TTL being 1 packet number: 0

TTL invalid packet number: 0

ICMP packet input number: 0

Echo reply: 0

Unreachable: 0

Source quench: 0

Routing redirect: 0

Echo request: 0

Router advert: 0

Router solicit: 0

Time exceed: 0

IP header bad: 0

Timestamp request: 0

Timestamp reply: 0

Information request: 0

Information reply: 0

Netmask request: 0

Netmask reply: 0

Unknown type: 0

You can see that VLANIF 101 on ACC2 has obtained a service IP address on the wired service network segment 192.168.101.0/24.

Use the ping tool to test the connectivity with the gateway.

```
[ACC2] ping -a 192.168.101.192 192.168.101.254

PING 192.168.101.254: 56 data bytes, press CTRL_C to break

Reply from 192.168.101.254: bytes=56 Sequence=1 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 192.168.101.254: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 192.168.101.254 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

Check address allocation of the interface address pool.

```
[Core1] display ip pool interface Vlanif3 used

Pool-name      : Vlanif3
Pool-No       : 0
Lease         : 1 Days 0 Hours 0 Minutes
Domain-name    : -
Option-code    : 43
Option-subcode : 2
Option-type    : ip-address
Option-value   : 192.168.11.2
```

```

DNS-server0      : -
NBNS-server0     : -
Netbios-type     : -
Position         : Interface
Status           : Unlocked
Gateway-0        : 192.168.3.254
Network          : 192.168.3.0
Mask             : 255.255.255.0
VPN instance     : --
Logging          : Disable

```

Conflicted address recycle interval: -

```

Address Statistic:  Total      :253      Used      :1
                   Idle       :252      Expired   :0
                   Conflict    :0        Disabled  :0

```

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
192.168.3.1	192.168.3.254	253	1	252(0)	0	0

Client-ID format as follows:

```

DHCP      : mac-address
IPSec     : user-id/portnumber/vrf
SSL-VPN   : user-id/session-id

```

Index	IP	Client-ID	Type	Left	Status
-------	----	-----------	------	------	--------

56	192.168.3.203	fa53-fa95-0062	DHCP	86377	Used

You can see that one IP address has been assigned from the address pool of VLANIF 3 on Core1 and the IP address used by WAC1 to establish a CAPWAP tunnel is specified.

[Core1] display ip pool interface Vlanif101 used

```

Pool-name          : Vlanif101
Pool-No            : 2
Lease               : 1 Days 0 Hours 0 Minutes
Domain-name        : -
DNS-server0        : -
NBNS-server0       : -
Netbios-type       : -
Position           : Interface
Status             : Unlocked
Gateway-0          : 192.168.101.254
Network            : 192.168.101.0
Mask               : 255.255.255.0
VPN instance       : --
Logging            : Disable
Conflicted address recycle interval: -

Address Statistic: Total      :253      Used      :1
                      Idle      :252      Expired    :1
                      Conflict   :0        Disabled   :0
  
```

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
192.168.101.1	192.168.101.254	253	1	252(0)	0	0
Client-ID format as follows:						
DHCP	: mac-address					
IPSec	: user-id/portnumber/vrf					
SSL-VPN	: user-id/session-id					
Index	IP	Client-ID	Type	Left	Status	
159	192.168.101.192	20df-73f5-4335	DHCP	85861	Used	

You can see that one IP address has been assigned from the address pool of VLANIF 101 on Core1.

Step 6 Configure OSPF.

The enterprise network uses OSPF to enable Layer 3 connectivity. Core1, R1, and WAC1 are Layer 3 devices. OSPF needs to be configured on these devices, and related interfaces need to be advertised to OSPF area 0. The LoopBack0 interface is used as the router ID.

Configure OSPF on Core1. Enable OSPF on the interconnection interface, wireless management interface, wireless and wired service interfaces, and LoopBack0.

```
[Core1] ospf 1 router-id 10.0.1.1
[Core1-ospf-1] area 0
[Core1-ospf-1-area-0.0.0.0] network 192.168.10.1 0.0.0.0
[Core1-ospf-1-area-0.0.0.0] network 192.168.11.1 0.0.0.0
[Core1-ospf-1-area-0.0.0.0] network 192.168.3.0 0.0.0.255
[Core1-ospf-1-area-0.0.0.0] network 192.168.100.0 0.0.0.255
[Core1-ospf-1-area-0.0.0.0] network 192.168.101.0 0.0.0.255
```

```
[Core1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0  
[Core1-ospf-1-area-0.0.0.0] quit  
[Core1-ospf-1] quit
```

Configure OSPF on R1. Enable OSPF on the interconnection interface and LoopBack0.

```
[R1] ospf 1 router-id 10.0.2.2  
[R1-ospf-1] area 0  
[R1-ospf-1-area-0.0.0.0] network 192.168.10.2 0.0.0.0  
[R1-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0  
[R1-ospf-1-area-0.0.0.0] quit  
[R1-ospf-1] quit
```

Configure OSPF on WAC1. Enable OSPF on the interconnection interface and LoopBack0.

```
[WAC1] ospf 1 router-id 10.0.3.3  
[WAC1-ospf-1] area 0  
[WAC1-ospf-1-area-0.0.0.0] network 192.168.11.2 0.0.0.0  
[WAC1-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0  
[WAC1-ospf-1-area-0.0.0.0] quit  
[WAC1-ospf-1] quit
```

Check the OSPF neighbor status.

```
[Core1] display ospf peer brief  
  
(M) Indicates MADJ neighbor  
  
OSPF Process 1 with Router ID 10.0.1.1  
Peer Statistic Information  
Total number of peer(s): 2
```

```
Peer(s) in full state: 2
```

```
-----
Area Id      Interface    Neighbor id  State
0.0.0.0      Vlanif10     10.0.2.2     Full
0.0.0.0      Vlanif11     10.0.3.3     Full
-----
```

The command output shows that Core1 has established OSPF neighbor relationships with R1 and WAC1.

Check the IP routing table.

```
[R1] display ip routing-table
```

```
Proto: Protocol
```

```
Pre: Preference
```

```
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
```

```
-----
Routing Table : _public_
```

```
Destinations : 14
```

```
Routes : 14
```

```
Destination/Mask    Proto    Pre  Cost  Flags    NextHop    Interface
10.0.1.1/32         OSPF     10   1     D        192.168.10.1  GE0/0/2
10.0.2.2/32         Direct   0     0     D        127.0.0.1    LoopBack0
10.0.3.3/32         OSPF     10   2     D        192.168.10.1  GE0/0/2
127.0.0.0/8         Direct   0     0     D        127.0.0.1    InLoopBack0
127.0.0.1/32        Direct   0     0     D        127.0.0.1    InLoopBack0
127.255.255.255/32  Direct   0     0     D        127.0.0.1    InLoopBack0
192.168.3.0/24       OSPF     10   2     D        192.168.10.1  GE0/0/2
192.168.10.0/30      Direct   0     0     D        192.168.10.2  GE0/0/2
192.168.10.2/32      Direct   0     0     D        127.0.0.1    GE0/0/2
192.168.10.3/32      Direct   0     0     D        127.0.0.1    GE0/0/2
192.168.11.0/30     OSPF     10   2     D        192.168.10.1  GE0/0/2
```

192.168.100.0/24	OSPF	10	2	D	192.168.10.1	GE0/0/2
192.168.101.0/24	OSPF	10	2	D	192.168.10.1	GE0/0/2
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

The command output shows that R1 has learned routes to each network segment through OSPF.

(Optional) Check the IP routing tables of WAC1 and Core1 in the same way.

On ACC2, use the ping tool to test the connectivity between ACC2 and R1.

```
[ACC2] ping -a 192.168.101.192 10.0.2.2

PING 10.0.2.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=254 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=254 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=3 ttl=254 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=4 ttl=254 time=1 ms

Reply from 10.0.2.2: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 10.0.2.2 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms
```

Note: The IP address of VLANIF 101 on ACC2 is dynamically obtained. You can replace it with the actual interface IP address based on the site requirements.

On WAC1, use the ping tool to test the connectivity between WAC1 and AP1.

```
[WAC1] ping 192.168.3.203

PING 192.168.3.203: 56 data bytes, press CTRL_C to break

Reply from 192.168.3.203: bytes=56 Sequence=1 ttl=254 time=42 ms

Reply from 192.168.3.203: bytes=56 Sequence=2 ttl=254 time=24 ms

Reply from 192.168.3.203: bytes=56 Sequence=3 ttl=254 time=20 ms
```



```

Reply from 192.168.3.203: bytes=56 Sequence=4 ttl=254 time=25 ms

Reply from 192.168.3.203: bytes=56 Sequence=5 ttl=254 time=21 ms

--- 192.168.3.203 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/26/42 ms

```

Note: The management IP address of AP1 is dynamically obtained. You can replace it with the actual management IP address based on the site requirements.

Step 7 Configure WLAN services.

The enterprise network needs to provide wireless access for guests. Therefore, wireless services need to be deployed and an SSID successfully released.

Table 17-3 WLAN data planning

Item	Setting
Management VLAN for APs	VLAN 3
Service VLAN for STAs	VLAN 100
DHCP server	Core1 functions as a DHCP server to assign a management IP address to AP1.
	Core1 functions as the DHCP server to assign a service IP address to PC1.
IP address pool for APs	192.168.3.1~192.168.3.253/24
IP address pool for STAs	192.168.100.1~192.168.100.253/24
IP address of the WAC's source interface	VLANIF 11: 192.168.11.2/30
AP group	Name: ap-group1
	Referenced profiles: VAP profile HCIA-WLAN and regulatory domain profile default
Regulatory domain profile	Name: default

Item	Setting
	Country code: CN
SSID profile	Name: HCIA-WLAN
	SSID: HCIA-Guest
Security profile	Name: HCIA-WLAN
	Security policy: open
VAP profile	Name: HCIA-WLAN
	Forwarding mode: direct forwarding
	Service VLAN: VLAN 100
	Referenced profiles: SSID profile HCIA-WLAN and security profile HCIA-WLAN
Radio parameters	Radio: radio 0
	Frequency band: 2.4G
	Channel: disable automatic channel selection

Configure the AP to go online.

```
[WAC1] wlan
[WAC1-wlan] regulatory-domain-profile name default
[WAC1-wlan-regulate-domain-default] country-code cn
[WAC1-wlan-regulate-domain-default] quit
[WAC1-wlan] ap-group name ap-group1
[WAC1-wlan-ap-group-ap-group1] regulatory-domain-profile default
Warning: This configuration change will clear the channel and power configurations of radios, and may
restart APs. Continue? [Y/N]: y
[WAC1-wlan-ap-group-ap-group1] quit
[WAC1-wlan] quit

[WAC1] capwap source interface vlanif 11
```

Set the DTLS PSK(contains 8-32 plain-text characters, or 128 or 148 cipher-text characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm PSK: **Huawei@123**

Info: Deliver DTLS PSK to devices using CAPWAP connections. It may take a few minutes.

Set the user name for FIT APs(The value is a string of 4 to 31 characters, which can contain letters, underscores, and digits, and must start with a letter): **admin**

Set the password for FIT APs(plain-text password of 8-128 characters or cipher-text password of 128-268 characters that must be a combination of at least three of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm password: **Huawei@123**

Set the PSK of the global offline management VAP(plain-text password of 8-63 characters or cipher-text password of 128-188 characters that must be a combination of at least two of the following: lowercase letters a to z, uppercase letters A to Z, digits, and special characters): **Huawei@123**

Confirm PSK: **Huawei@123**

Warning: Ensure that the management VLAN and service VLAN are different. Otherwise, services may be interrupted.

Info: Ensure that the management VLAN for devices is within the VLAN range mapping the source interface. Otherwise, the devices cannot go online.

```
[WAC1] wlan
```

```
[WAC1-wlan] ap auth-mode mac-auth
```

```
[WAC1-wlan] ap-id 1 ap-mac fa53-fa95-0060
```

```
[WAC1-wlan-ap-1] ap-name AP1
```

```
[WAC1-wlan-ap-1] ap-group ap-group1
```

Warning: This operation may cause AP reset. If the country code changes, it will clear channel, power and antenna gain configurations of the radio, Whether to continue? [Y/N]: **y**

```
[WAC1-wlan-ap-1] quit
```

```
[WAC1-wlan] quit
```

Note: The MAC address of AP1 is obtained by running the **display lldp neighbor interface GE 1/0/4** command on the access switch ACC1.

Wait for several minutes and check the AP information.

```
[WAC1] display ap all
```

Total AP information:

nor : normal [1]

ExtralInfo : Extra information

Total: 1

ID	MAC	Name	Group	IP	Type	State	STA	Uptime	ExtralInfo
1	fa53-fa95-0060	AP1	ap-group1	192.168.3.203	AirEngine5773-21	nor	0	14S	-

The command output shows that AP1 is in **nor** state, indicating that AP1 has successfully gone online on WAC1.

Configure WLAN service parameters.

```
[WAC1] wlan
```

```
[WAC1-wlan] security-profile name HCIA-WLAN
```

```
[WAC1-wlan-sec-prof-HCIA-WLAN] security open
```

```
[WAC1-wlan-sec-prof-HCIA-WLAN] quit
```

```
[WAC1-wlan] ssid-profile name HCIA-WLAN
```

```
[WAC1-wlan-ssid-prof-HCIA-WLAN] ssid HCIA-Guest
```

```
[WAC1-wlan-ssid-prof-HCIA-WLAN] quit
```

```
[WAC1-wlan] vap-profile name HCIA-WLAN
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] forward-mode direct-forward
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] service-vlan vlan-id 100
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] security-profile HCIA-WLAN
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] ssid-profile HCIA-WLAN
```

```
[WAC1-wlan-vap-prof-HCIA-WLAN] quit
```

```
[WAC1-wlan] ap-group name ap-group1
```

```
[WAC1-wlan-ap-group-ap-group1] vap-profile HCIA-WLAN wlan 1 radio 0
```

```
[WAC1-wlan-ap-group-ap-group1] radio 0
```

```
[WAC1-wlan-ap-group-ap-group1-radio-0] calibrate auto-channel-select disable
[WAC1-wlan-ap-group-ap-group1-radio-0] quit
[WAC1-wlan-ap-group-ap-group1] quit
[WAC1-wlan] quit
```

Display VAP information.

```
[WAC1] display vap all
WID : WLAN ID
Total: 1
```

AP ID	AP name	Rfid	WID	BSSID	Status	Auth type	STA	SSID	Radio Type
1	AP1	0	1	FA53-FA95-0060	ON	Open	0	HCIA-Guest	11be

You can see that AP1 has successfully broadcast an SSID named **HCIA-Guest**.

Step 8 Configure outbound NAT.

The enterprise has obtained the public IP address 1.2.3.4/24. To allow only wired users to access the public network, configure Easy IP in source NAT.

Configure a public IP address for R1.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] ip address 1.2.3.4 24
[R1-GE0/0/1] quit
```

Configure a public IP address for R2.

```
[R2] interface GE 0/0/1
[R2-GE0/0/1] ip address 1.2.3.100 24
[R2-GE0/0/1] quit
```

Note: In this lab, GE0/0/1 of R2 is used to simulate the public service IP address, and GE0/0/1 of R2 is directly connected to GE0/0/1 of R1.

Configure a NAT policy.

```
[R1] nat-policy
[R1-policy-nat] rule name policy_nat1
[R1-policy-nat-rule-policy_nat1] source-address 192.168.101.0 24
[R1-policy-nat-rule-policy_nat1] action source-nat easy-ip
[R1-policy-nat-rule-policy_nat1] quit
[R1-policy-nat] quit
```

Enable the NAT function.

```
[R1] interface GE 0/0/1
[R1-GE0/0/1] nat enable
[R1-GE0/0/1] quit
```

Advertise the default route in the OSPF process on R1 so that private traffic can be forwarded to the public network through R1.

```
[R1] ospf 1
[R1-ospf-1] default-route-advertise always
[R1-ospf-1] quit
```

On ACC2, use the ping tool to test the connectivity between ACC2 and the public network service IP address.

```
[ACC2] ping -a 192.168.101.192 1.2.3.100

PING 1.2.3.100: 56 data bytes, press CTRL_C to break

Reply from 1.2.3.100: bytes=56 Sequence=1 ttl=252 time=1 ms

Reply from 1.2.3.100: bytes=56 Sequence=2 ttl=252 time=1 ms

Reply from 1.2.3.100: bytes=56 Sequence=3 ttl=252 time=1 ms

Reply from 1.2.3.100: bytes=56 Sequence=4 ttl=252 time=1 ms
```

```

Reply from 1.2.3.100: bytes=56 Sequence=5 ttl=252 time=1 ms

--- 1.2.3.100 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 1/1/1 ms

```

Note: The IP address of VLANIF 101 on ACC2 is dynamically obtained. You can replace it with the actual interface IP address based on the site requirements.

Check the NAT session table on R1.

```

[R1] display session all verbose

Session Table Information:

Protocol                : 1 (ICMP)
SrcAddr VPN             : 192.168.101.192
DestAddr VPN            : 1.2.3.100
Type Code IcmpIid       : 8 0 4885
Time To Live            : 8 s
NAT Info
  New SrcAddr           : 1.2.3.4
  New DestAddr          : -
  New IcmpIid           : 2050

Total :      1

```

The command output shows that the private IP address of VLANIF 101 on ACC2 is translated into the public IP address 1.2.3.4.

Step 9 Configure SSH.

The administrator needs to remotely log in to other devices from any device for O&M and commissioning. The following example demonstrates how to configure the STelnet

function, using the scenario where an administrator remotely logs in to other access switches from Core1.

Configure the VTY user interface of the SSH servers (ACC1 and ACC2).

```
[ACC1] user-interface vty 0 4
[ACC1-ui-vty0-4] authentication-mode aaa
[ACC1-ui-vty0-4] protocol inbound ssh
[ACC1-ui-vty0-4] user privilege level 3
[ACC1-ui-vty0-4] quit
```

```
[ACC2] user-interface vty 0 4
[ACC2-ui-vty0-4] authentication-mode aaa
[ACC2-ui-vty0-4] protocol inbound ssh
[ACC2-ui-vty0-4] user privilege level 3
[ACC2-ui-vty0-4] quit
```

On ACC1, create a user account with the username **admin1** and password **Huawei@123** and set the authentication mode to **password**.

```
[ACC1] aaa
[ACC1-aaa] local-user admin1 password
Please configure the login password (8-128)
It is recommended that the password consist of four types of characters, including lowercase letters,
uppercase letters, numerals and special characters.
Please enter new password: Huawei@123
Please confirm new password: Huawei@123
Info: The initial password of admin1 must be changed during login.
[ACC1-aaa] local-user admin1 service-type ssh
Warning: If the service type of local users is configured multiple times, the new configuration overwrites
the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: y
```


Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

[ACC1-aaa] local-user admin1 privilege level 3

Warning: This operation may affect online users and will change the user privilege level, Continue?
[Y/N]: **y**

[ACC1-aaa] quit

[ACC1] ssh user admin1

Info: Succeeded in adding a new SSH user.

[ACC1] ssh user admin1 authentication-type password

On ACC2, create a user account with the username **admin2** and password **Huawei@123** and set the authentication mode to **password**.

[ACC2] aaa

[ACC2-aaa] local-user admin2 password

Please configure the login password (8-128)

It is recommended that the password consist of four types of characters, including lowercase letters, uppercase letters, numerals and special characters.

Please enter password: **Huawei@123**

Please confirm password: **Huawei@123**

Info: The initial password of admin2 must be changed during login.

[ACC2-aaa] local-user admin2 service-type ssh

Warning: If the service type of local users is configured multiple times, the new configuration overwrites the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: **y**

Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory) of a local user, the rights of users already online do not change. The change takes effect to users who go online after the change.

[ACC2-aaa] local-user admin2 privilege level 3

Warning: This operation may affect online users and will change the user privilege level, Continue?
[Y/N]: **y**

[ACC2-aaa] quit

[ACC2] ssh user admin2

Info: Succeeded in adding a new SSH user.

[ACC2] ssh user admin2 authentication-type password

Enable STelnet, set the service type to STelnet, and set the service port number to 1025.

[ACC1] stelnet server enable

[ACC1] ssh server-source all-interface

Warning: SSH server source configuration will take effect in the next login. Continue? [Y/N]: **y**

Warning: It expands the range of accessed IP.

[ACC1] ssh user admin1 service-type stelnet

[ACC1] ssh server port 1025

Warning: The operation will disconnect all online users. Continue? [Y/N]: **y**

Info: Succeeded in changing SSH listening port.

[ACC2] stelnet server enable

[ACC2] ssh server-source all-interface

Warning: SSH server source configuration will take effect in the next login. Continue? [Y/N]: **y**

Warning: It expands the range of accessed IP.

[ACC2] ssh user admin2 service-type stelnet

[ACC2] ssh server port 1025

Warning: The operation will disconnect all online users. Continue? [Y/N]: **y**

Info: Succeeded in changing SSH listening port.

Configure the first login function for an SSH client.

[Core1] ssh client first-time enable

[Core1] quit

Log in to the SSH server ACC1 from the SSH client Core1.

<Core1> stelnet 192.168.2.1 1025

```
Trying 192.168.2.1 ...
Press CTRL + K to abort
Connected to 192.168.2.1 ...
The server's key fingerprint is ssh-eccdsa 521 QPNMCBM8TFBCWtdi3Z2EliwL6fNLXnBvdjxmC24COsg.
The server is not authenticated. Continue to access it? [Y/N]: y
Save the server's public key? [Y/N]: y
The server's public key will be saved with the name 192.168.2.1. Please wait...

Please input the username: admin1
Enter password: Huawei@123
Warning: The initial password poses security risks.
The password needs to be changed, Continue? [Y/N]: y
Please enter old password: Huawei@123
Please enter new password: Huawei@1234
Please confirm new password: Huawei@1234
The password has been changed successfully.

Info: The max number of VTY users is 5, the number of current VTY users online is 1, and total number
of terminal users online is 2.

    The current login time is XXXX-XX-XX 09:31:04.
<ACC1> system-view
Enter system view, return user view with return command.
```

Check SSH user information on ACC1.

```
[ACC1] display users

NOTE:

User-Intf: The absolute number and the relative number of user interface

Authen: Whether the authentication passes

Author: Command line authorization flag
```

User-Intf		Delay	Type	Network Address	Authen	Author	Username
* 0	CON 0	00:00:00	--	1	pass	no	Unspecified
34	VTY 0	00:00:28	SSH	192.168.2.254	pass	no	admin1

On ACC1, you can see that Core1 logs in to the local device using the username **admin1** through **SSH**.

Log in to the SSH server ACC2 from the SSH client Core1.

```
[ACC1] quit
<ACC1> quit

Info: The max number of VTY users is 5, and the number of current VTY users on line is 0.

Info: The connection was closed by the remote host.

<Core1> stelnet 192.168.2.2 1025

Trying 192.168.2.2 ...

Press CTRL + K to abort

Connected to 192.168.2.2 ...

The server's key fingerprint is ssh-eccsa 521 gD5y+2E2URLPIMG2JdQ/wfE4bQyyNH1+NwRGxEyzokY.

The server is not authenticated. Continue to access it? [Y/N]: y

Save the server's public key? [Y/N]: y

The server's public key will be saved with the name 192.168.2.2. Please wait...

Please input the username: admin2

Enter password: Huawei@123

Warning: The initial password poses security risks.

The password needs to be changed, Continue? [Y/N]: y

Please enter old password: Huawei@123

Please enter new password: Huawei@1234

Please confirm new password: Huawei@1234
```

The password has been changed successfully.

Info: The max number of VTY users is 5, the number of current VTY users online is 1, and total number of terminal users online is 2.

The current login time is XXXX-XX-XX 09:40:41.

<ACC2> system-view

Enter system view, return user view with return command.

Check SSH user information on ACC2.

[ACC2] display users

NOTE:

User-Intf: The absolute number and the relative number of user interface

Authen: Whether the authentication passes

Author: Command line authorization flag

	User-Intf	Delay	Type	Network Address	Authen	Author	Username
* 0	CON 0	00:00:00	--	1	pass	no	Unspecified
34	VTY 0	00:00:27	SSH	192.168.2.254	pass	no	admin2

On ACC2, you can see that Core1 logs in to the local device using the username **admin2** through **SSH**.

Log out of the SSH client Core1.

[ACC2] return

<ACC2> quit

Info: The max number of VTY users is 5, and the number of current VTY users on line is 0.

Info: The connection was closed by the remote host.

<Core1>

Step 10 Configure the FTP function.

Core1 is the core device on the entire campus network. Its configuration file needs to be backed up. You can configure WAC1 as an FTP server to store the configuration file of Core1, and allow only Core1 to access the FTP server.

Save the configuration file of Core1.

```
<Core1> save test1.cfg

Warning: Are you sure to save the configuration to flash:/test1.cfg? [Y/N]: y

Now saving the current configuration to the slot 1

Info: Save the configuration successfully.
```

Display the current file list.

```
<Core1> dir

Directory of flash:/

Idx      Attr      Size(Byte)  Date        Time        FileName
.....
17      -rw-      8,387      Nov 25 XXXX 09:45:08  test1.cfg
.....

1,070,832 KB total (326,768 KB free)
```

Note: The preceding command output contains only key information.

On WAC1, configure the FTP server function and create a user account with the username **admin3** and password **Huawei@123**.

```
[WAC1] quit

<WAC1> install feature-software WEAKEA

<WAC1> system-view

[WAC1] ftp server enable

[WAC1] ftp server source all-interface
```

```
Warning: FTP server source configuration will take effect in the next login. Continue? [Y/N]: y
Warning: It expands the range of accessed IP.

[WAC1] aaa
[WAC1-aaa] local-user admin3 password
Please configure the login password (8-128)
It is recommended that the password consist of four types of characters, including lowercase letters,
uppercase letters, numerals and special characters.
Please enter password: Huawei@123
Please confirm password: Huawei@123
Info: The initial password of admin3 must be changed during login.
[WAC1-aaa] local-user admin3 privilege level 3
Warning: This operation may affect online users and will change the user privilege level, Continue?
[Y/N]: y
[WAC1-aaa] local-user admin3 service-type ftp
Warning: If the service type of local users is configured multiple times, the new configuration overwrites
the previous one. The new configuration may cause local user authentication to fail. Continue? [Y/N]: y
Warning: The user access modes include Telnet, FTP, or HTTP, so security risks exist.
Info: After changing the rights (including the password, access type, FTP directory, and HTTP directory)
of a local user, the rights of users already online do not change. The change takes effect to users who
go online after the change.
[WAC1-aaa] local-user admin3 ftp-directory flash:/
[WAC1-aaa] quit
```

On WAC1, configure the FTP server access permission.

```
[WAC1] acl number 2001
[WAC1-acl4-basic-2001] rule 5 permit source 192.168.11.1 0
[WAC1-acl4-basic-2001] rule 10 deny
[WAC1-acl4-basic-2001] quit
[WAC1] ftp server acl 2001
```

```
[WAC1] quit
```

On the FTP client (Core1), connect to WAC1 through FTP.

```
<Core1> install feature-software WEAKEA
<Core1> ftp 192.168.11.2
Warning: FTP is not secure. Using SFTP is recommended.
Trying 192.168.11.2 ...
Press CTRL + K to abort
Connected to 192.168.11.2.
220 FTP service ready.
User(192.168.11.2:(none)): admin3
331 Password required for admin3.
Enter password: Huawei@123
230 User logged in.
[ftp] ascii
200 Type set to A.
```

Upload the configuration file to the FTP server.

```
[ftp] put test1.cfg
Warning: The file may not transfer correctly in ASCII mode.
200 Port command okay.
150 Opening ASCII mode data connection for /test1.cfg.
/ 100% [*****]
226 Transfer complete.

FTP: 8387 byte(s) send in 0.010 second(s) 819.000Kbyte(s)/sec.
[ftp] quit
221 Server closing.
```


<Core1>

Check whether the backup file has been uploaded to WAC1.

<WAC1> dir

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time	FileName
.....					
17	-rw-	8,387	Nov 25 XXXX	09:57:27	test1.cfg
.....					

1,070,832 KB total (359,260 KB free)

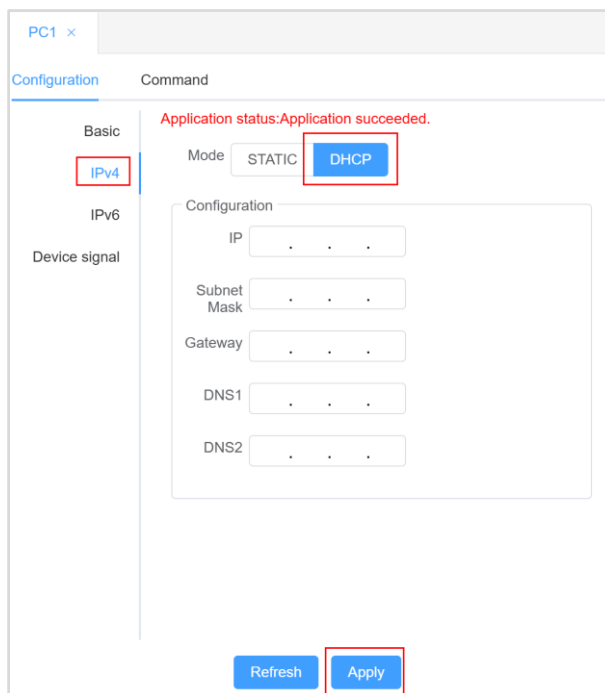
Note: The preceding command output contains only key information.

17.3 Verification

Use the wireless terminal PC1 to perform the following verification:

1. Configure PC1.

Double-click the activated PC1. In the dialog box that is displayed, select **IPv4**, select **DHCP**, and click **Apply**.



2. PC1 connects to the SSID.

Drag PC1 into the wireless signal coverage of AP1, and then run the following commands on the CLI of PC1 to search for and associate with the SSID.

```
PC> wnmcli device wifi list --rescan yes
```

SSID	Status
HCIA-Guest	down

```
PC> wnmcli device wifi connect HCIA-Guest
```

```
Connect is done
```

wnmcli device wifi list --rescan yes is used to list available wireless signals.

wnmcli device wifi connect ~~xx~~ is used to connect to a specific wireless signal.

3. After PC1 connects to the SSID, view the IP address obtained by PC1 on the CLI of PC1, and run the **ping 192.168.100.254** command on PC1.

```
PC> ifconfig
```

```
Local IPv6 address..... : fe80::b001:19ff:fedb:a30e
```

```
IPv6 address..... : ::
IPv6 gateway..... : ::
IPv4 address..... : 192.168.100.123
Subnet mask..... : 255.255.255.0
Gateway..... : 192.168.100.254
Physical address..... : b2:01:19:db:a3:0e
DNS server..... :
```

```
PC> ping 192.168.100.254

PING 192.168.100.254 (192.168.100.254) 56(84) bytes of data.

64 bytes from 192.168.100.254: icmp_seq=1 ttl=255 time=20.5 ms
64 bytes from 192.168.100.254: icmp_seq=2 ttl=255 time=48.5 ms
64 bytes from 192.168.100.254: icmp_seq=3 ttl=255 time=16.4 ms
64 bytes from 192.168.100.254: icmp_seq=4 ttl=255 time=30.2 ms
64 bytes from 192.168.100.254: icmp_seq=5 ttl=255 time=47.3 ms

--- 192.168.100.254 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 16.410/32.577/48.490/13.276 ms
```

4. After PC1 connects to the SSID, run the **display station all** command on WAC1 to check information about PC1.

```
[WAC1] display station all

Rf/WLAN: Radio ID/WLAN ID

Rx/Tx: link receive rate/link transmit rate(Mbps)

Total: 1 2.4G: 1 5G: 0

-----
STA MAC      AP ID Ap name  Rf/WLAN  Band   Type   VLAN  IP address  SSID
```

```
-----
b201-19db-a30e 1      AP1      0/1      2.4G      11g      100      192.168.100.123  HCIA-Guest
-----
```

5. After PC1 obtains the IP address, run the **display ip pool interface Vlanif100 used** command on Core1 to check the address allocation in the interface address pool.

```
[Core1] display ip pool interface Vlanif100 used
```

```
Pool-name       : Vlanif100
Pool-No         : 1
Lease           : 1 Days 0 Hours 0 Minutes
Domain-name     : -
DNS-server0     : -
NBNS-server0    : -
Netbios-type    : -
Position        : Interface
Status          : Unlocked
Gateway-0       : 192.168.100.254
Network         : 192.168.100.0
Mask            : 255.255.255.0
VPN instance    : --
Logging         : Disable
Conflicted address recycle interval: -

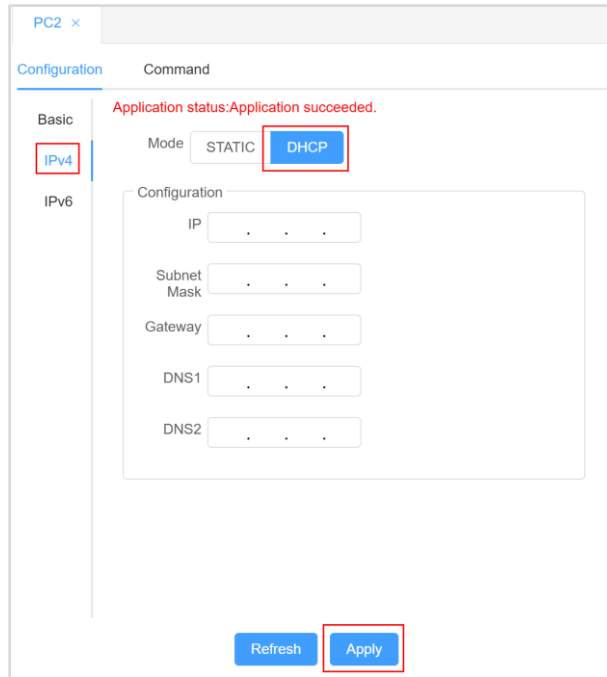
Address Statistic:  Total    :253      Used      :1
                   Idle      :252      Expired   :0
                   Conflict   :0        Disabled  :0
                   Ping       :0        Decline   :0
```

Network section						
Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
192.168.100.1	192.168.100.254	253	1	252(0)	0	0
Client-ID format as follows:						
DHCP	: mac-address					
IPSec	: user-id/portnumber/vrf					
SSL-VPN	: user-id/session-id					
Index	IP	Client-ID	Type	Left	Status	
61	192.168.100.123	b201-19db-a30e	DHCP	85822	Used	

Use the wired terminal PC2 to perform the following verification:

1. Configure PC2.

Double-click the activated PC2. In the dialog box that is displayed, select **IPv4**, select **DHCP**, and click **Apply**.



PC2 x

Configuration Command

Basic

IPv4

Mode STATIC DHCP

Configuration

IP . . .

Subnet Mask . . .

Gateway . . .

DNS1 . . .

DNS2 . . .

Refresh Apply

Application status: Application succeeded.

2. On the CLI of PC2, view the IP address obtained by PC2 and run the **ping 192.168.101.254** command on PC2.

```
PC> ifconfig

Local IPv6 address..... :      fe80::6252:b7ff:fe31:cf94
IPv6 address.....       :      ::
IPv6 gateway.....       :      ::
IPv4 address.....       :      192.168.101.18
Subnet mask.....       :      255.255.255.0
Gateway.....           :      192.168.101.254
Physical address.....   :      60:52:b7:31:cf:94
DNS server.....       :
```

```
PC> ping 192.168.101.254

PING 192.168.101.254 (192.168.101.254) 56(84) bytes of data.
64 bytes from 192.168.101.254: icmp_seq=1 ttl=255 time=32.6 ms
64 bytes from 192.168.101.254: icmp_seq=2 ttl=255 time=13.5 ms
```

```

64 bytes from 192.168.101.254: icmp_seq=3 ttl=255 time=14.6 ms
64 bytes from 192.168.101.254: icmp_seq=4 ttl=255 time=10.5 ms
64 bytes from 192.168.101.254: icmp_seq=5 ttl=255 time=11.4 ms

--- 192.168.101.254 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 10.530/16.503/32.563/8.158 ms

```

3. After PC2 obtains the IP address, run the **display ip pool interface Vlanif101 used** command on Core1 to check the address allocation in the interface address pool.

```
[Core1] display ip pool interface Vlanif101 used
```

```

Pool-name       : Vlanif101
Pool-No         : 2
Lease           : 1 Days 0 Hours 0 Minutes
Domain-name     : -
DNS-server0     : -
NBNS-server0    : -
Netbios-type    : -
Position        : Interface
Status          : Unlocked
Gateway-0       : 192.168.101.254
Network         : 192.168.101.0
Mask            : 255.255.255.0
VPN instance    : --
Logging         : Disable
Conflicted address recycle interval: -

Address Statistic:  Total          :253          Used          :2

```

Idle	:251	Expired	:0
Conflict	:0	Disabled	:0
Ping	:0	Decline	:0

Network section

Start	End	Total	Used	Idle(Expired)	Conflict	Disabled
-------	-----	-------	------	---------------	----------	----------

192.168.101.1	192.168.101.254	253	2	251(0)	0	0
---------------	-----------------	-----	---	--------	---	---

Client-ID format as follows:

DHCP : mac-address

IPSec : user-id/portnumber/vrf

SSL-VPN : user-id/session-id

Index	IP	Client-ID	Type	Left	Status
-------	----	-----------	------	------	--------

28	192.168.101.18	6052-b731-cf94	DHCP	85061	Used
112	192.168.101.192	20df-73f5-4335	DHCP	86207	Used

After PC1 and PC2 obtain IP addresses, perform the following verification:

1. Ping the dynamic IP address of PC2 from PC1. The ping operation is successful.

```
PC> ping 192.168.101.18
```

```
PING 192.168.101.18 (192.168.101.18) 56(84) bytes of data.
```

```
64 bytes from 192.168.101.18: icmp_seq=1 ttl=63 time=70.6 ms
```

```
64 bytes from 192.168.101.18: icmp_seq=2 ttl=63 time=21.7 ms
```

```
64 bytes from 192.168.101.18: icmp_seq=3 ttl=63 time=38.2 ms
```



```
64 bytes from 192.168.101.18: icmp_seq=4 ttl=63 time=25.1 ms
```

```
64 bytes from 192.168.101.18: icmp_seq=5 ttl=63 time=46.6 ms
```

```
--- 192.168.101.18 ping statistics ---
```

```
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
```

```
rtt min/avg/max/mdev = 21.656/40.434/70.624/17.561 ms
```

2. Ping the public network service address 1.2.3.100 from PC1 and PC2. The ping operation from PC1 fails, but the ping operation from PC2 is successful.

PC1 pings the public network service address.

```
PC> ping 1.2.3.100
```

```
PING 1.2.3.100 (1.2.3.100) 56(84) bytes of data.
```

```
--- 1.2.3.100 ping statistics ---
```

```
42 packets transmitted, 0 received, 100% packet loss, time 42018ms
```

PC2 pings the public network service address.

```
PC> ping 1.2.3.100
```

```
PING 1.2.3.100 (1.2.3.100) 56(84) bytes of data.
```

```
64 bytes from 1.2.3.100: icmp_seq=1 ttl=253 time=112 ms
```

```
64 bytes from 1.2.3.100: icmp_seq=2 ttl=253 time=78.2 ms
```

```
64 bytes from 1.2.3.100: icmp_seq=3 ttl=253 time=95.2 ms
```

```
64 bytes from 1.2.3.100: icmp_seq=4 ttl=253 time=76.2 ms
```

```
64 bytes from 1.2.3.100: icmp_seq=5 ttl=253 time=82.6 ms
```

```
--- 1.2.3.100 ping statistics ---
```

```
5 packets transmitted, 5 received, 12.5% packet loss, time 7007ms
```

```
rtt min/avg/max/mdev = 76.173/94.215/116.022/14.825 ms
```

3. After PC1 and PC2 ping the public network service address, run the **display session all verbose** command on R1 to check the NAT session table. You can see that only the private address of PC2 is translated into the public address 1.2.3.4.

```
[R1] display session all verbose
```

Session Table Information:

Protocol	: 1 (ICMP)
SrcAddr VPN	: 192.168.101.18
DestAddr VPN	: 1.2.3.100
Type Code IcmpId	: 8 0 3
Time To Live	: 19 s
NAT Info	
New SrcAddr	: 1.2.3.4
New DestAddr	: -
New IcmpId	: 2049

Total : 1

17.4 Reference Configuration

17.4.1 R1 Configuration

```
#
sysname R1

#
interface GE0/0/1

 ip address 1.2.3.4 255.255.255.0

 nat enable

#
interface GE0/0/2
```

```
ip address 192.168.10.2 255.255.255.252
#
interface LoopBack0
 ip address 10.0.2.2 255.255.255.255
#
ospf 1 router-id 10.0.2.2
 default-route-advertise always
 area 0.0.0.0
  network 10.0.2.2 0.0.0.0
  network 192.168.10.2 0.0.0.0
#
lldp enable
#
nat-policy
 rule name policy_nat1
  source-address 192.168.101.0 mask 255.255.255.0
  action source-nat easy-ip
#
return
```

17.4.2 R2 Configuration

```
#
sysname R2
#
interface GE0/0/1
 ip address 1.2.3.100 255.255.255.0
#
```

[return](#)

17.4.3 WAC1 Configuration

```
#
sysname WAC1

#
ftp server enable
ftp server acl 2001
ftp server source all-interface

#
wlan

security-profile name HCIA-WLAN
security open
ssid-profile name HCIA-WLAN
ssid HCIA-Guest
vap-profile name HCIA-WLAN
service-vlan vlan-id 100
ssid-profile HCIA-WLAN
security-profile HCIA-WLAN
regulatory-domain-profile name default
ap-group name ap-group1
radio 0
vap-profile HCIA-WLAN wlan 1
calibrate auto-channel-select disable

ap-id 1 type-id 220 ap-mac fa53-fa95-0060 ap-sn 21500868341234500106
ap-name AP1
ap-group ap-group1
```

```
provision-ap

#
capwap source interface vlanif 11

#
vlan batch 11 100

#
acl number 2001

rule 5 permit source 192.168.11.1 0

rule 10 deny

#
aaa

local-user admin3 password irreversible-cipher Huawei@123

local-user admin3 privilege level 3

local-user admin3 ftp-directory flash:/ read execute write

local-user admin3 service-type ftp

#
interface Vlanif11

ip address 192.168.11.2 255.255.255.252

#
interface GE0/0/2

port link-type access

port default vlan 11

#
interface LoopBack0

ip address 10.0.3.3 255.255.255.255

#
ospf 1 router-id 10.0.3.3

area 0.0.0.0
```

```
network 10.0.3.3 0.0.0.0

network 192.168.11.2 0.0.0.0

#

return
```

17.4.4 Core1 Configuration

```
#

sysname Core1

#

dhcp enable

#

vlan batch 2 to 3 10 to 11 100 to 101

#

stp mode stp

stp instance 0 root primary

#

rsa peer-public-key 192.168.2.1

public-key-code begin

3082018A

02820181

00B87277 492658CE 5A36B4AF A7B6CB64 35280856 0CDB1680 142BDC45 78B16592

FD655189 2DDE2BEC 6DC11452 3F05B00F 25C5A944 7888CE63 79321DAA CA65B8C9

CE4FB525 8910FB65 54E466C7 2A958365 1BFF10A8 0EB38DD5 B9F8F40F 2D34F3C2

D09CF999 F938B06E D8D8A69E 500BC7C9 B8308297 D738069B B787CC3F E8EA2387

EBC21071 E8D3F3EB 8DF539E7 E9917FA7 FBAE90D3 5FE91349 93D76787 3868E7B1

4DAC1B1F 4214D2F1 3811F1F7 7922DFEC AB7E7AA7 4EBBD1AB 0C80B9C6 02421085

E42AC189 6AAA0676 7E7313C6 D6102D0C 8430AFF2 F005906B EDF96C91 217D88E1
```

```
A6ABD7CF 1A8422D7 09170C8B 1A0C1F37 FE44E7E6 F19C4AB4 917AD5BD FDB81EB7
E2EE24C1 467D7204 3940651C AA729742 DD4713DA A6A815F8 CCFE9B54 1CBAF796
AB399B1B E00E0B1C AC4BA531 6AEB9660 20A2735A 48535E75 2F2A6282 9ADD5537
9849B09B 5E32EE69 1A111692 C0FEF581 235DB752 0591BE0A 01DA14DB D066C41F
D348B1E0 1B3BFA2D 07F377ED 7583EA77 A5D0F10D 92F01E1F E06C8B8B 20E30868
AF
0203
010001
public-key-code end
peer-public-key end
#
rsa peer-public-key 192.168.2.2
public-key-code begin
3082018A
02820181
00A848C2 232E3430 AA9E88CF D66198C2 5076CE1D 0DFED7F1 E9E93451 F6244F1E
6FDB1EA9 57089964 C9D91601 BD1B94DA C864523C 77D1311E E6B1ED64 0812BAF9
125FF8DF 30EA102A 9F1E89BD BFD02380 56E375F7 510C69D6 FD274AB1 1DBB4315
0A1CA352 DF173320 26FAC21F FBACC414 8FE3B0C6 F258E775 1C2CBDF4 BE8CBDC2
183C78C5 E705C4ED 1F387A42 7556FB9A 7F0C8C78 D4E0C409 6208FF6B 9A0A5549
E218FEB4 DF5EA73C BF89E87B 9AF905C4 83F53A70 B9BC36C2 EC760C13 F5433D47
E3EE7B55 3951E5C8 23139359 A3A5ABC5 AFDE2112 E25146BB F3D69F59 97AA55BA
62F8F3C4 630B8964 FA55BFCD C9691BFD 568A060D 475412FC B36A9F7D B21A5CF8
8B9D8DA7 1994DF08 9BD78861 4BA063A0 48880676 65A45FC8 C98CB630 48B54A07
D5A859A3 C7F9339C 78BFBC77 CE8871A2 591EB2A2 7E7F58CF 6F7FF7CF 0E267D4F
5E6B3BE8 A6650FA3 75F87CB4 51767DA7 C72651C4 C65242C4 3387E039 EF5E0820
1BC93CED 8FCC304D DE0ED57F EC675108 1CB201A5 07F44BF9 6E0F185B A4155CE7
1D
```

```
0203

    010001

public-key-code end

peer-public-key end

#

interface Vlanif2

    ip address 192.168.2.254 255.255.255.0

#

interface Vlanif3

    ip address 192.168.3.254 255.255.255.0

    dhcp select interface

    dhcp server gateway-list 192.168.3.254

    dhcp server option 43 sub-option 2 ip-address 192.168.11.2

#

interface Vlanif10

    ip address 192.168.10.1 255.255.255.252

#

interface Vlanif11

    ip address 192.168.11.1 255.255.255.252

#

interface Vlanif100

    ip address 192.168.100.254 255.255.255.0

    dhcp select interface

    dhcp server gateway-list 192.168.100.254

#

interface Vlanif101

    ip address 192.168.101.254 255.255.255.0

    dhcp select interface
```



```
dhcp server gateway-list 192.168.101.254

#
interface GE1/0/1
port link-type access
port default vlan 10
#
interface GE1/0/2
port link-type access
port default vlan 11
#
interface GE1/0/5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 to 3 100
#
interface GE1/0/6
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 101
#
interface LoopBack0
ip address 10.0.1.1 255.255.255.255
#
ospf 1 router-id 10.0.1.1
area 0.0.0.0
network 10.0.1.1 0.0.0.0
network 192.168.3.0 0.0.0.255
network 192.168.10.1 0.0.0.0
```

```
network 192.168.11.1 0.0.0.0

network 192.168.100.0 0.0.0.255

network 192.168.101.0 0.0.0.255

#

ssh client first-time enable

ssh client peer 192.168.2.1 assign rsa-key 192.168.2.1

ssh client peer 192.168.2.2 assign rsa-key 192.168.2.2

#

return
```

17.4.5 ACC1 Configuration

```
#

sysname ACC1

#

dhcp enable

#

dhcp snooping enable

#

vlan batch 2 to 3 100

#

stp mode stp

#

aaa

local-user admin1 password irreversible-cipher Huawei@123

local-user admin1 privilege level 3

local-user admin1 service-type ssh

#
```

```
interface Vlanif2

ip address 192.168.2.1 255.255.255.0

#

interface GE1/0/2

port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 2 to 3 100

dhcp snooping enable

dhcp snooping trusted

#

interface GE1/0/4

port link-type trunk

port trunk pvid vlan 3

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 3 100

dhcp snooping enable

#

stelnet server enable

ssh ipv4 server port 1025

ssh user admin1

ssh user admin1 authentication-type password

ssh user admin1 service-type stelnet

ssh server-source all-interface

#

user-interface vty 0 4

authentication-mode aaa

user privilege level 3

protocol inbound ssh
```

```
#  
return
```

17.4.6 ACC2 Configuration

```
#  
sysname ACC2  
  
#  
dhcp enable  
  
#  
dhcp snooping enable  
  
#  
vlan batch 2 101  
  
#  
stp mode stp  
  
#  
aaa  
    local-user admin2 password irreversible-cipher Huawei@123  
    local-user admin2 privilege level 3  
    local-user admin2 service-type ssh  
  
#  
interface Vlanif2  
    ip address 192.168.2.2 255.255.255.0  
  
#  
interface Vlanif101  
    ip address dhcp-alloc  
  
#  
interface GE1/0/2
```

```
port link-type trunk

undo port trunk allow-pass vlan 1

port trunk allow-pass vlan 2 101

dhcp snooping enable

dhcp snooping trusted

#

interface GE1/0/4

port link-type access

port default vlan 101

dhcp snooping enable

#

stelnet server enable

ssh ipv4 server port 1025

ssh user admin2

ssh user admin2 authentication-type password

ssh user admin2 service-type stelnet

ssh server-source all-interface

#

user-interface vty 0 4

authentication-mode aaa

user privilege level 3

protocol inbound ssh

#

return
```

17.5 Quiz

1. In step 7, if WLAN data is forwarded in tunnel mode, which devices and interfaces must permit packets from VLAN 100 so that the STA (wireless terminal PC1) can obtain the service IP address on the network segment 192.168.100.0/24?

Answer:

If tunnel forwarding is used, wireless service traffic is forwarded from AP1 to WAC1 through the CAPWAP tunnel. Therefore, GE0/0/2 of WAC1 and GE1/0/2 of Core1 need to allow packets from VLAN 100 to pass through.